

**FEDERAL HIGHWAY ADMINISTRATION
TURNER-FAIRBANK HIGHWAY RESEARCH CENTER REPORT**

I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation:

Adhesive Anchor Load and Load-Displacement Testing Results

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CHAPTER 1. INTRODUCTION

On July 10, 2006 the ceiling system of the eastbound I-90 Seaport tunnel in Boston, Massachusetts partially collapsed. In support of the National Transportation Safety Board's (NTSB) investigation of this failure, the Federal Highway Administration (FHWA) dispatched a team of engineers and technicians from the Turner-Fairbank Highway Research Center (TFHRC) to conduct static tension load testing and load-displacement testing, and sustained load testing of the adhesive anchor rods used to support the ceiling system. The load testing and load-displacement testing that was conducted on 230 anchor rods during 9 days between July 15, 2006 and July 28, 2006 is the subject of this report.

All testing was conducted in full or substantial compliance with the static tension testing provisions of ASTM E 488 *Standard Test Method for Strength of Anchors in Concrete and Masonry Elements* (hereafter referred as the ASTM Specification). Load testing was conducted between July 15 and July 20 on 188 anchor rods in the eastbound tunnel. Load-displacement testing was conducted on 42 anchors rods July 20, 26, 27 and 28 in the westbound tunnel.

This report summarizes the equipment and procedures used, and the data produced during the testing performed by the FHWA team on the aforementioned dates.

ADHESIVE ANCHOR IDENTIFYING NOMENCLATURE

An identifying nomenclature for the adhesive anchors was established by NTSB personnel. This nomenclature has been consistently used throughout the field investigation and testing, and is illustrated using the following example adhesive anchor descriptor:

EB082-M1-3/4/5-NE

Tunnel:.....*EB* in this example indicates the East Bound tunnel (*EBD* also used).

WB indicates the West Bound tunnel (*WBD* also used).

Panel:.....*082* in this example indicates the panel module (module numbers were taken from the contract plans).

EB panel numbers include *080*, *081*, *082* and *083*.

WB panel numbers include *081*, *082*, *083* and *084*.

Line:.....*M1* in this example indicates hanger line within the tunnel.

EB hanger lines were labeled *N* for northern, *M1* for northern middle, *M2* for southern middle and *S* for southern.

WB hanger lines include *N* for northern, *M* for middle and *S* for southern.

Location:.....Location *3/4/5* in the example indicates the hanger to which clevis and turnbuckle suspension rods 3, 4 and 5 (*3,4,5* or *3-4-5* also used) were attached. Hanger

locations were assigned a label that identified which suspension rods were attached to them. Suspension rods were numbered by panel starting with 1 on the eastern side and incrementally progressing west along a hanger line.

Bolt:.....*NE* in the example indicates the northeastern most adhesive anchor at this location. Locations indicated by a single number (representing a single suspension rod) had a pair of adhesive anchors securing the hanger to the ceiling. Adhesive anchors at these locations were labeled *N* for northern and *S* for southern.

Locations indicated by more than a single number (as in our example) had four adhesive anchors securing the hanger to the ceiling. These adhesive anchors were labeled *NE* for northeastern, *SE* for southeastern, *NW* for northwestern and *SW* for southwestern.

CHAPTER 2. LOAD TESTING

LOAD TEST APPARATUS

The load testing conducted on July 15, 2006 utilized the loading apparatus depicted in Figure 1. A coupler was used to attach an additional length of threaded rod to the in-situ adhesive anchor. Spacer plates, a hollow core 12 ton Enerpac (Model #RCH-123) hydraulic jack and a Wagner Instruments 50-kip load cell were all centered on the threaded rod and held in place with a heavy hex nut. As the spacer plates react off the ceiling at a distance from the anchor of less than the embedment length of the anchor, this system is used for confined testing. A hand pump (Enerpac Model# P392) was used to pressurize and extend the jack transferring load to the epoxy anchor by creating a reaction between the spacer plates and the concrete ceiling. A Vishay Model P3 strain indicator box was used to monitor the load cell and capture the peak load resisted by the adhesive anchor rod prior to failure. The weight of the loading apparatus experienced by the adhesive anchor rod was approximately 30 lbs.

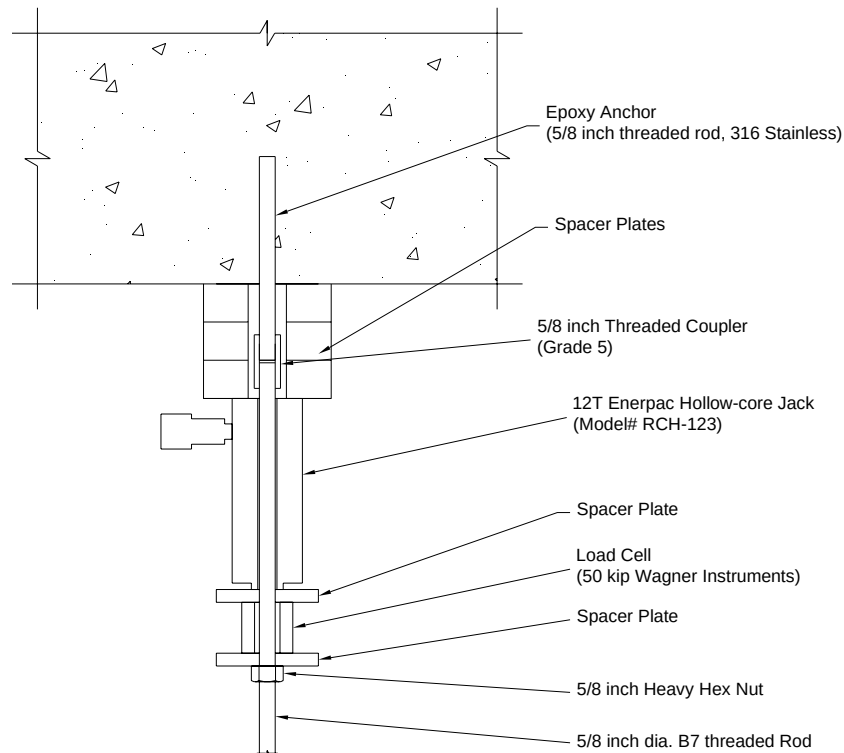


Figure1. Loading apparatus for Confined Testing (July 15, 2006).

Load testing conducted on other than July 15, 2006 used the loading apparatus shown in Figure 2. This system employed two steel wide-flange sections and a “needle-beam” to react the jack off the ceiling and transfer load to the adhesive anchor. As the wide-flanges reacted off the ceiling at a distance from the anchor greater than the depth of anchor embedment, this system is used for unconfined testing. While confined testing is an adequate method for determining adhesive anchor strength when failure occurs in the adhesive, unconfined testing is the preferred

method as it allows failure to occur in the concrete and does not bias any failures occurring in the adhesive by imparting localized compressive stresses.

Referring to Figure 2, the W8 wide-flange sections were attached to the ceiling using 3/8" mechanical anchors and positioned such that there was a minimum clear distance of 11 inches between the side of the anchor rod and the flange tip of each wide-flange section. The needle-beam is comprised of two steel C-channel sections placed back-to-back but welded to end plates that space the webs approximately 2 inches apart. The needle-beam spanned between the two wide-flange sections and was held in place with C-clamps (not shown). Before it was clamped tight, the gap between the needle-beam webs was centered over an anchor rod. A coupler was used to attach an additional length of threaded rod passed through the web gap of the needle-beam to the in-situ adhesive anchor. Washers, the hollow core 12 ton Enerpac hydraulic jack and the Wagner Instruments 50-kip load cell were all centered on the threaded rod and held in place with a heavy hex nut. A hand pump was used to pressurize and extend the jack and a Vishay Model P3 strain indicator box was used to monitor the load cell and capture the peak load resisted by the adhesive anchor prior to failure. The weight of the loading apparatus experienced by the adhesive anchor was approximately 20 lbs.

Two different needle-beams were used. In general, a needle-beam comprised of 4-inch deep C-channels was used to test anchor rods at locations where two anchors were present, and a needle-beam comprised of 6-inch deep C-channels was used to test anchor rods at locations where 4 anchors were present. The deeper channels were required to keep the system elastic while spanning the entire 4 bolt group and maintaining the necessary offsets for unconfined testing.

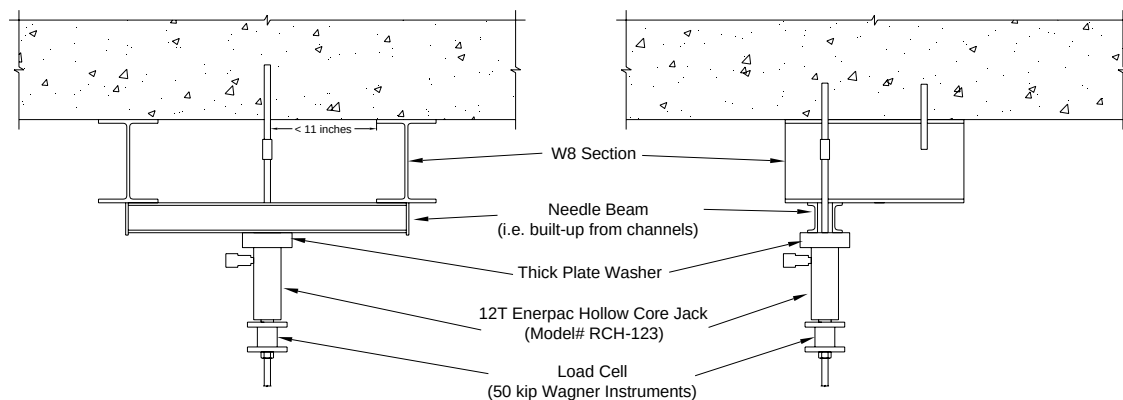


Figure 2. Loading apparatus for Unconfined Testing.

LOAD TEST PROCEDURE

The testing procedures outlined in Appendix A were generally used to conduct all static tension testing. Briefly, the initial adhesive anchor in-situ protrusion was measured and photographed. The load apparatus was then connected to the adhesive anchor rod and the spacer plates, hydraulic jack, and load cell were all aligned. The Vishay Model P3 strain indicator was balanced and zeroed, and then configured to display peak minimum load. Then, loading proceeded in accordance with the ASTM Specification.

Load was applied at an approximate rate of one stroke from the hand pump every 5 seconds. This typically caused the first peak load to occur in 45 seconds or less. At this point, peak pressure in the hydraulic jack and peak load from the strain indicator were recorded. The peak load was reset on the strain indicator and loading continued until a second post failure peak load was attained. The post failure peak load was then recorded. Then the adhesive anchor was removed from the ceiling by continued jacking. Once out of the ceiling, custody of the adhesive anchor was taken by an accompanying crime scene investigator (FBI Special Agent or MA State Policeman). This process was consistently repeated for all load tests conducted July 15-19, 2006.

LOAD TEST RESULTS

The tables in this section review the initial protrusion, pre-test displacement (if any) and peak load measured during each load test reported. The data contained in these tables summarizes that reported for each adhesive anchor individually in Appendix B. A pre-test photograph of the in-situ condition of each anchor is also included in Appendix B.

Initial protrusion and pre-test displacement measurements were made using an adjustable hand-square. If the tables in this section do not contain an entry for pre-test displacement then the anchor displayed no visible sign of displacement prior to testing. Protrusion and displacement were measured in fractions of an inch to the nearest 1/32 inches and reported in decimal form to 3 places. These measurements do not intend to report accuracy to 3 decimal places; rather the numbers are solely the result of a truncated conversion from fraction to decimal form.

Initial protrusion measurements were made by placing the end of the hand-square rule against the tunnel ceiling and adjusting the headpiece so that the 90-degree leg rested on the end of the adhesive anchor as shown in the figures of Appendix B. Pre-test displacement measurements were made in a similar manner except the headpiece was adjusted to rest on the exposed face of the adhesive retaining (red) cap flange and a measurement was made on both the east and west sides of the anchor.

Integral to most all pre-test displacement measurements reported herein is the thickness of the adhesive retaining cap flange which is approximately 1/32 of an inch thick. However, there were occasions when the cap was obviously displaced from its initial position on the anchor. In those instances a measurement to the approximate pre-displaced position of the cap was made that did not include the thickness of the cap flange. In other cases separation between the ceiling and the cap flange existed but was not easily identified as displacement from the initial condition or quantified because of its relatively small magnitude.

Table 1 contains data for the 16 adhesive anchor rods load tested with the confined test method on July 15, 2006.

Table 1. Summary of Load Test Data Collected on July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
EB080-M2-6-N	3.063	a	a	8682
EB080-M2-6-S	3.031	a	a	9117
EB080-M2-3/4/5-NW	2.875			14300
EB080-M2-3/4/5-SW	3.000	a	a	11904
EB080-M2-3/4/5-NE	3.125			16443
EB080-M2-3/4/5-SE	3.031	a	a	6695
EB080-M2-2-N	3.125			14580
EB080-M2-2-S	3.094			11648
EB080-M2-1-N	3.000			16679
EB080-M2-1-S	3.000			14562
EB081-M2-12-N	2.938			11312
EB081-M2-12-S	3.563			12258
EB081-M2-9/10/11-NW	3.000			14126
EB081-M2-9/10/11-SW	3.063			7093
EB081-M2-9/10/11-NE	3.000			15803
EB081-M2-9/10/11-SE	3.188	a	a	6043
^a – Slight pre-test displacement observed but not quantified.				

Table 2 contains data for the 22 adhesive anchor rods load tested with the unconfined test method on July 16, 2006.

Table 2. Summary of Load Test Data Collected on July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-Test Displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
EB081-M2-8-N	2.875			11846
EB081-M2-8-S	2.688			13117
EB081-M2-7-N	2.875	a	a	11011
EB081-M2-7-S	3.000	a	a	12175
EB081-M2-6-N	2.875			10368
EB081-M2-6-S	2.750			22770
EB081-M2-5-N	2.813			12045
EB081-M2-5-S	2.813			3179
EB081-M2-1-N	2.813			13030
EB081-M2-1-S	2.813			15285
EB082-M2-12-N	3.250	0.375	0.469	4755
EB082-M2-12-S	2.813			17196
EB082-M2-11-N	3.188	0.250	0.250	6437
EB082-M2-11-S	2.813			14215
EB082-M2-7.5	2.875			7485
EB082-M2-7-N	3.125			6966
EB082-M2-7-S	3.031			14155
EB082-M2-6.5	2.938			13450
EB083-M2-10-N	3.500	0.438	0.375	1676
EB083-M2-10-S	2.938			20626
EB083-M2-11.5-N	3.000	a	a	6041
EB083-M2-11.5-S	2.938			7450
^a – Slight pre-test displacement observed but not quantified.				

Table 3 contains data for the 22 adhesive anchor rods load tested with the unconfined test method on July 17, 2006.

Table 3. Summary of Load Test Data Collected on July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-Test Displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
EB083-M2-5-N	3.031			17793
EB083-M2-5-S	3.000			19069
EB083-M2-11-N	3.438	0.500	0.438	6608
EB083-M2-11-S	2.969			17558
EB083-S-10-N	3.000			14515
EB083-S-10-S	2.906			15022
EB082-S-7-N	3.000			12522
EB082-S-7-S	2.938			11107
EB082-S-7.5-N	3.063			3329
EB082-S-7.5-S	3.625	0.625	0.625	2768
EB081-S-9-N	3.094			12058
EB081-S-9-S	3.313	0.438	0.406	4489
EB081-S-10-N	3.250			5633
EB081-S-10-S	4.125	1.375	1.375	1639
EB080-M1-2-N	4.688	1.906	1.906	1600
EB080-M1-2-S	3.000			19972
EB080-M1-1-N ^a	3.438	0.500	0.438	9437
EB080-M1-1-S	5.375	2.625	2.625	1121
EB082-M1-12-N ^a	3.000	0.313	0.375	8969
EB082-M1-12-S ^a	5.500	2.500	2.500	1559
EB082-M1-11-N	2.875			16732
EB082-M1-11-S	4.000	1.250	1.250	3448
^a – Slight bend north.				

Table 4 contains data for the 50 adhesive anchor rods load tested with the unconfined test method on July 18, 2006.

Table 4. Summary of Load Test Data Collected on July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-Test Displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
EB080-S-3-N ^a	3.000			12324
EB080-S-3-S	3.031			14736
EB080-S-2-N	3.000			10104
EB080-S-2-S	3.063			13428
EB080-S-4/5/6-NW	2.938			12673
EB080-S-4/5/6-NE	2.906			16552
EB080-S-4/5/6-SW	3.156			8021
EB080-S-4/5/6-SE	2.906			9914
EB081-S-8-N	2.781			18622
EB081-S-8-S	3.156	0.531	0.563	2793
EB080-S-1-N	3.156			15039
EB080-S-1-S	3.000			10692
EB081-S-4/5/6-NW	2.906			12375
EB081-S-4/5/6-NE	3.031			12618
EB081-S-4/5/6-SW ^b	2.938			8000 ^c
EB081-S-4/5/6-SE	3.000			7440
EB081-S-3-N	3.000			11166
EB081-S-3-S	2.969			8220
EB081-S-7-N	2.781			10613
EB081-S-7-S	2.813			8165
EB081-S-2-N	3.000			2683
EB081-S-2-S	3.063			8568
EB081-S-1-N	2.906			10855
EB081-S-1-S	2.781			9184
EB082-S-10-N	2.781			9894
EB082-S-10-S	2.750			12526
EB082-S-9-N	2.844			14964
EB082-S-9-S	2.938			13454
EB082-S-9.5-N	2.969			10156
EB082-S-9.5-S	3.094			4928
EB082-S-4/5/6-NW	3.000			9497
EB082-S-4/5/6-NE	2.969			12191
EB082-S-4/5/6-SW	2.875			14872
EB082-S-4/5/6-SE	2.906			12820
EB082-S-3-N	3.000			14185

EB082-S-3-S	3.063			23115
EB082-S-2-N	2.969			10040
EB082-S-2-S	3.000			9576
EB082-S-1-N	3.000			18021
EB082-S-1-S	3.125			1851
EB083-S-9-N	2.938			11535
EB083-S-9-S	3.000			18876
EB083-S-8-N	2.875			17684
EB083-S-8-S	3.063			8251
EB083-S-7-N	2.906			9208
EB083-S-7-S	2.906			12811
EB083-S-4/5/6-NW	2.875			13373
EB083-S-4/5/6-NE	2.875			10932
EB083-S-4/5/6-SW	2.969			13328
EB083-S-4/5/6-SE ^b	2.938			17395
^a – Slight bend south. ^b – Shallow cone failure. ^c – Unreliable peak load due to load cell cable separation.				

Table 5 contains data for the 70 adhesive anchor rods load tested with the unconfined test method on July 19, 2006.

Table 5. Summary of Load Test Data Collected on July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-Test Displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
EB083-S-3-N	2.938			9014
EB083-S-3-S	2.938			14928
EB083-S-2-N	3.125			15296
EB083-S-2-S	3.125			5379
EB083-S-1-N	3.031			8389
EB083-S-1-S	3.438	0.250	0.219	9511
EB082-S-8-N	3.156			13387
EB082-S-8-S	3.063			7566
EB082-S-8.5-N	2.813			17323
EB082-S-8.5-S	2.813			20264
EB083-M2-1-N	3.125	0.281	0.625 ^e	8867
EB083-M2-1-S	3.031	0.125	0.125	12977
EB083-M2-2-N	2.875			18356
EB083-M2-2-S	2.938			16914
EB083-M2-3/4-NW	3.031			20198
EB083-M2-3/4-NE	3.000			15516
EB083-M2-3/4-SW	2.875			21725
EB083-M2-3/4-SE	3.094			18260
EB083-M2-6-N	3.063			9622
EB083-M2-6-S	3.063			12766
EB083-M2-7/8/9-NW	3.063			11340
EB083-M2-7/8/9-NE ^a	3.000			18854
EB083-M2-7/8/9-SW	3.031	0.281	0.281	9130
EB083-M2-7/8/9-SE	3.156	0.094	0.125	9925
EB082-M2-1-N	3.063			10998
EB082-M2-1-S	3.063			15552
EB082-M2-2-N	3.000			19933
EB082-M2-2-S	3.188			6304
EB082-M2-3/4/5-NW ^b	3.000			20367
EB082-M2-3/4/5-NE	3.094	^d	^d	13083
EB082-M2-3/4/5-SW	3.000			17207
EB082-M2-3/4/5-SE	3.156	^d	^d	9740
EB082-M2-6-N	3.063			17061
EB082-M2-6-S	3.125			13888
EB082-M2-8/9/10-NW	3.125	0.313	0.281	4152

EB082-M2-8/9/10-NE	3.094	0.188	0.219	7444
EB082-M2-8/9/10-SW	3.000			12995
EB082-M2-8/9/10-SE	2.938			12956
EB081-M2-2/3/4-NW	2.719	0.063	0.063	16119
EB081-M2-2/3/4-NE	2.781			15732
EB081-M2-2/3/4-SW	2.781			13192
EB081-M2-2/3/4-SE	2.813			14278
EB083-N-3-N	2.969			16830
EB083-N-3-S	3.000			24242
EB083-M1-1-N	2.969			14935
EB083-M1-1-S	3.031			12524
EB083-M1-2-N	3.000			16930
EB083-M1-2-S	3.000	^d	^d	9297
EB082-M1-8/9/10-NW	3.438	0.625	0.594	6021
EB082-M1-8/9/10-NE	3.219	0.344	0.344	10204
EB082-M1-8/9/10-SW	3.281	0.438	0.406	9979
EB082-M1-8/9/10-SE	3.188	0.156	0.188	10678
EB080-M1-3/4/5-NW	3.719	0.750	0.750	3562
EB080-M1-3/4/5-NE	3.938	0.875	0.875	3762
EB080-M1-3/4/5-SW	3.156	0.313	0.344	10740
EB080-M1-3/4/5-SE	3.250	0.438	0.438	7369
EB080-N-4/5/6-NW	2.969			14716
EB080-N-4/5/6-NE	3.250	0.313	0.281	1854
EB080-N-4/5/6-SW	3.344			19955
EB080-N-4/5/6-SE	3.188	0.313	0.313	5737
EB081-N-10-N ^b	5.125	2.250	2.219	8605
EB081-N-10-S ^b	3.063			5020
EB081-N-9-N ^c	2.969			18822
EB081-N-9-S	3.000			19721
EB081-N-4/5/6-NW	3.000	0.125	0.125	17658
EB081-N-4/5/6-NE	2.938	0.125	0.125	8264
EB081-N-4/5/6-SW	2.813			15017
EB081-N-4/5/6-SE	2.938			11653
EB082-N-10-N	2.813			13595
EB082-N-10-S	2.875			16962
^a – Shallow cone failure. ^b – Slight bend south. ^c – Slight bend north. ^d – Slight pre-test displacement observed but not quantified. ^e – Possible measurement recording error as magnitude does not match photograph.				

Table 6 contains data for the 8 adhesive anchor rods load tested with the unconfined test method on July 20, 2006.

Table 6. Summary of Load Test Data Collected on July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-Test Displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
WBD083-M-9/10/11-NW	3.063	^b	^b	9280
WBD083-M-9/10/11-SW	4.688	1.875	1.875	1403
WBD083-M-9/10/11-SE	4.625	1.813	1.813	1679
WBD082-S-2-N	2.875			12201
WBD082-S-2-S	2.469 ^c			12164
WBD082-M-6-N	3.000	0.125	0.125	5598
WBD082-M-6-S	4.000	1.188	1.188	2727
WBD082-M-5-S	3.813	1.031	1.031	470 ^a
^a – Suspect value, load was much lower than corresponding pressure value (approximately 3500 lbs based on recorded pressure of 1400 psi). ^b – Slight pre-test displacement observed but not quantified. ^c – Possible measurement recording error as magnitude does not match photograph.				

CHAPTER 3. LOAD-DISPLACEMENT TESTING

LOAD-DISPLACEMENT TEST APPARATUS

For the load-displacement testing conducted on July 20, 2006, load and displacement measurements were recorded by hand as the adhesive anchor rods were loaded using the setup shown in Figure 2. Figure 3 shows the displacement measurement system employed on this date. A steel plate was bolted to the concrete ceiling with an expansion anchor. A base fitted with an articulating arm which held a digital dial indicator was then magnetized to the plate. The steel cantilever plate shown was double-nutted to the adhesive anchor with a heavy-hex nut and a threaded coupler. The cantilever was located with the digital dial indicator by adjusting the articulating arm and functioned as the reference for movement of the adhesive anchor rod.

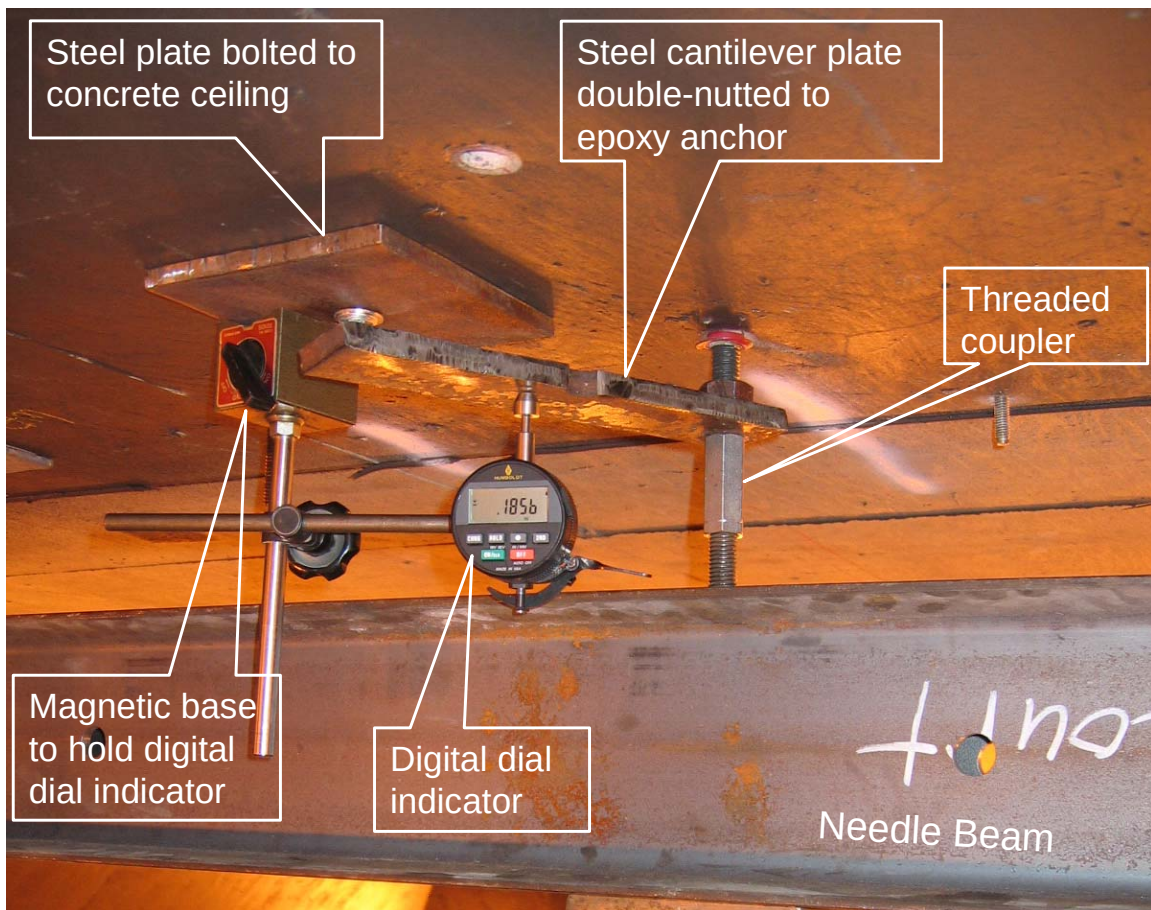


Figure 3. Displacement measurement system for July 20, 2006.

Load-displacement testing conducted July 26-28, 2006, also used the loading setup shown in Figure 2. However, displacement measurements were made using three linear variable differential transformers (LVDT). For these measurements, the LVDTs were mounted to a circular aluminum plate which was double-nutted to the adhesive anchor as shown in Figure 4. The 7-inch diameter aluminum plate had a 0.75-inch hole in the middle. The three LVDTs were

spaced evenly on a 6-inch diameter circle and were held into the aluminum plate with nylon thumb screws.

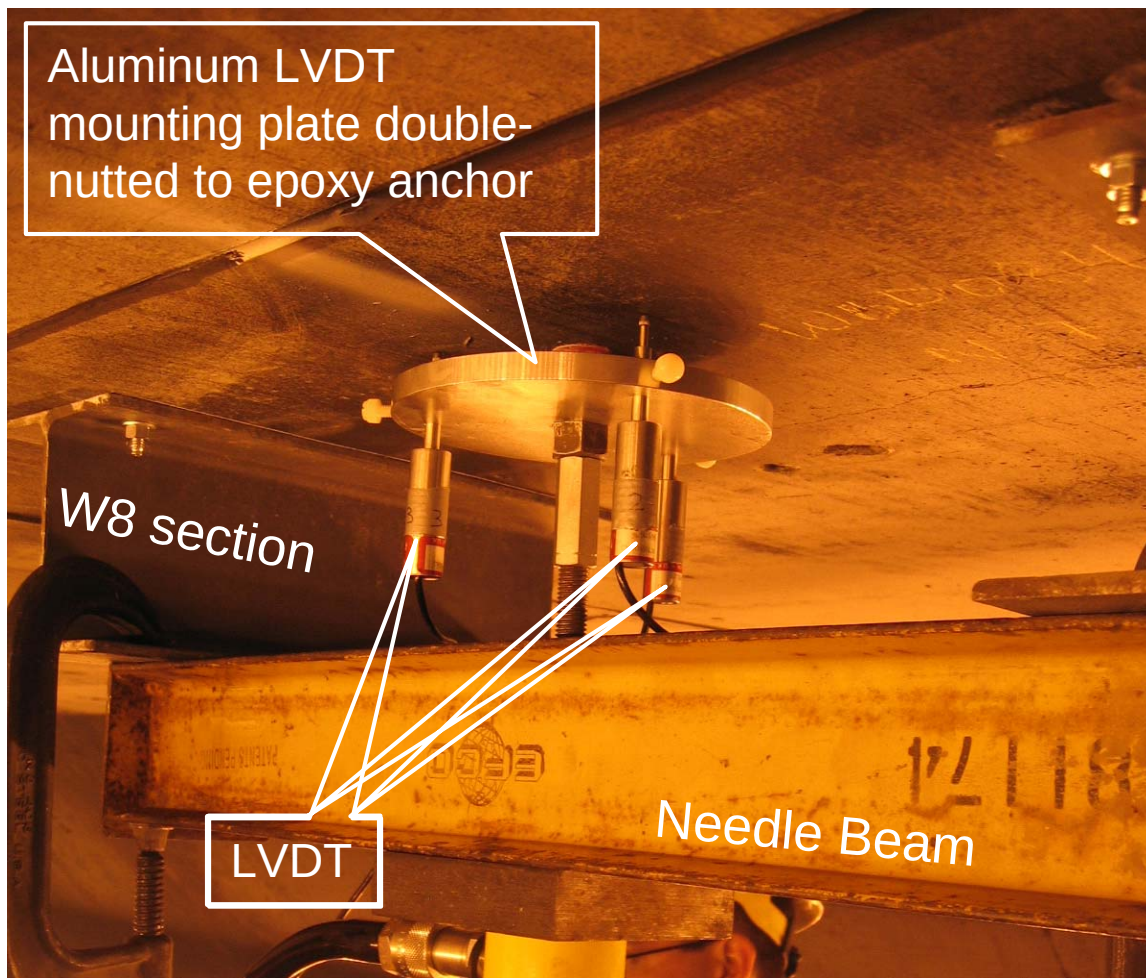


Figure 4. LVDT mounting plate.

Also for the load-displacement testing conducted July 26-28, 2006, concurrent load and displacement measurements were recorded using computerized high-speed data acquisition systems. Two distinct data acquisition systems were used for measurement collection, an Optum Electronic MEGADAC 3008 and a Data Translation model 9804 USB Data Acquisition Function Module.

LOAD-DISPLACEMENT TEST PROCEDURE

In general, the load-displacement tests were conducted using the procedure detailed in Appendix A and in accordance with the ASTM Specifications. Briefly, after the load and displacement measurement instruments were properly attached to the adhesive anchor, a small load was applied to the system to eliminate any seating displacements from the loading apparatus from effecting the test measurement. Then, an initial load of 500 lbs was applied and held for two minutes. This was followed by applying load at a rate of one stroke from the hand pump every 5

seconds until an approximate magnitude of 2000 lbs was reached. Again, the load was maintained at this level for approximately 2 minutes. After two minutes, the load was increased by approximately 1500 lbs to 3500 lbs at the usual rate of loading and held for another two minutes. This process was repeated by increasing the last pre-pause load level by 1500 lbs until the adhesive anchor was no longer capable of resisting the applied load. Load and displacement measurements were captured continuously during these tests using electronic data acquisition systems except those tests conducted on July 20, 2006 during which measurements were recorded manually at specific increments of time. Finally, a post peak load was attained and the rod was removed from the tunnel ceiling using the hydraulic jack.

The ASTM protocol for load–displacement testing specifies load increments of 15% of the expected ultimate strength of the adhesive anchor rod. The 1500 pound increment chosen for these tests was based on the results from the load tests conducted as a part of this investigation. The field data from those 180 load tests indicated a mean pull-out strength of approximately 10,000 lbs for adhesive anchors in the eastbound tunnel.

LOAD–DISPLACEMENT TEST RESULTS

July 20, 2006

Load–displacement testing July 20, 2006 was conducted on 7 adhesive anchors. Displacement and load measurements were recorded by hand. For the anchors tested this date, no attempt was made to sustain load during the load level increments pauses. The load drop during each pause associated with the compliance of the system and the performance of the jack is evident in the test records.

The peak load data reported for each anchor individually in Appendix B is summarized in Table 7. Plots of the load vs. displacement and time vs. displacement records for each anchor tested are shown in Figures 5 through 11.

Table 7. Summary of Load–Displacement Test Data Collected on July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-Test Displacement		Peak Load (lbs)
		East Side (inches)	West Side (inches)	
WBD084-N-4/5/6-NW	2.938			11336
WBD084-N-4/5/6-NE	3.063			7447
WBD084-N-4/5/6-SW	3.031			12733
WBD084-N-4/5/6-SE	3.000			8844
WBD082-M-5-N	2.875			12701
WBD083-N-9-N	3.156	0.469	0.438	3233
WBD083-N-9-S	2.875			11309

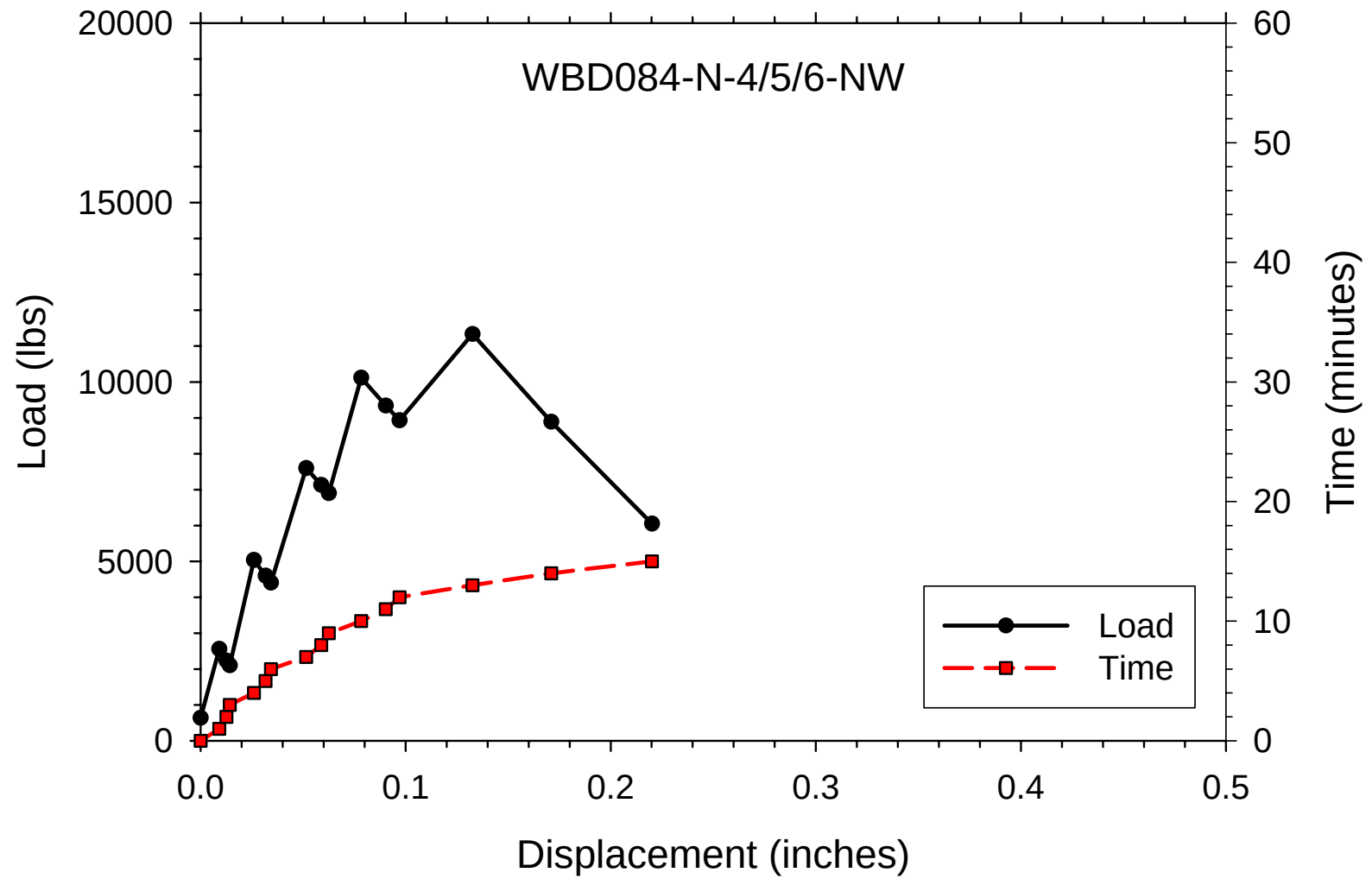


Figure 5. Load and Time vs. Displacement Plot for Anchor WBD084-N-4/5/6-NW.

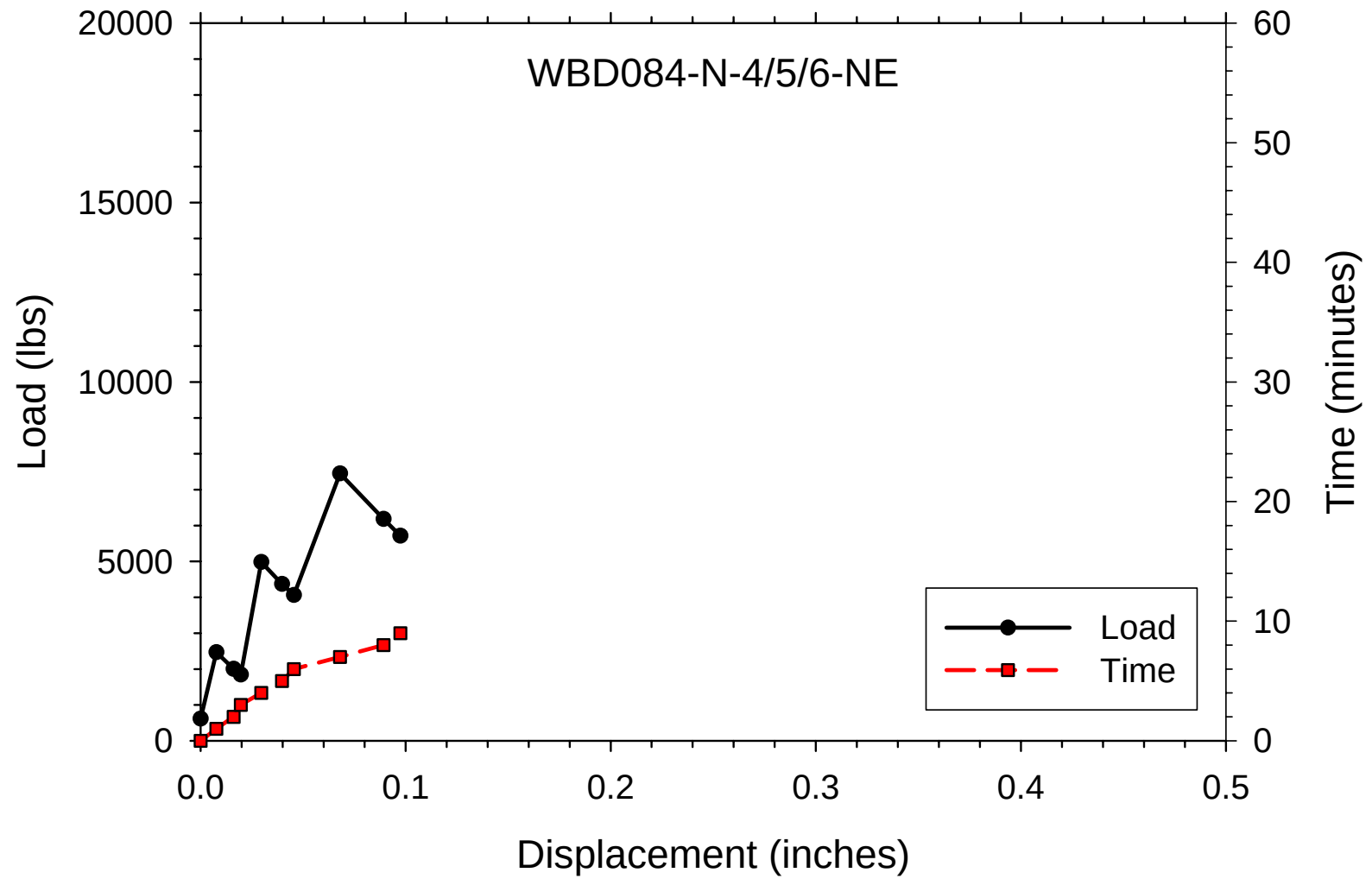


Figure 6. Load and Time vs. Displacement Plot for Anchor WBD084-N-4/5/6-NE.

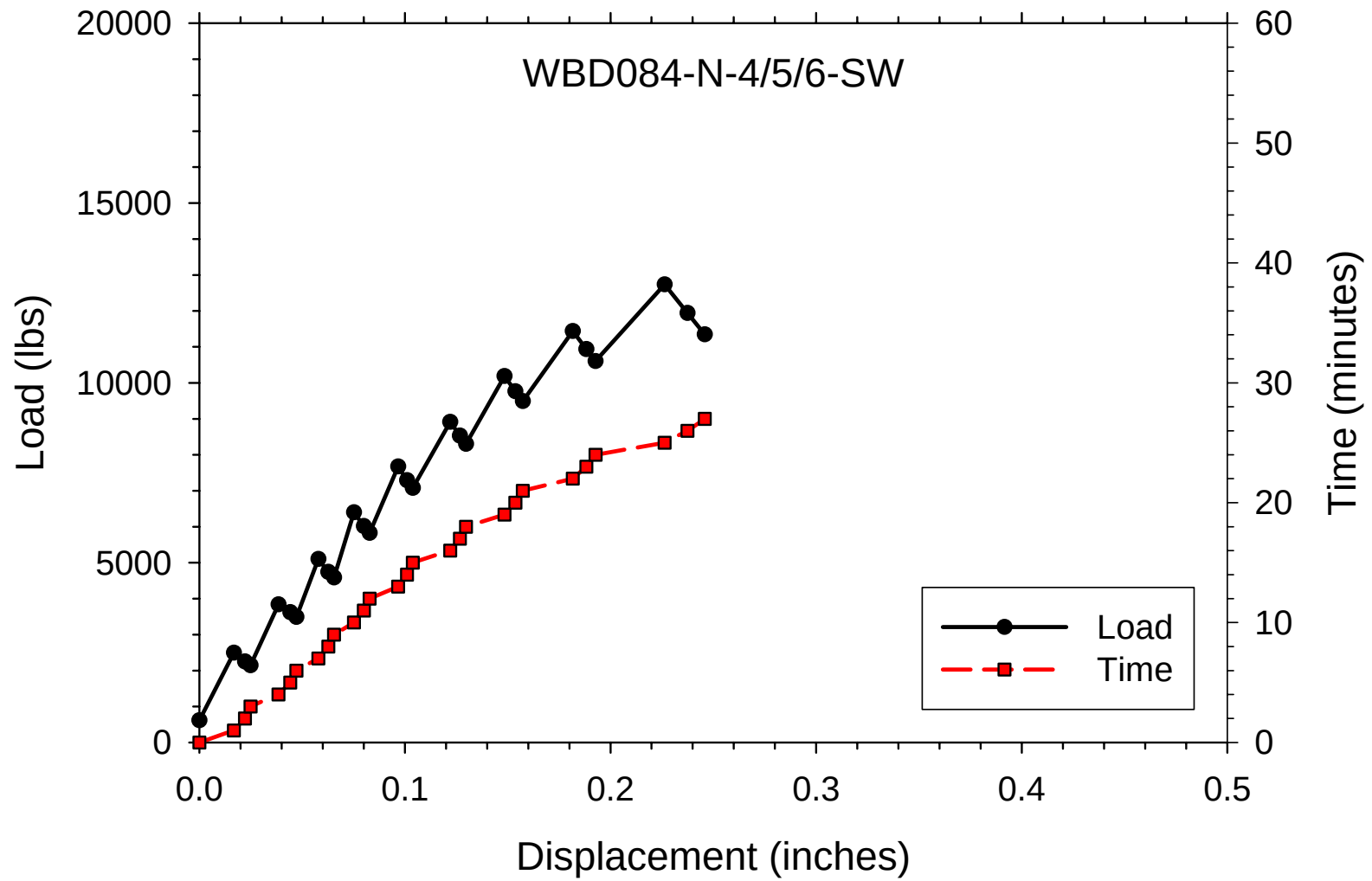


Figure 7. Load and Time vs. Displacement Plot for Anchor WBD084-N-4/5/6-SW.

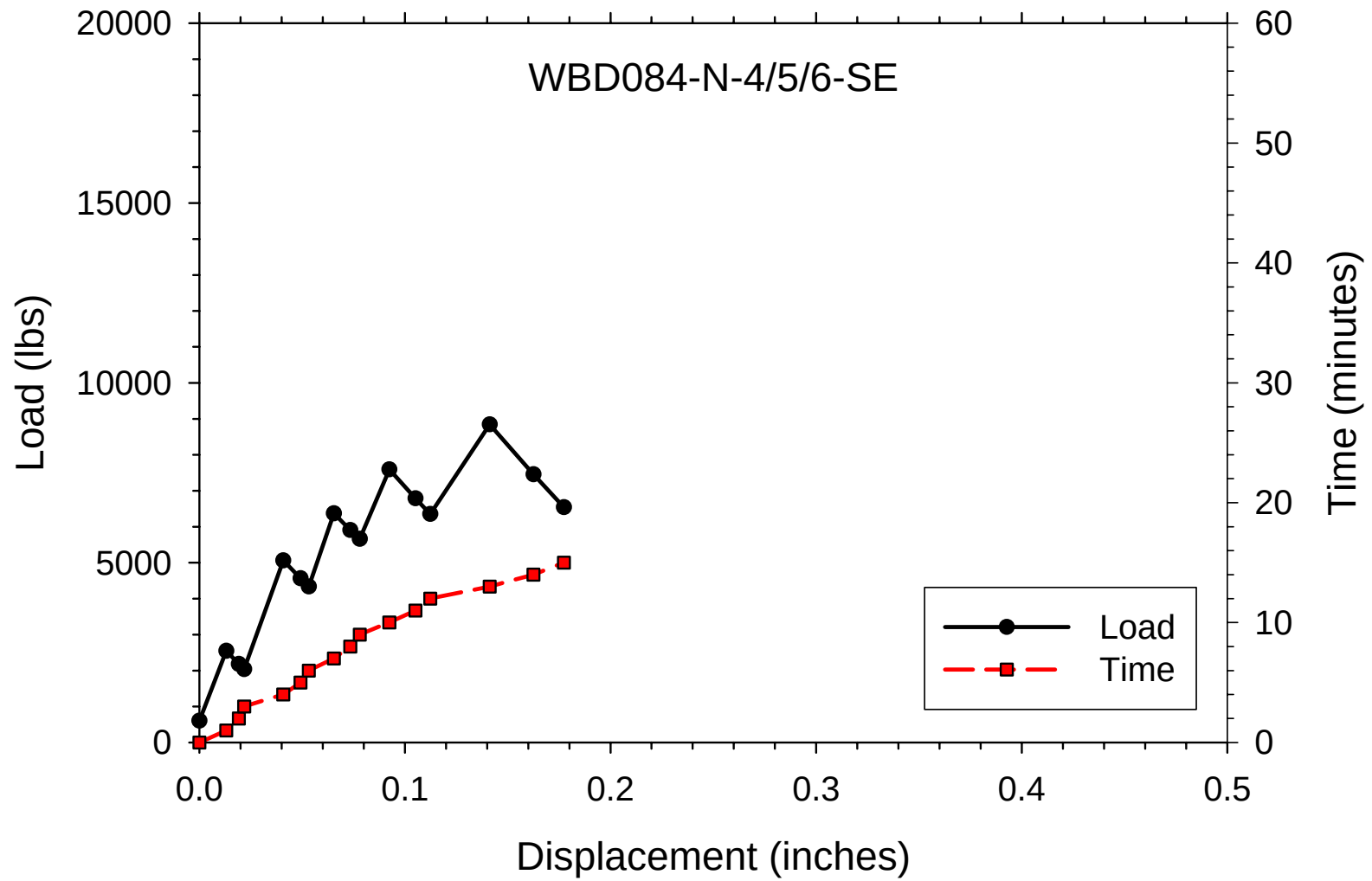


Figure 8. Load and Time vs. Displacement Plot for Anchor WBD084-N-4/5/6-SE.

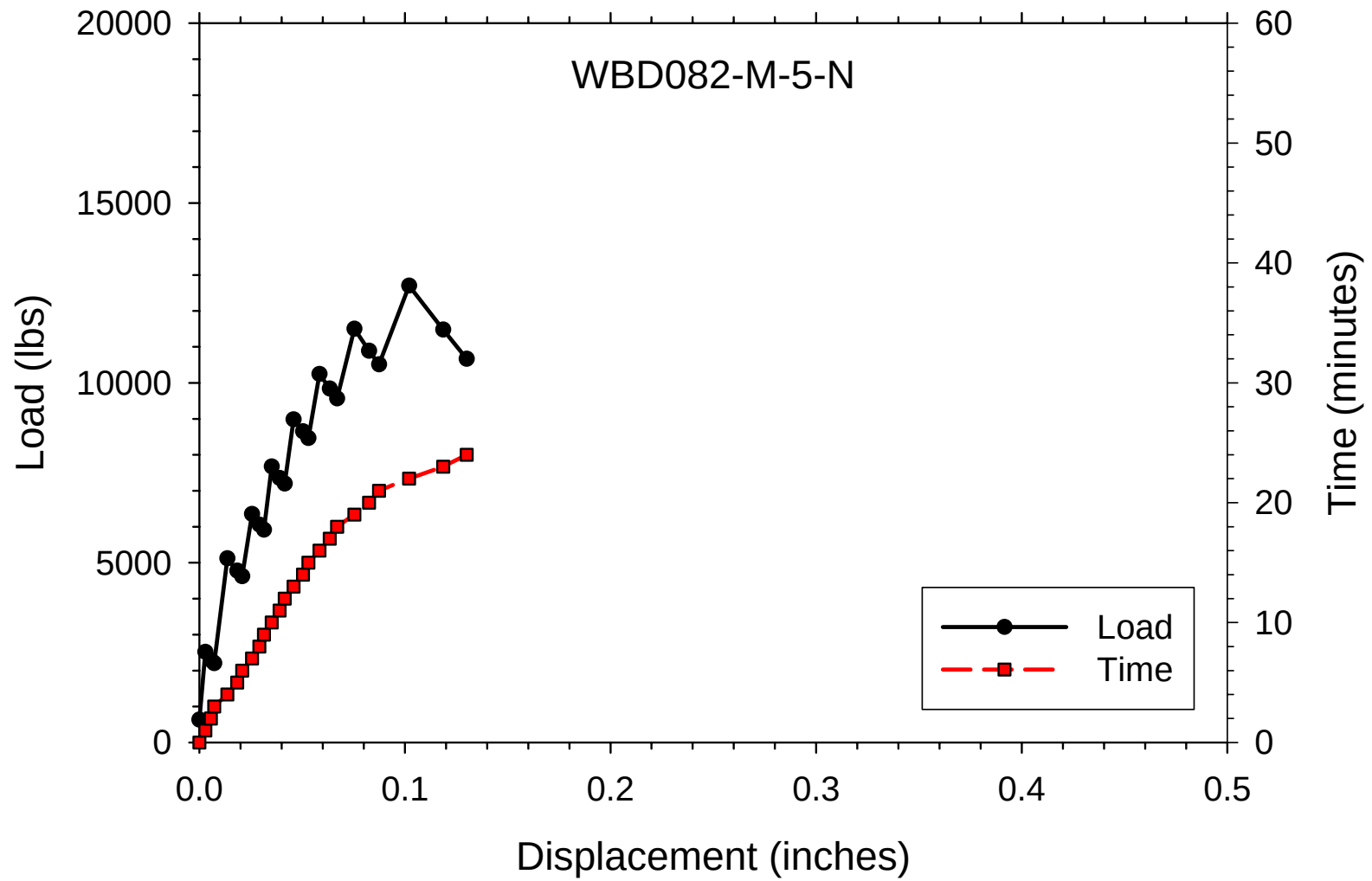


Figure 9. Load and Time vs. Displacement Plot for Anchor WBD082-M-5-N.

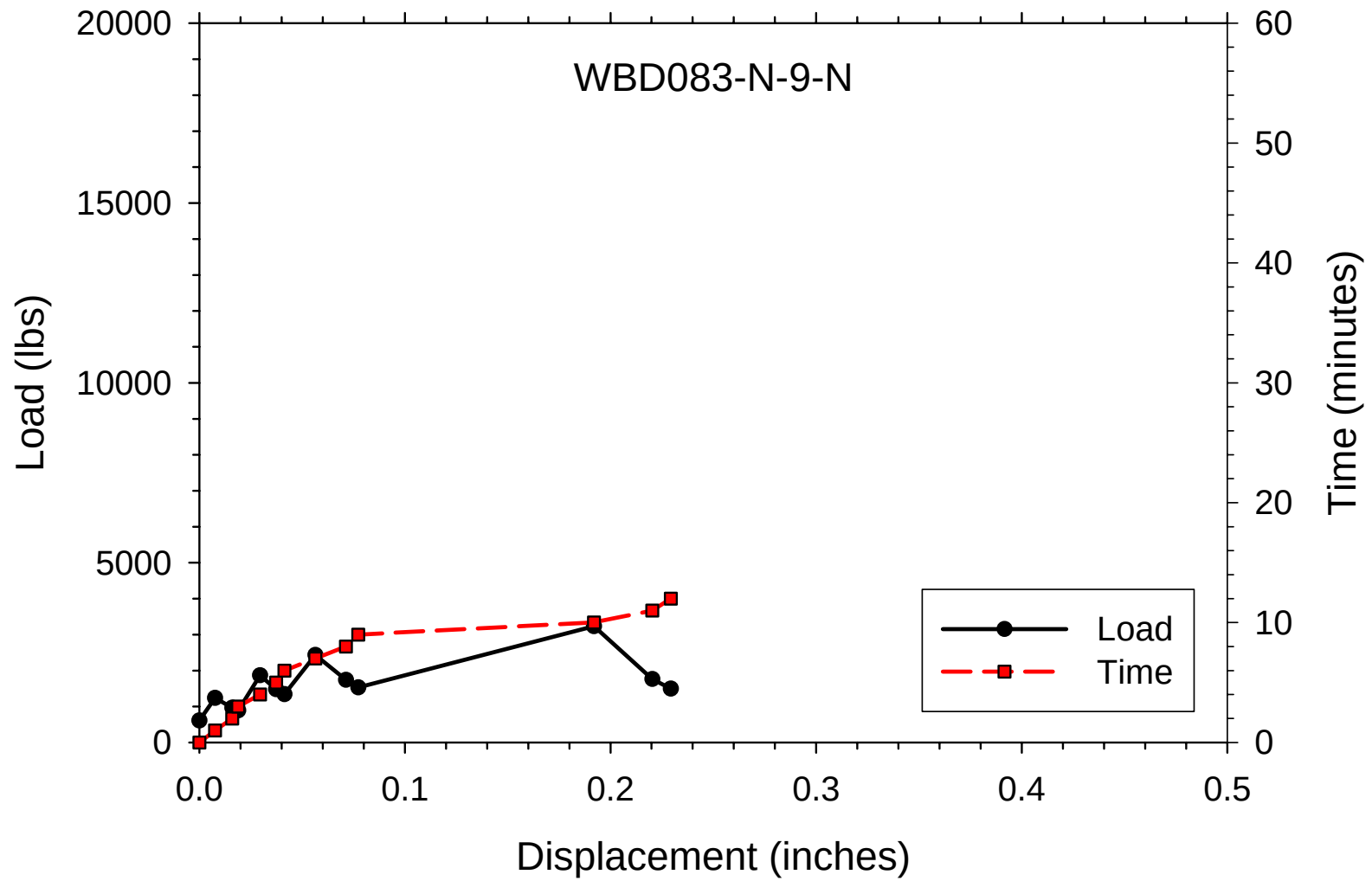


Figure 10. Load and Time vs. Displacement Plot for Anchor WBD083-N-9-N.

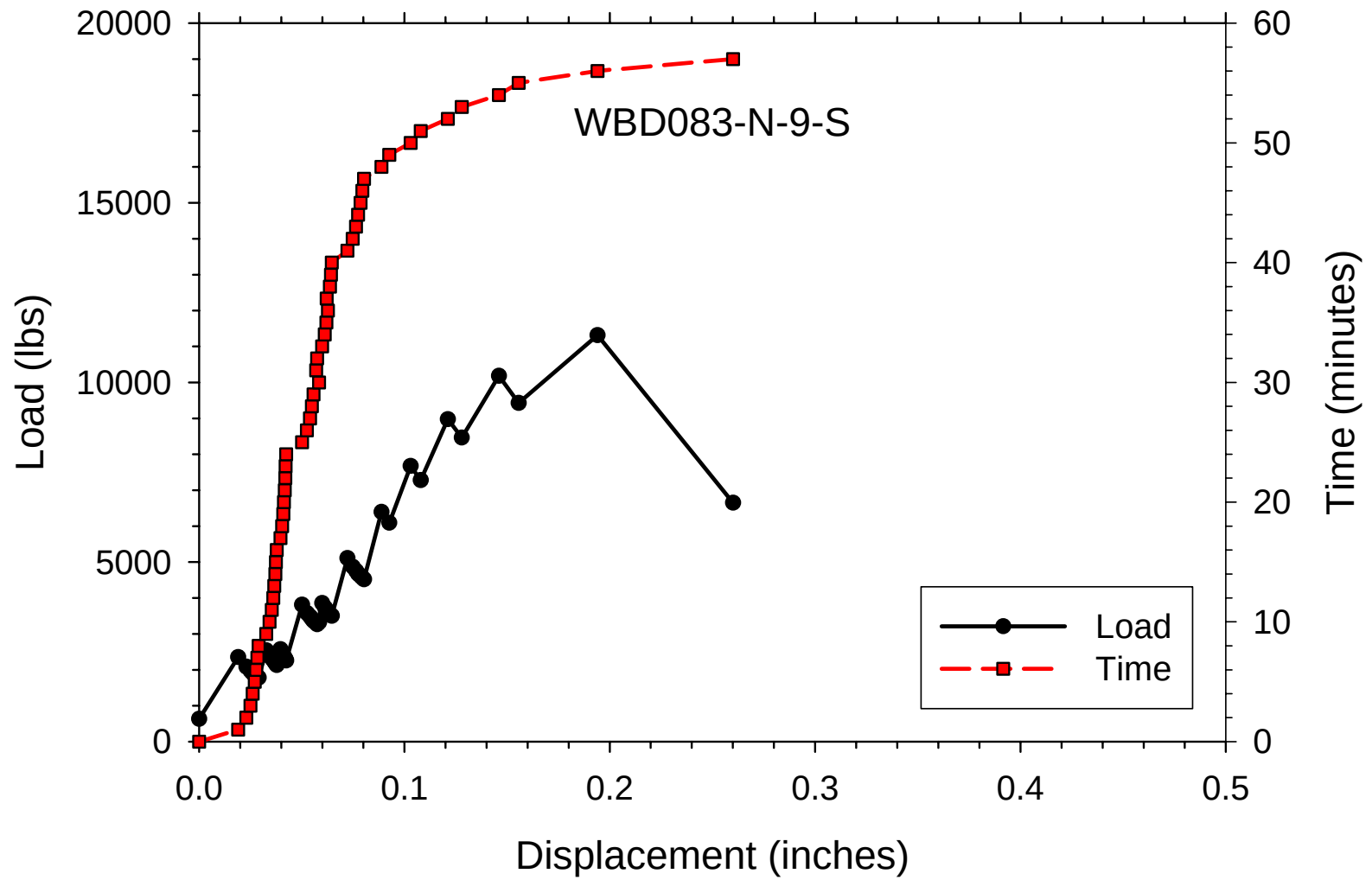


Figure 11. Load and Time vs. Displacement Plot for Anchor WBD083-N-9-S.

July 26-28, 2006

Load–displacement testing July 26-28, 2006 was conducted on 35 adhesive anchors. Displacement and load measurements were recorded using electronic data acquisition systems detailed elsewhere in this report. The peak load data reported for each anchor individually in Appendix B is summarized in Tables 8 through 10. Plots of the load vs. displacement and time vs. displacement records for each anchor tested are shown in Figures 12 through 45.

The majority of these tests were conducted in accordance with the ASTM Specifications. However, during 9 of these tests, investigators varied hold times or loading from strict adherence to the ASTM Specifications in an attempt to more broadly identify the behavior of the anchor system. These variations are seen in the test records of WBD082-M-2/3/4-SE, WBD082-M-2/3/4-SW, WBD082-M-2/3/4-NW, WBD082-M-2/3/4-NE, WBD082-M-8-N and WBD082-M-8-S conducted on July 26, 2006, and in the test records of WBD082-N-8-N, WBD082-N-8-S, WBD082-N-9-S and WBD082-N-9-N conducted on July 27, 2006. In addition, the load-displacement testing conducted July 28, 2006 concerned adhesive anchors that sustained an approximate 1-ton load for 48 hours prior to testing.

Table 8. Summary of Load–Displacement Test Data Collected on July 26, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-6-S	3.625	0.5	0.625	7000 / 2
WBD084-M-8-N	3.5625	0.75	0.75	5700 / 2
WBD084-M-8-S	3.0625			16500 / 2
WBD084-N-7-N	2.96875			10000 / 2
WBD084-N-7-S	2.90625			7000 / 1
WBD084-M-2/3/4-NW	3.96875	1.28125	1.28125	1500 / 2
WBD084-M-2/3/4-NE	4	1.1875	1.1875	5800 / 2
WBD084-M-2/3/4-SW	3.1875	0.1875	0.1875	4200 / 0
WBD084-M-2/3/4-SE	3.3125	0.34375	0.34375	5800 / 1
WBD084-S-10-N	3.28125	0.1875	0.1875	7000 / 1
WBD084-S-10-S	4	1	1	1000 / 2
WBD082-M-2/3/4-SE	2.875 ^a			6000 / 10
WBD082-M-2/3/4-SW	3.5	0.625	0.625	3000 / 3.5
WBD082-M-2/3/4-NW	3.54	0.433	0.433	3000 / 1.5
WBD083-M-8-N ^b	4.5	1.5	1.5	1500 / 1
WBD083-M-8-S	3.125			7000 / 8
^a – Measurement scaled from photograph.				
^b – Anchor threads were covered in epoxy preventing use of LVDTs. While, load-displacement test procedures were followed, only load and time were recorded. Therefore, a load-displacement figure could not be created.				

Table 9. Summary of Load–Displacement Test Data Collected on July 27, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD081-M-2/3/4-NW	2.9375			7000 / 1
WBD081-M-2/3/4-NE	3.03125	0.21875	0.21875	12500 / 1.5
WBD081-M-2/3/4-SW	2.9375	0.15625	0.15625	12500 / 1.5
WBD081-M-2/3/4-SE	2.875	0.1875	0.1875	11000 / 1.5
WBD081-N-1-N	2.875			11000 / 1.5
WBD081-N-1-S	2.84375			12500 / 1.5
WBD083-M-12-N	3.5	0.4375	0.40625	4500 / 1
WBD083-M-12-S	4.25	1.1875	1.1875	4500 / 0.33
WBD084-N-10-N	3.0625			12500 / 1
WBD084-N-10-S	3.03125			5800 / 1
WBD084-N-9-N	3.125			16500 / 1.5
WBD084-N-9-S	3.03125	0.125	0.125	15000 / 1.5
WBD084-M-6-N	3.0625	0.125	0.125	17500 / 1
WBD082-N-8-N	3.0625			10000 / 10
WBD082-N-8-S	3.0625			6000 / 5
WBD082-N-9-S	3.125			5000 / 9
WBD082-N-9-N	3.09375			5000 / 25

Table 10. Summary of Load–Displacement Test Data Collected on July 28, 2006.

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) / (min.)
		East Side (inches)	West Side (inches)	
WBD082-S-1-S	3.0625			9500 / 1
WBD083-S-10-S	2.96875			9500 / 0

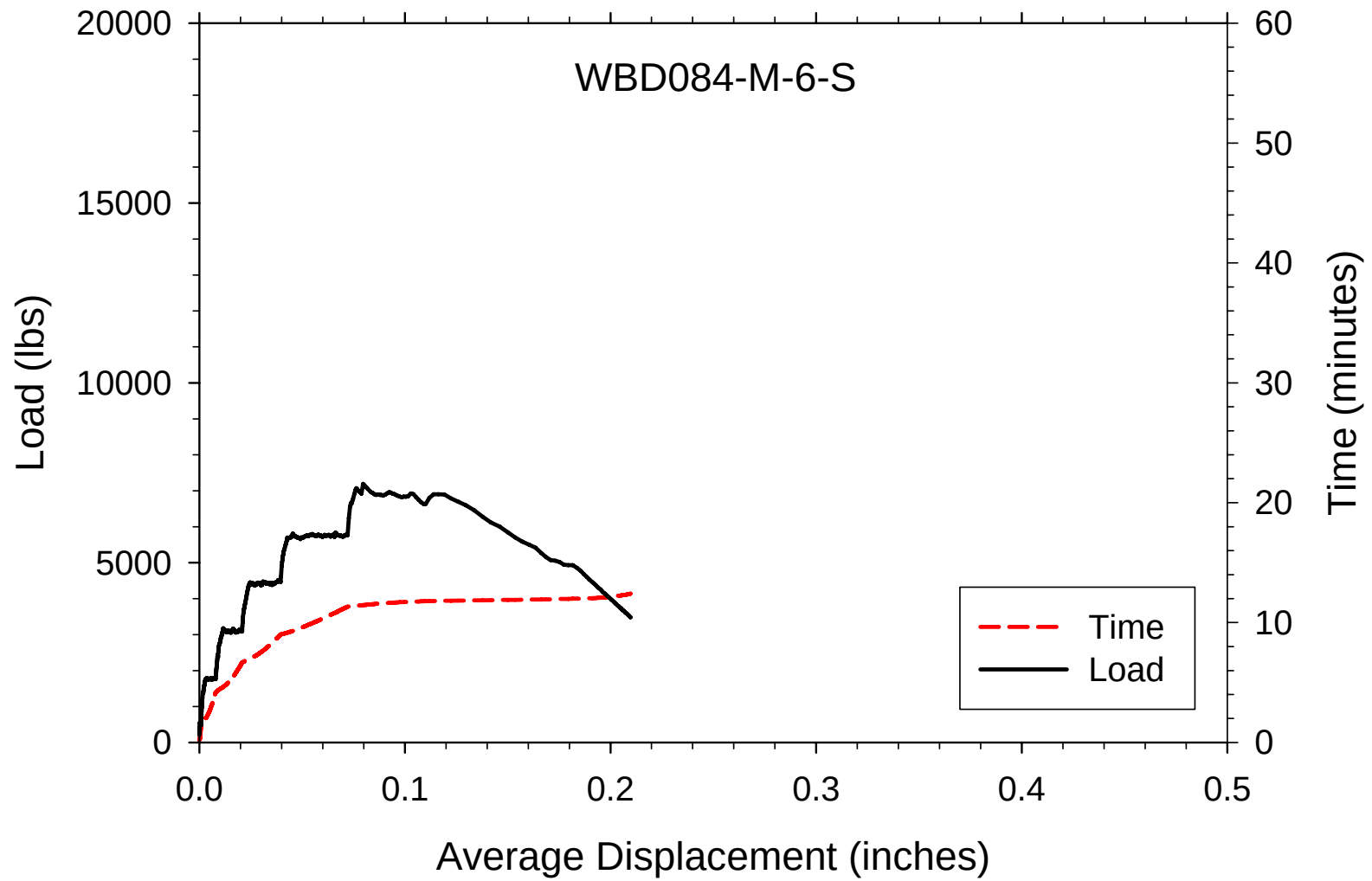


Figure 12. Load and Time vs. Displacement Plot for Anchor WBD084-M-6-S.

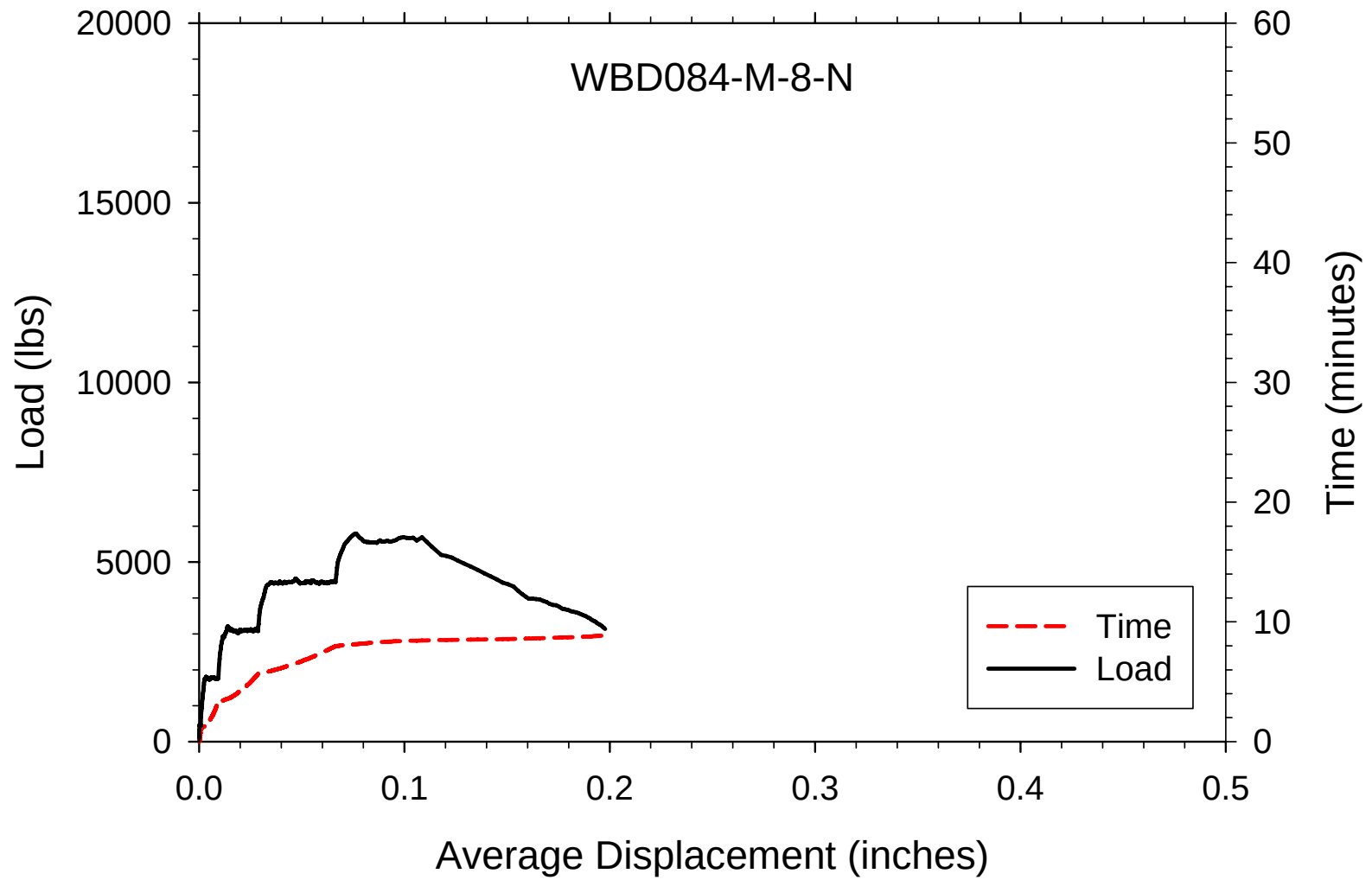


Figure 13. Load and Time vs. Displacement Plot for Anchor WBD084-M-8-N.

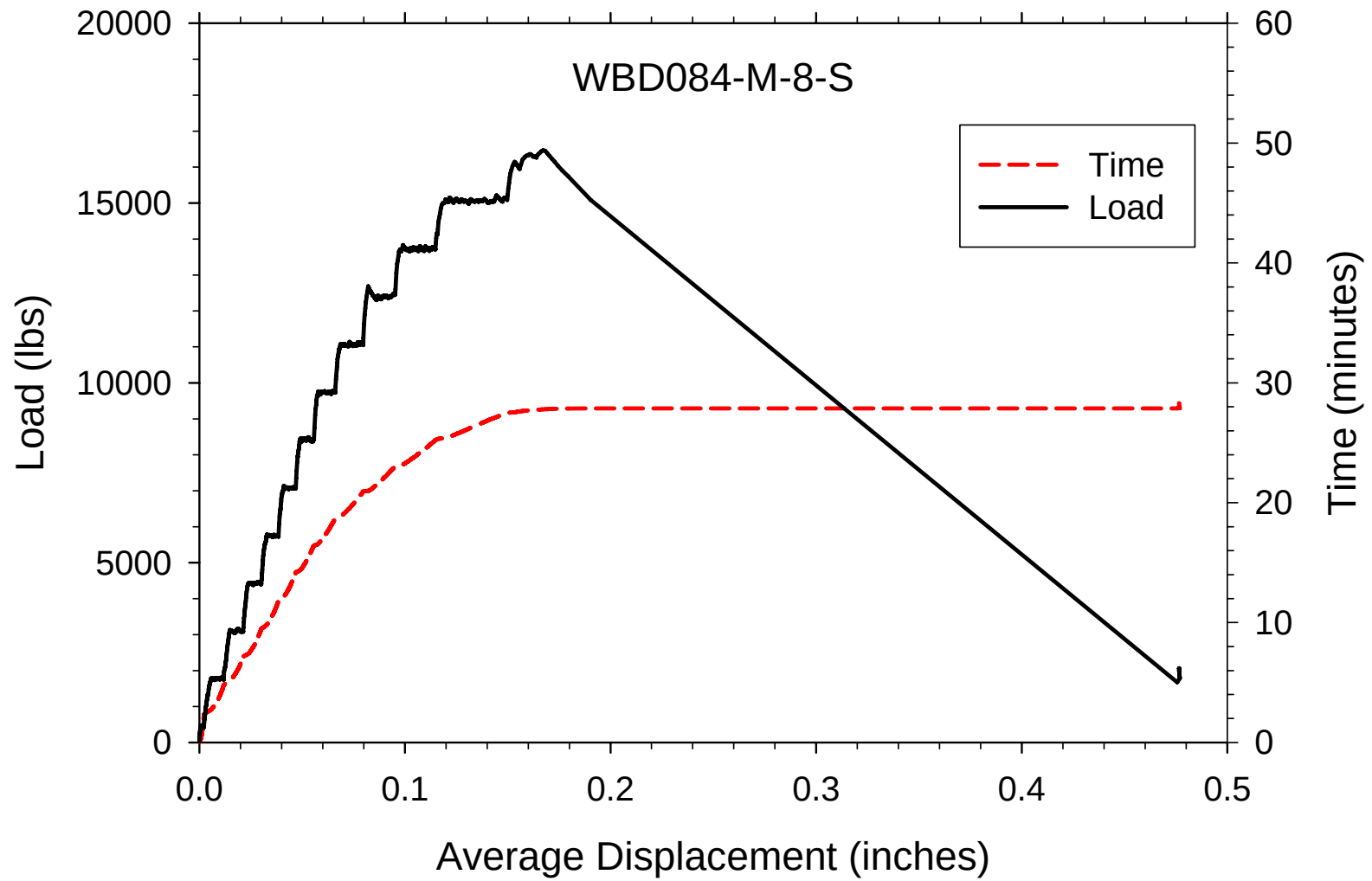


Figure 14. Load and Time vs. Displacement Plot for Anchor WBD084-M-8-S.

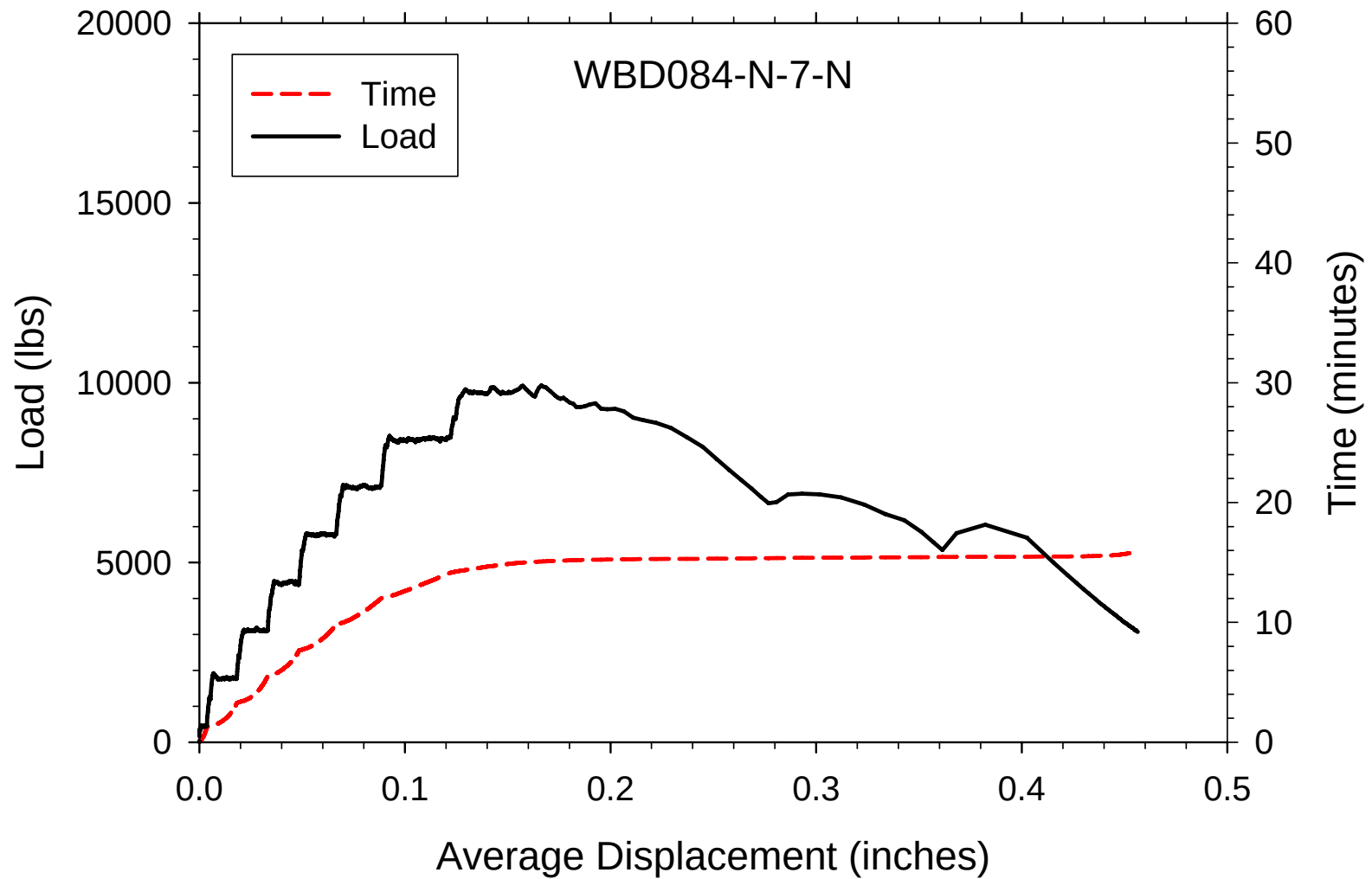


Figure 15. Load and Time vs. Displacement Plot for Anchor WBD084-N-7-N.

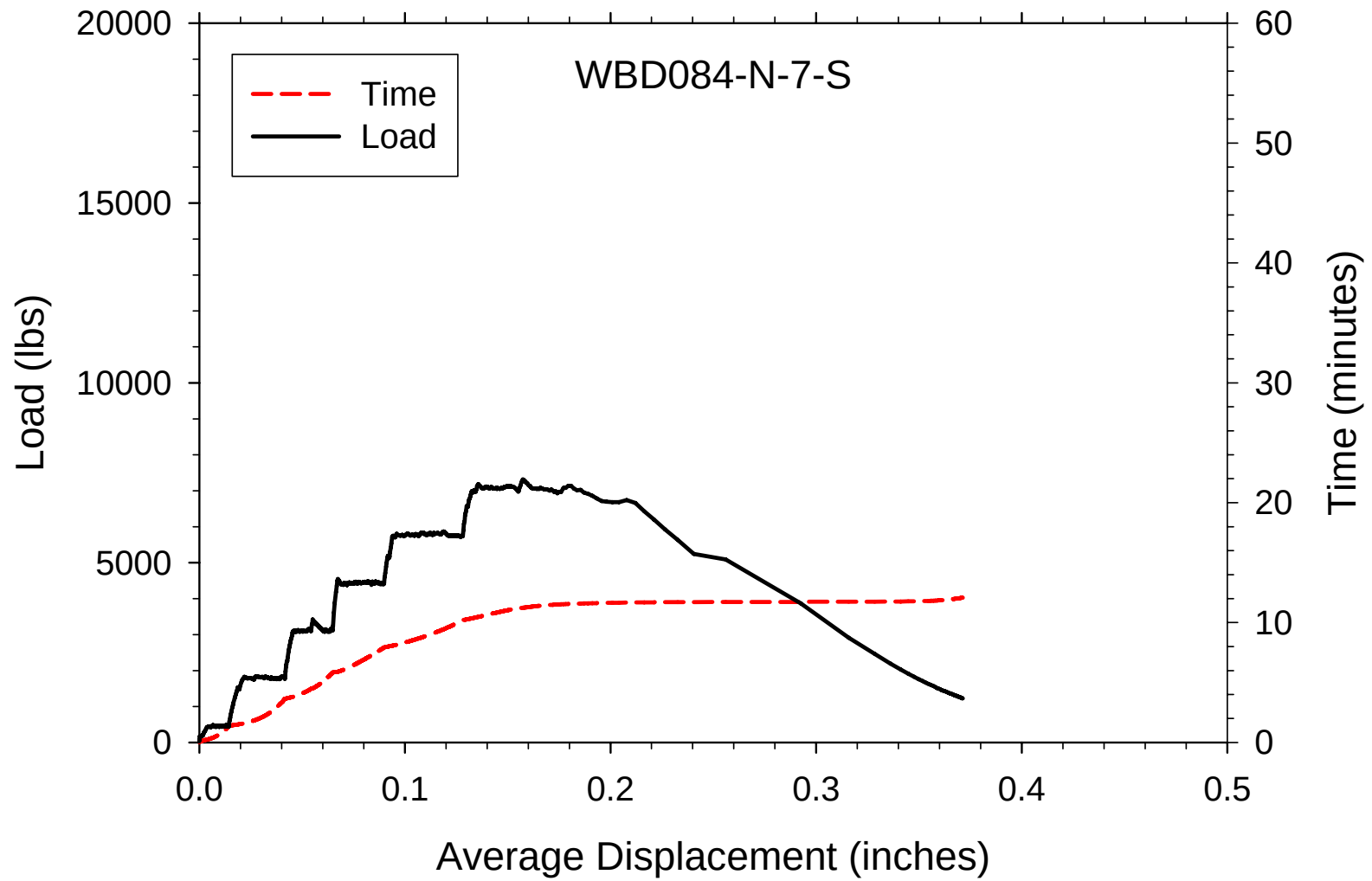


Figure 16. Load and Time vs. Displacement Plot for Anchor WBD084-N-7-S.

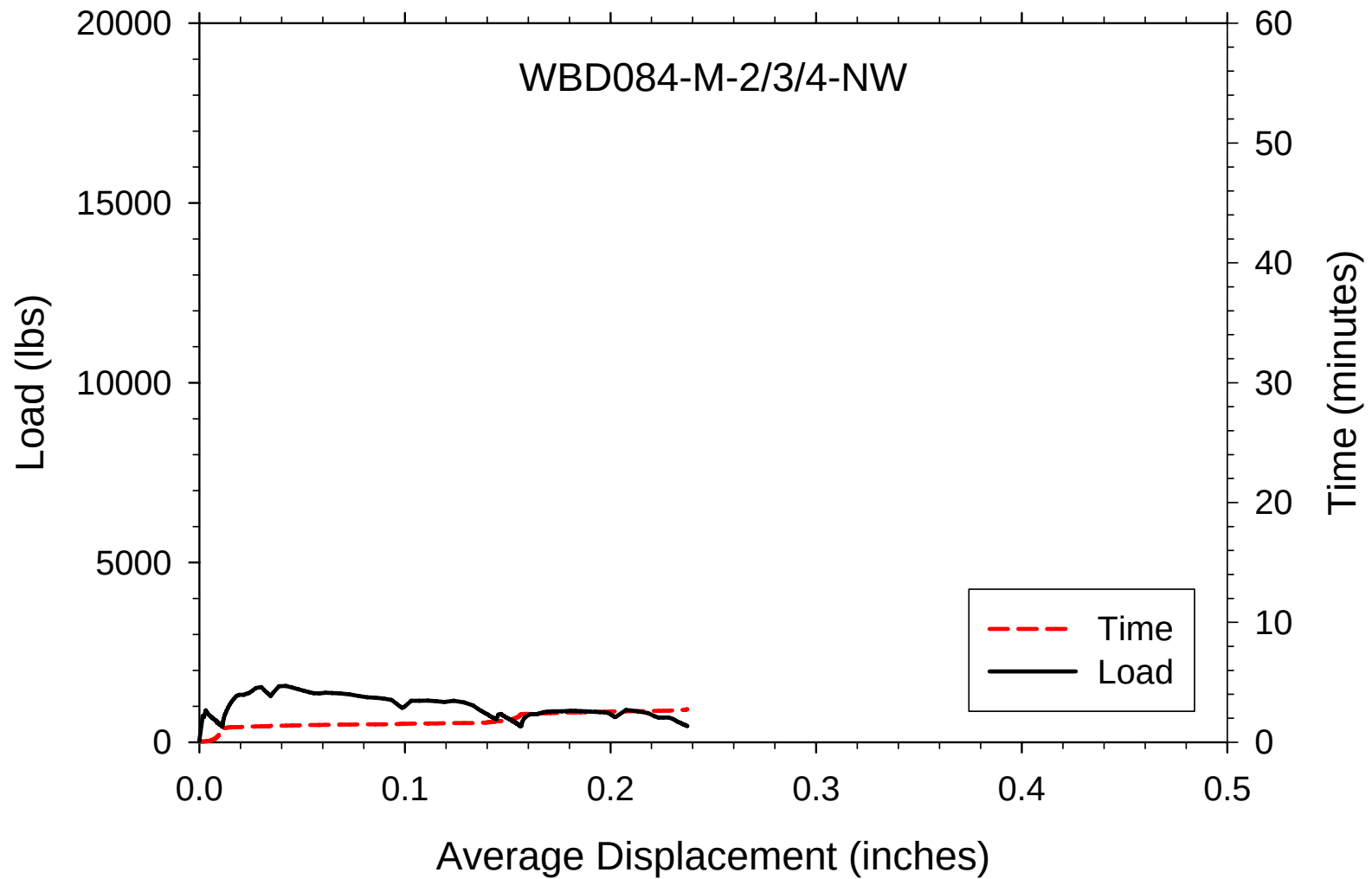


Figure 17. Load and Time vs. Displacement Plot for Anchor WBD084-M-2/3/4-NW.

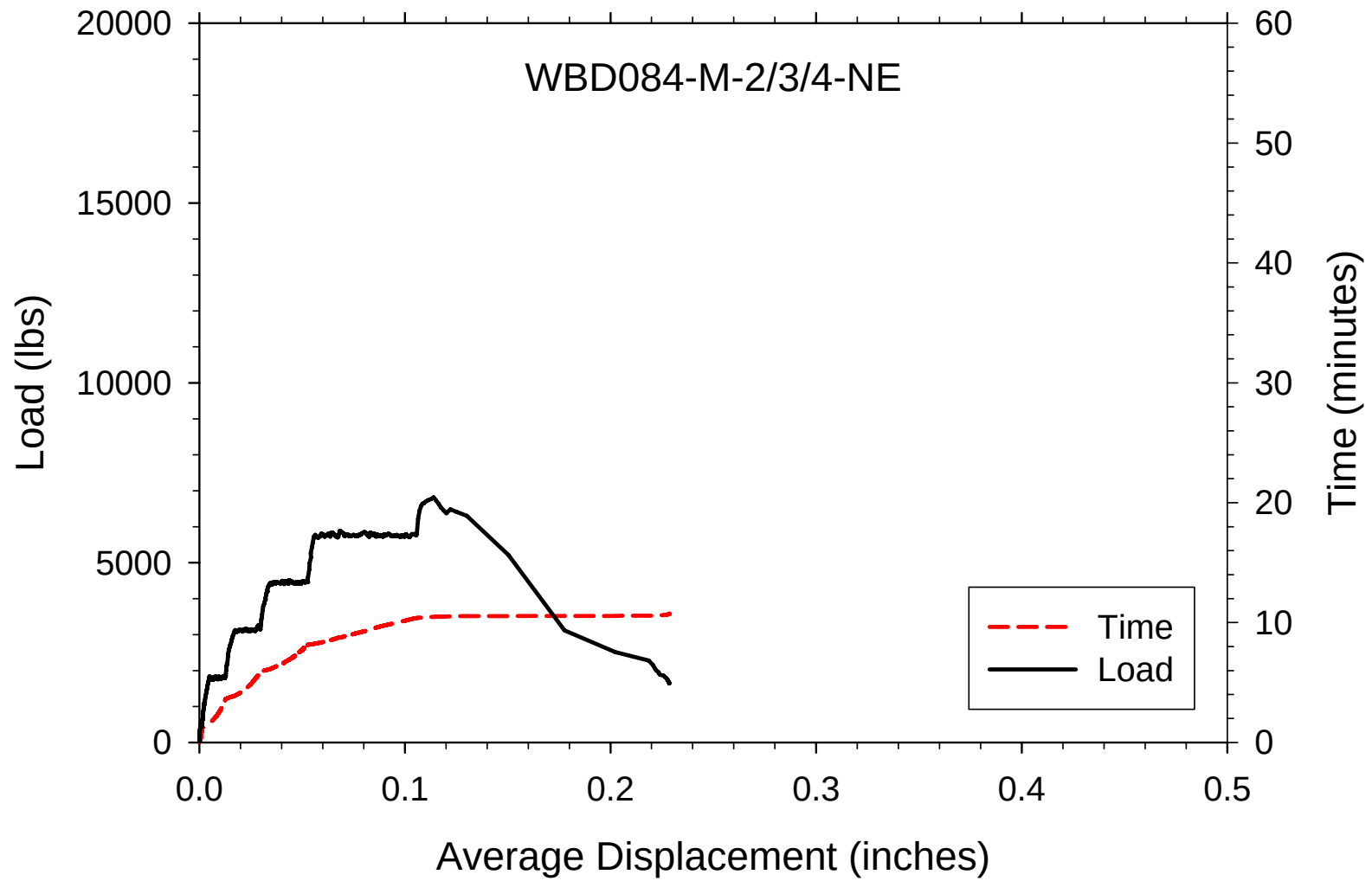


Figure 18. Load and Time vs. Displacement Plot for Anchor WBD084-M-2/3/4-NE.

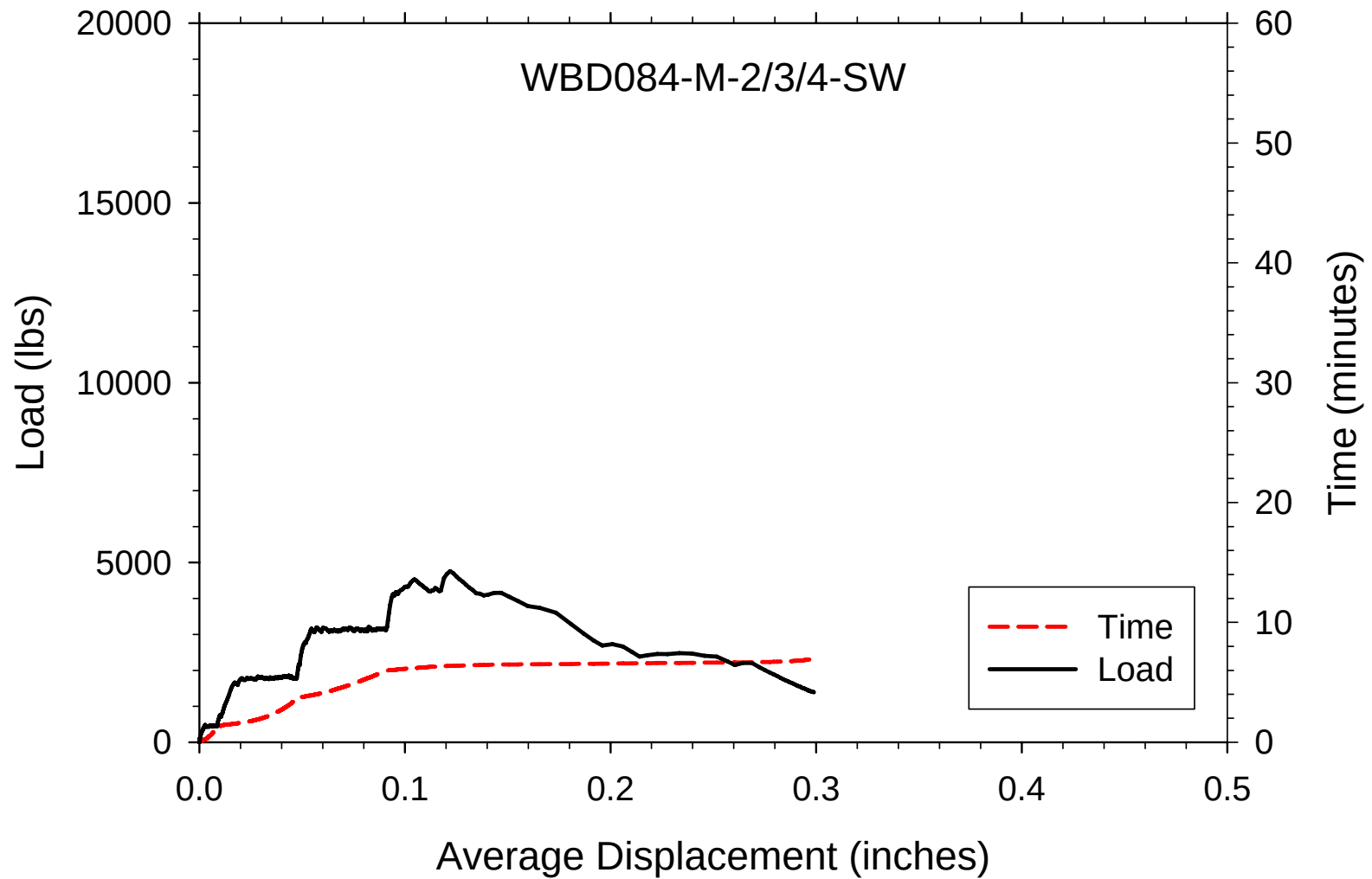


Figure 19. Load and Time vs. Displacement Plot for Anchor WBD084-M-2/3/4-SW.

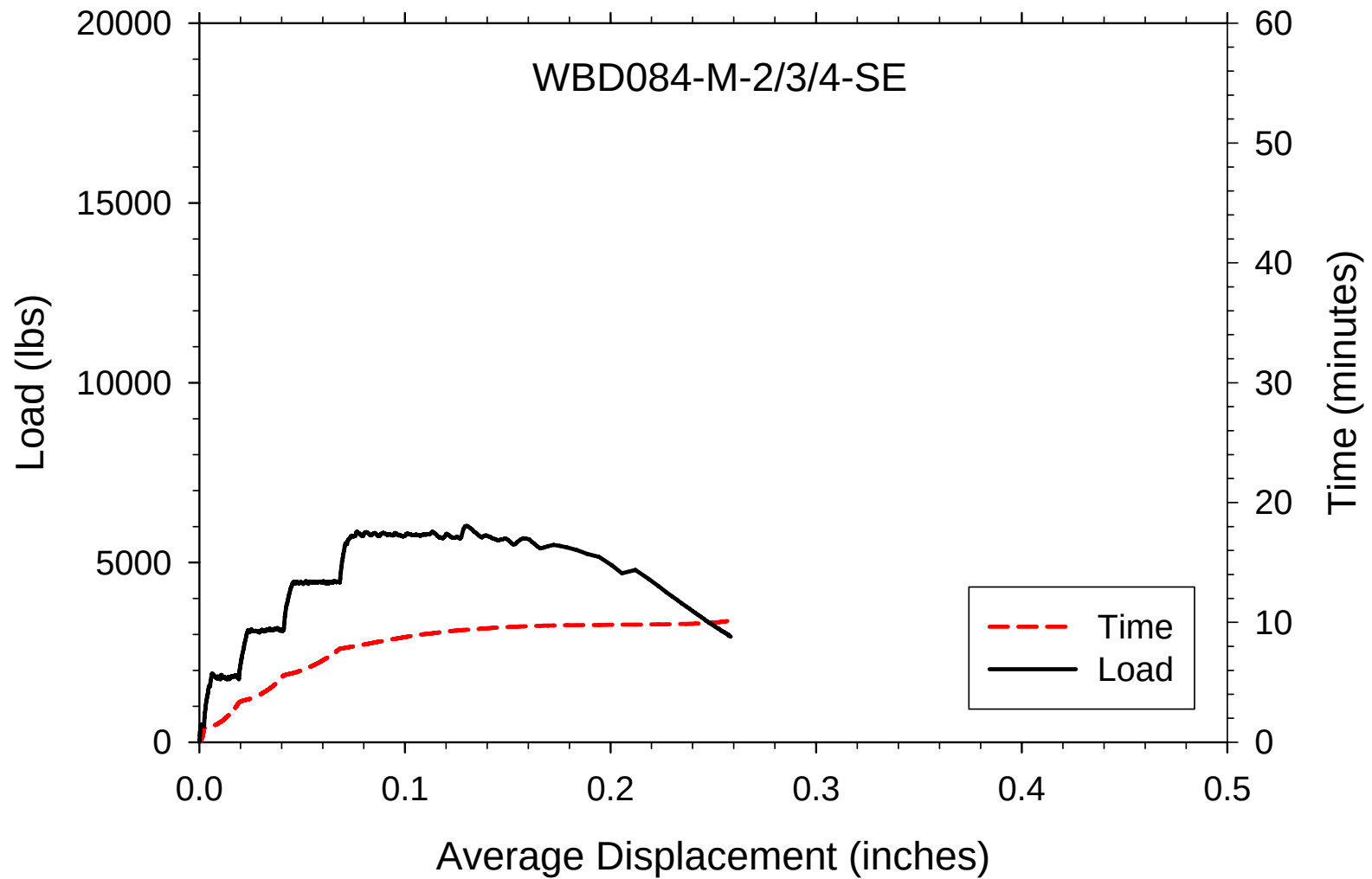


Figure 20. Load and Time vs. Displacement Plot for Anchor WBD084-M-2/3/4-SE.

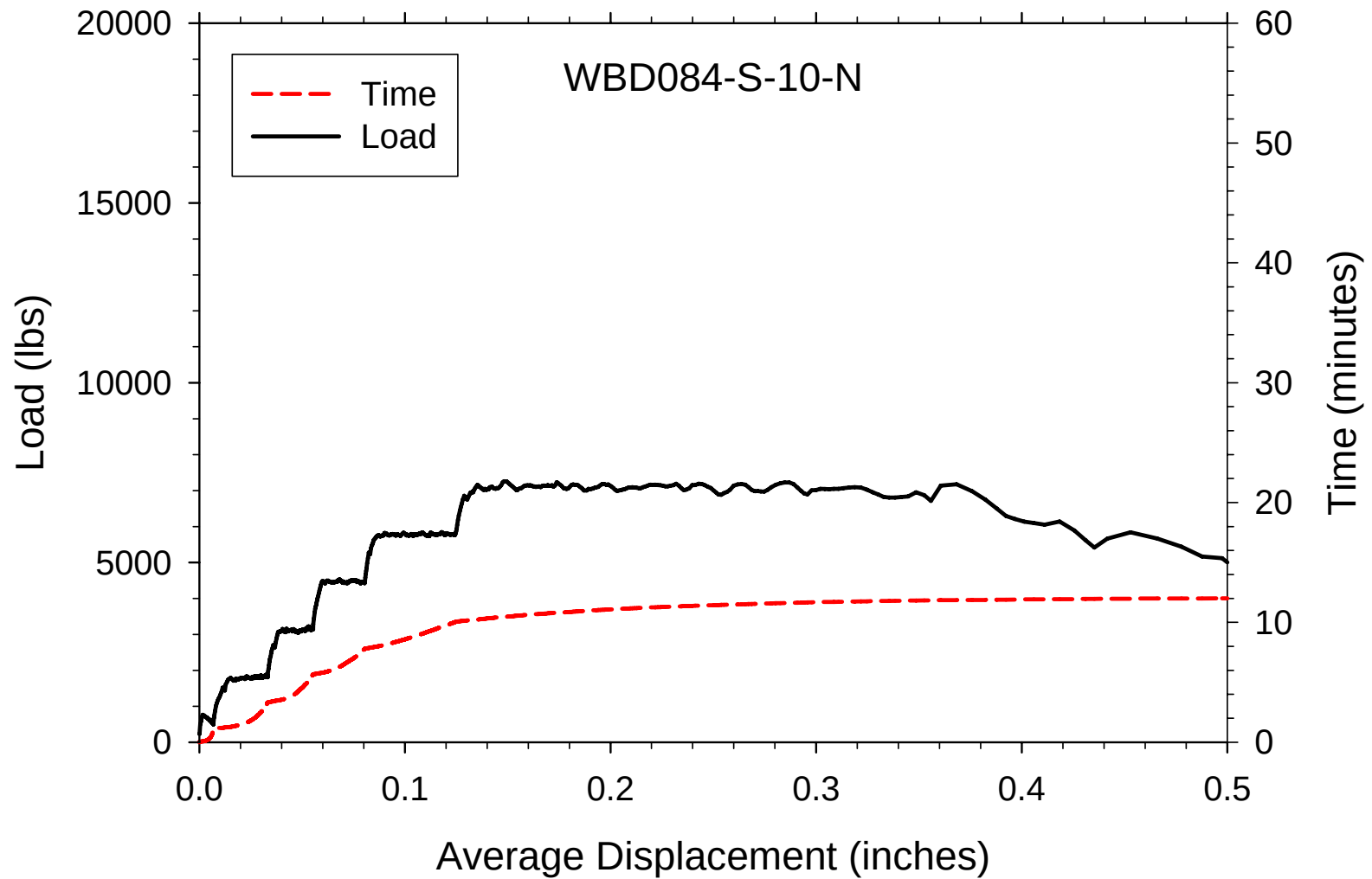


Figure 21. Load and Time vs. Displacement Plot for Anchor WBD084-S-10-N.

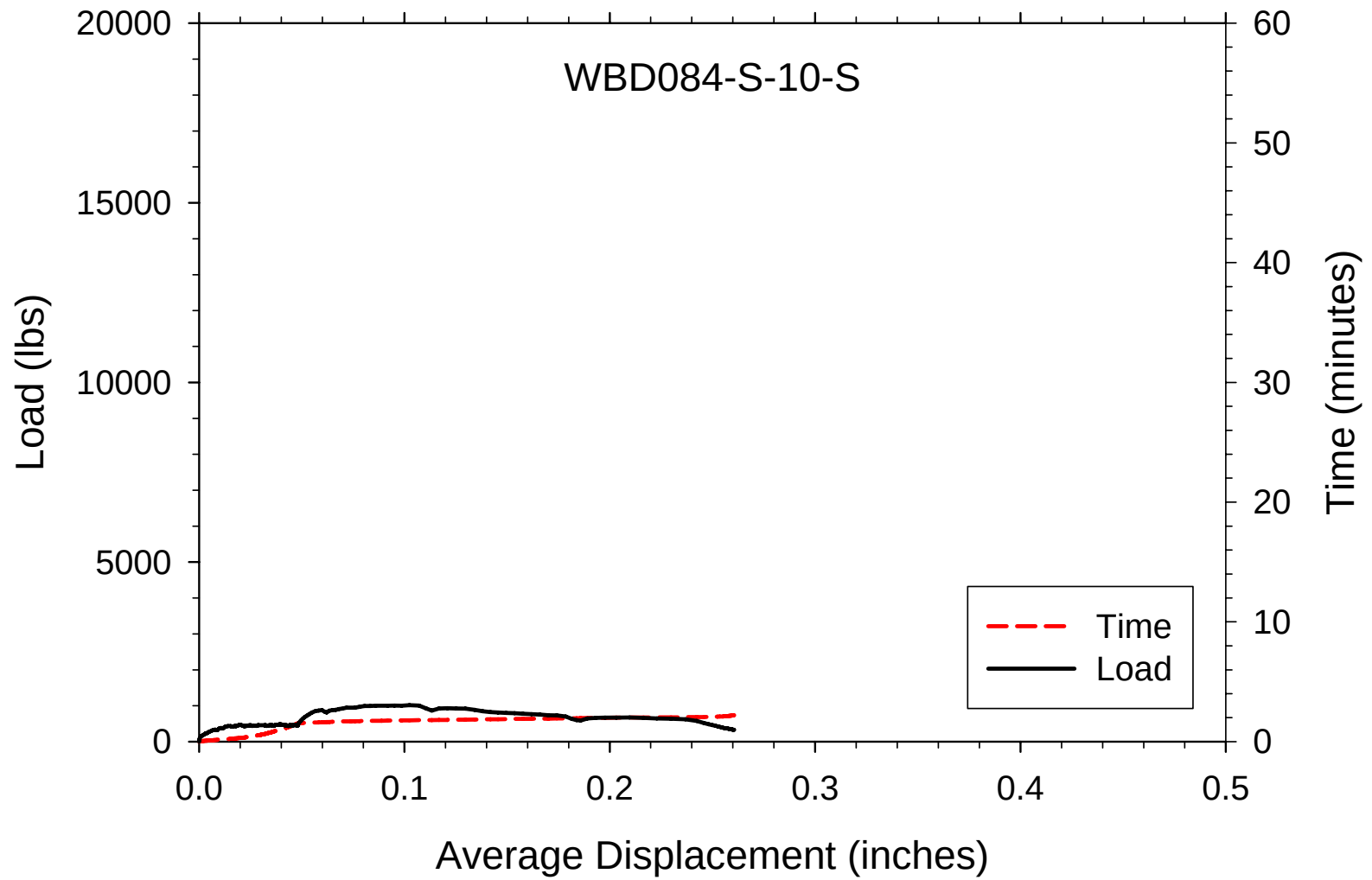


Figure 22. Load and Time vs. Displacement Plot for Anchor WBD084-S-10-S.

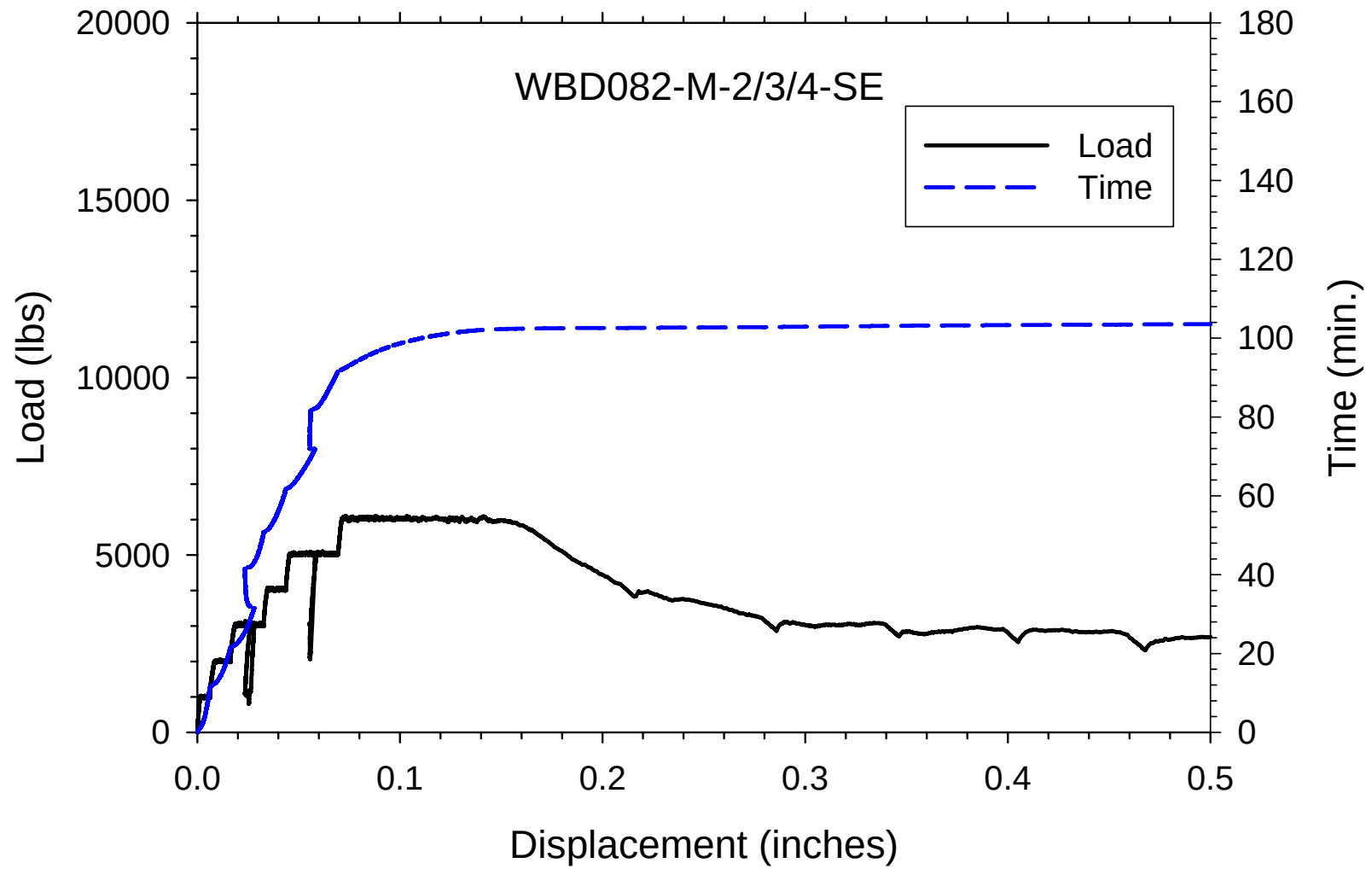


Figure 23. Load and Time vs. Displacement Plot for Anchor WBD082-M-2/3/4-SE.

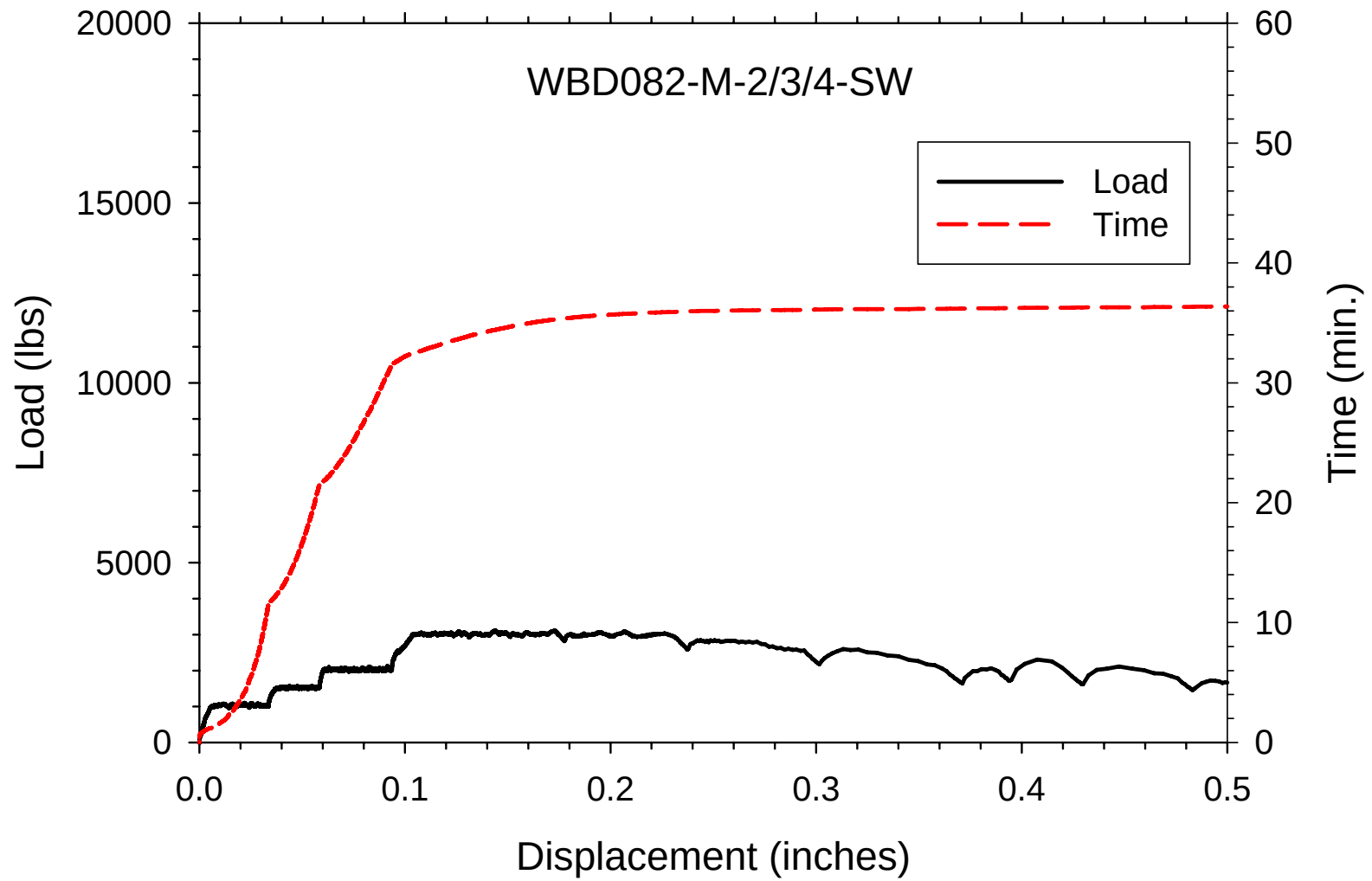


Figure 24. Load and Time vs. Displacement Plot for Anchor WBD082-M-2/3/4-SW.

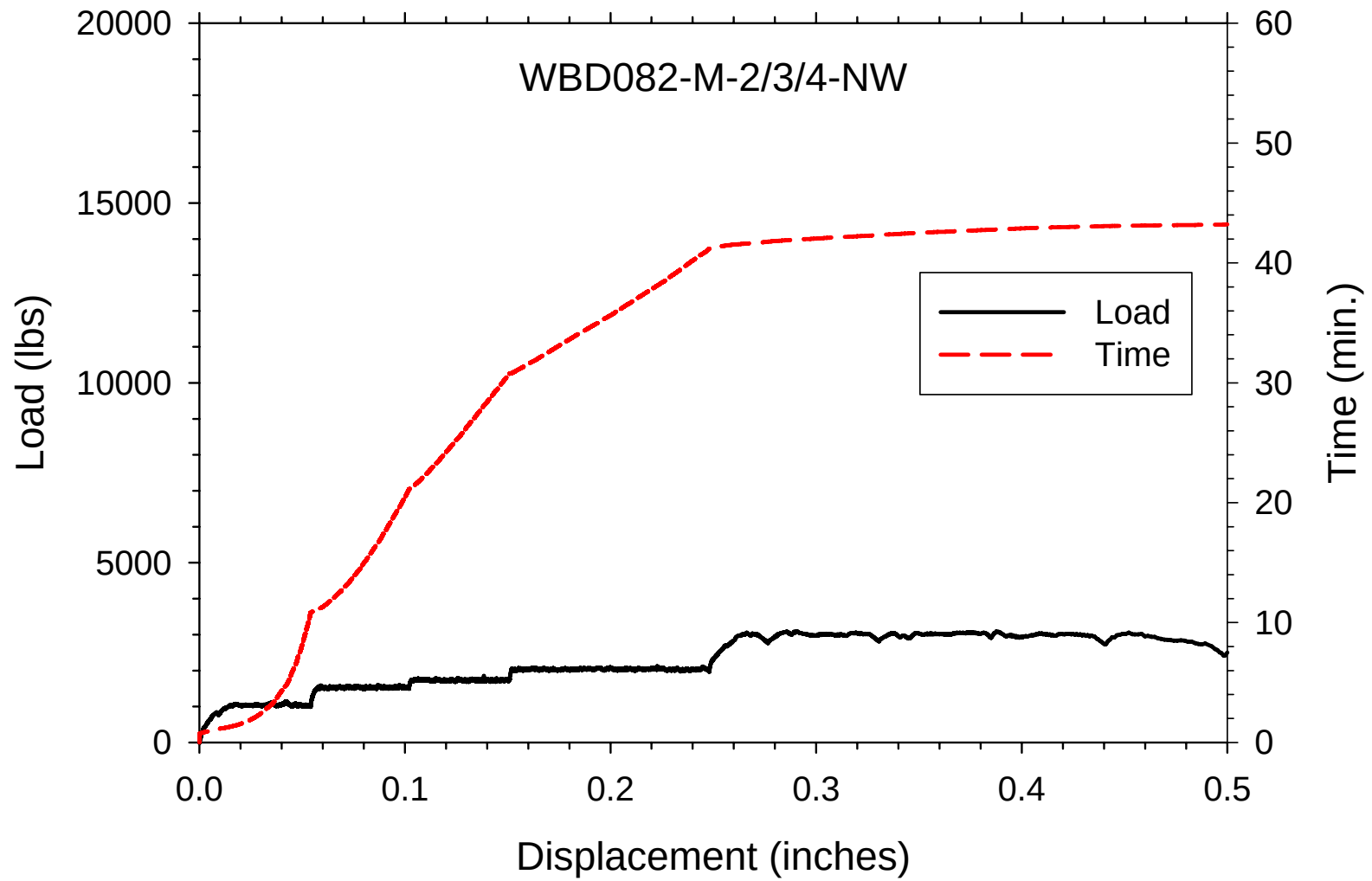


Figure 25. Load and Time vs. Displacement Plot for Anchor WBD082-M-2/3/4-NW.

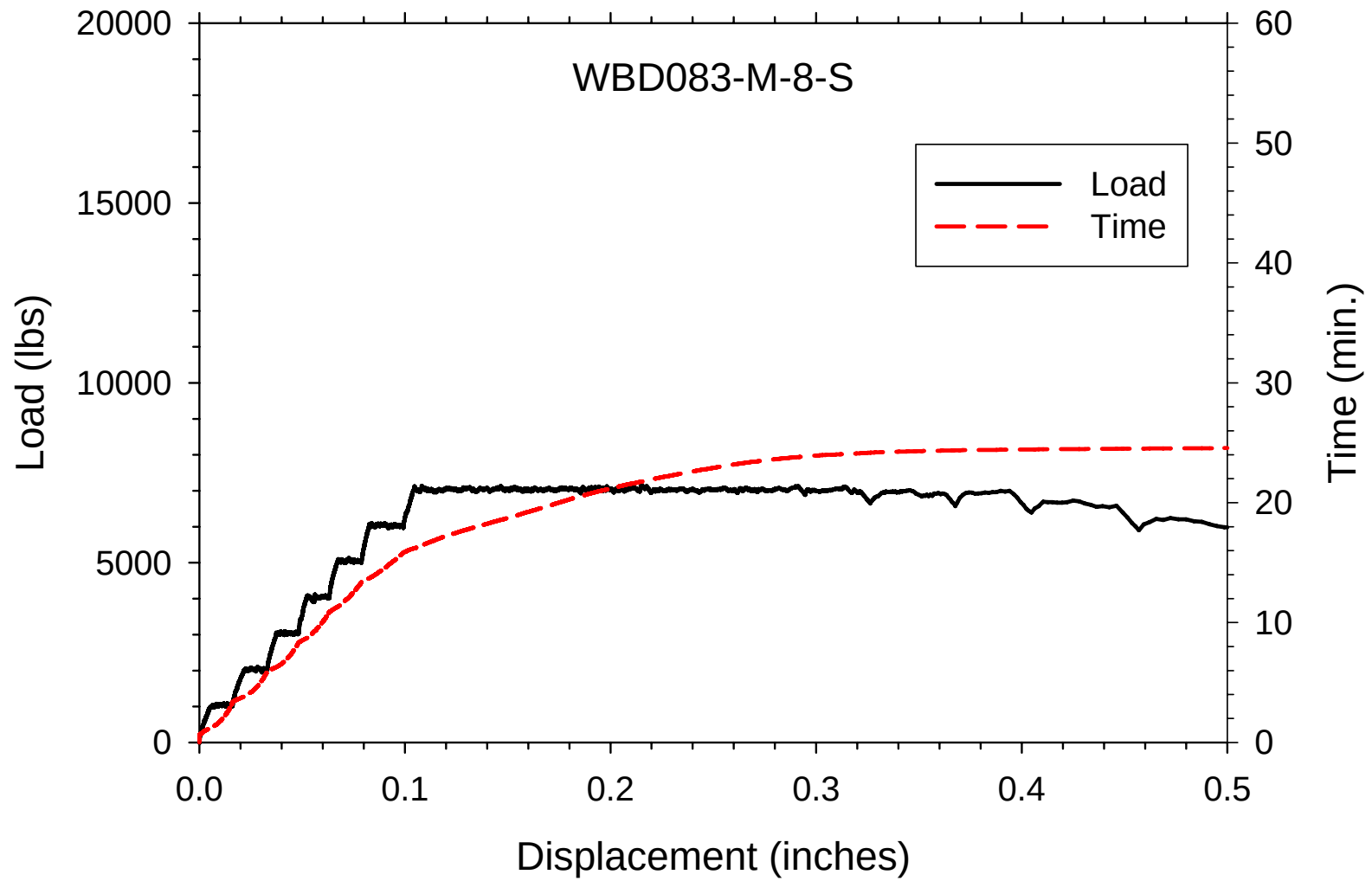


Figure 26. Load and Time vs. Displacement Plot for Anchor WBD083-M-8-S.

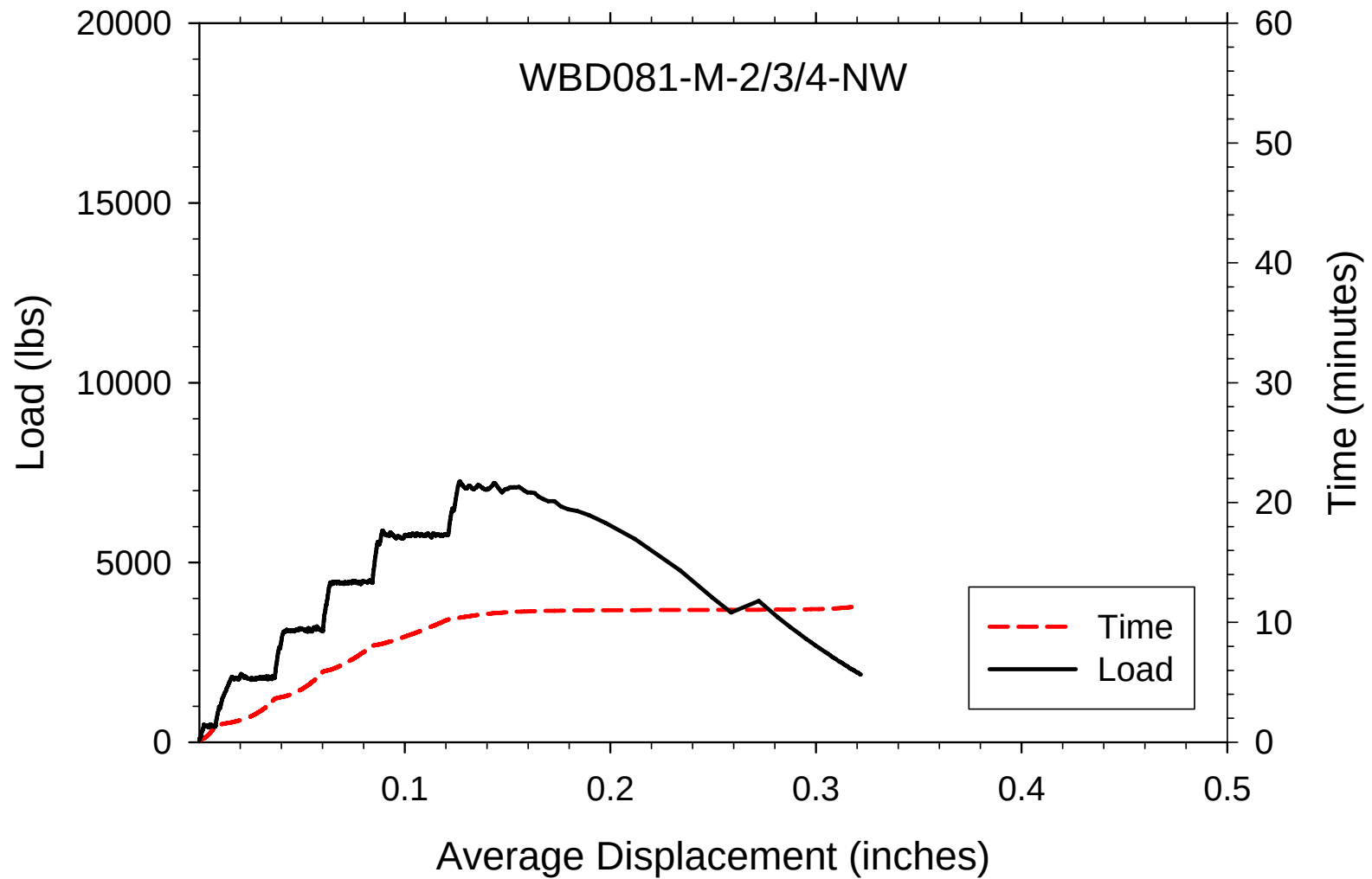


Figure 27. Load and Time vs. Displacement Plot for Anchor WBD081-M-2/3/4-NW.

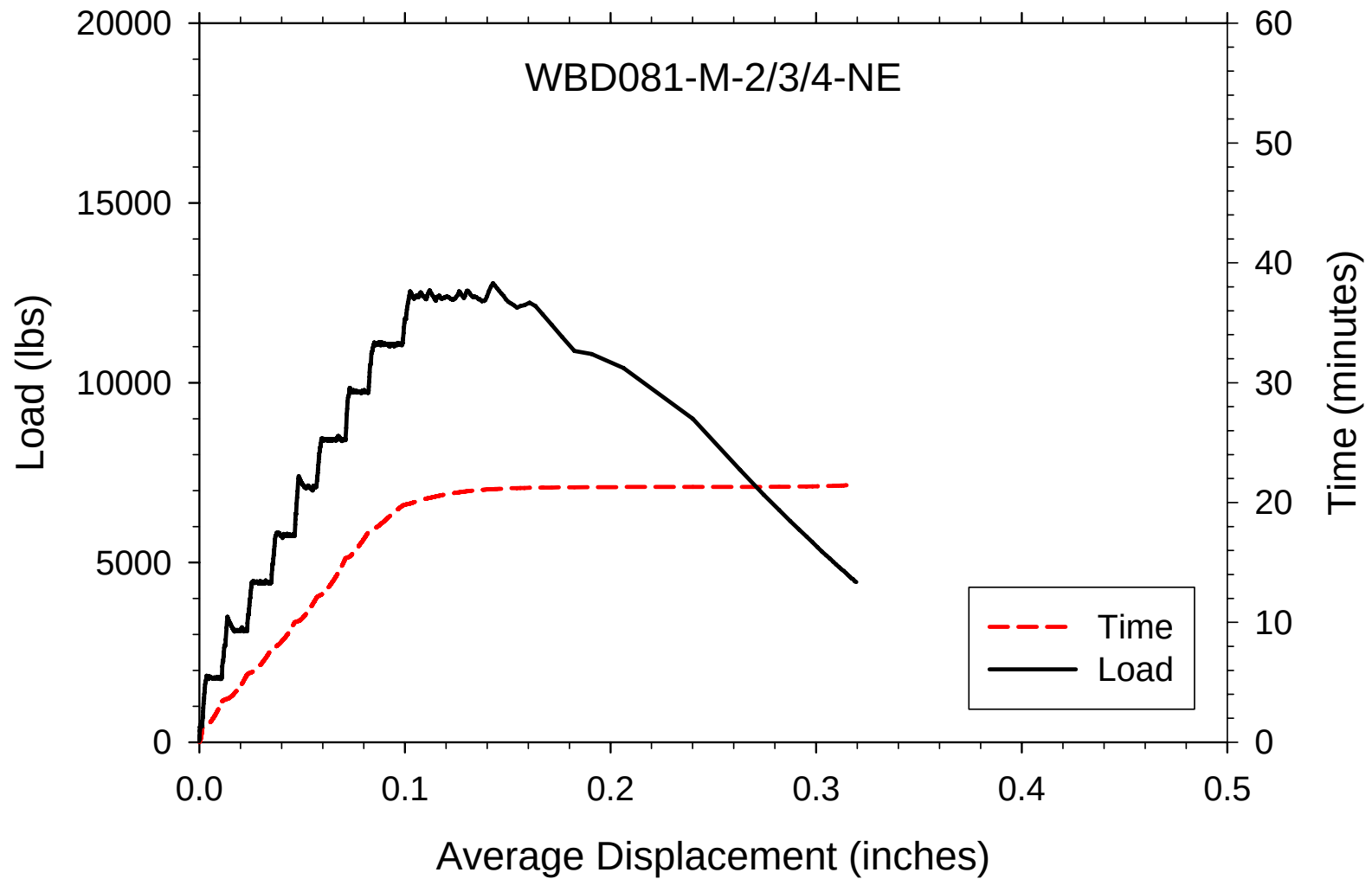


Figure 28. Load and Time vs. Displacement Plot for Anchor WBD081-M-2/3/4-NE.

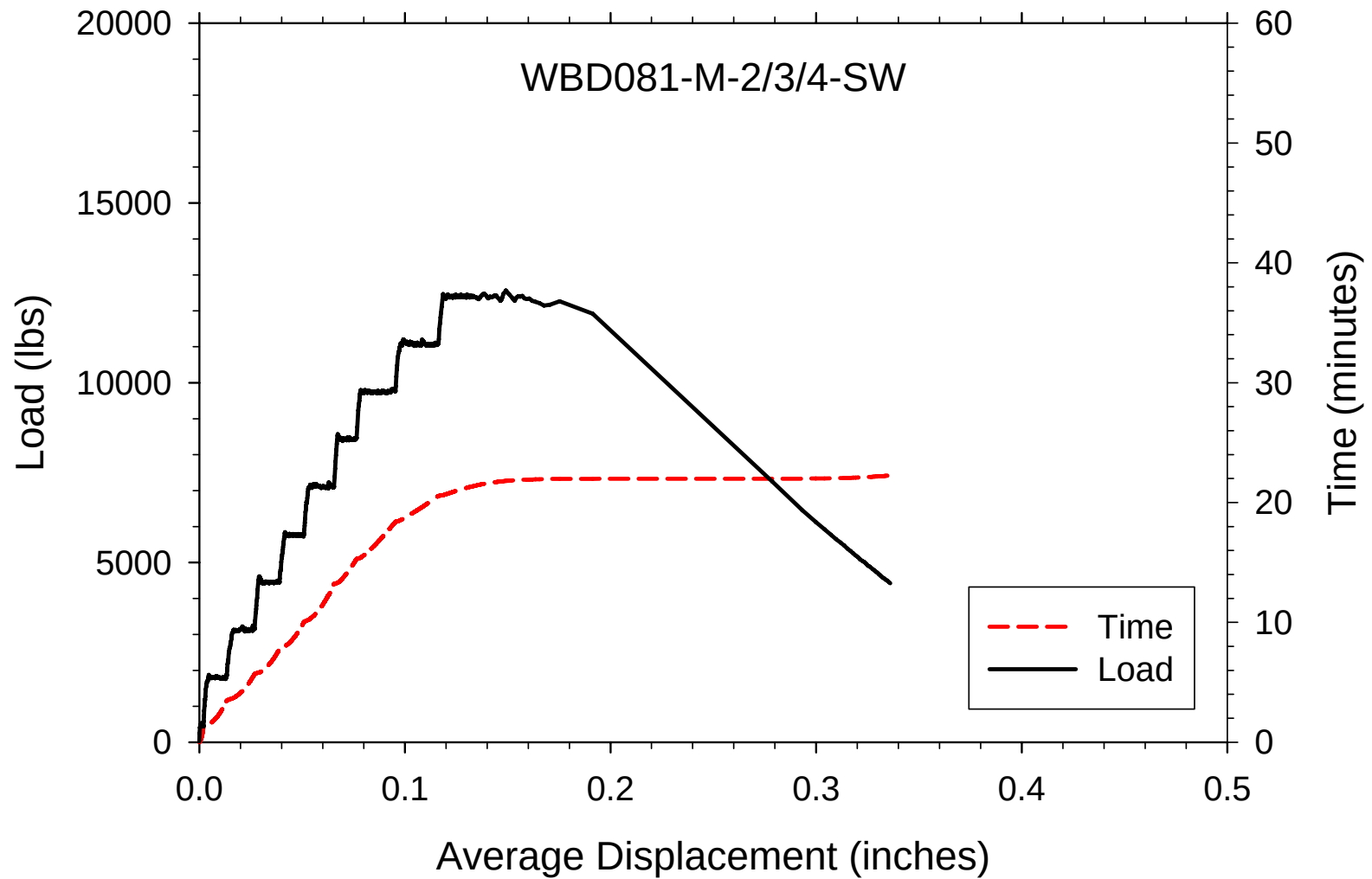


Figure 29. Load and Time vs. Displacement Plot for Anchor WBD081-M-2/3/4-SW.

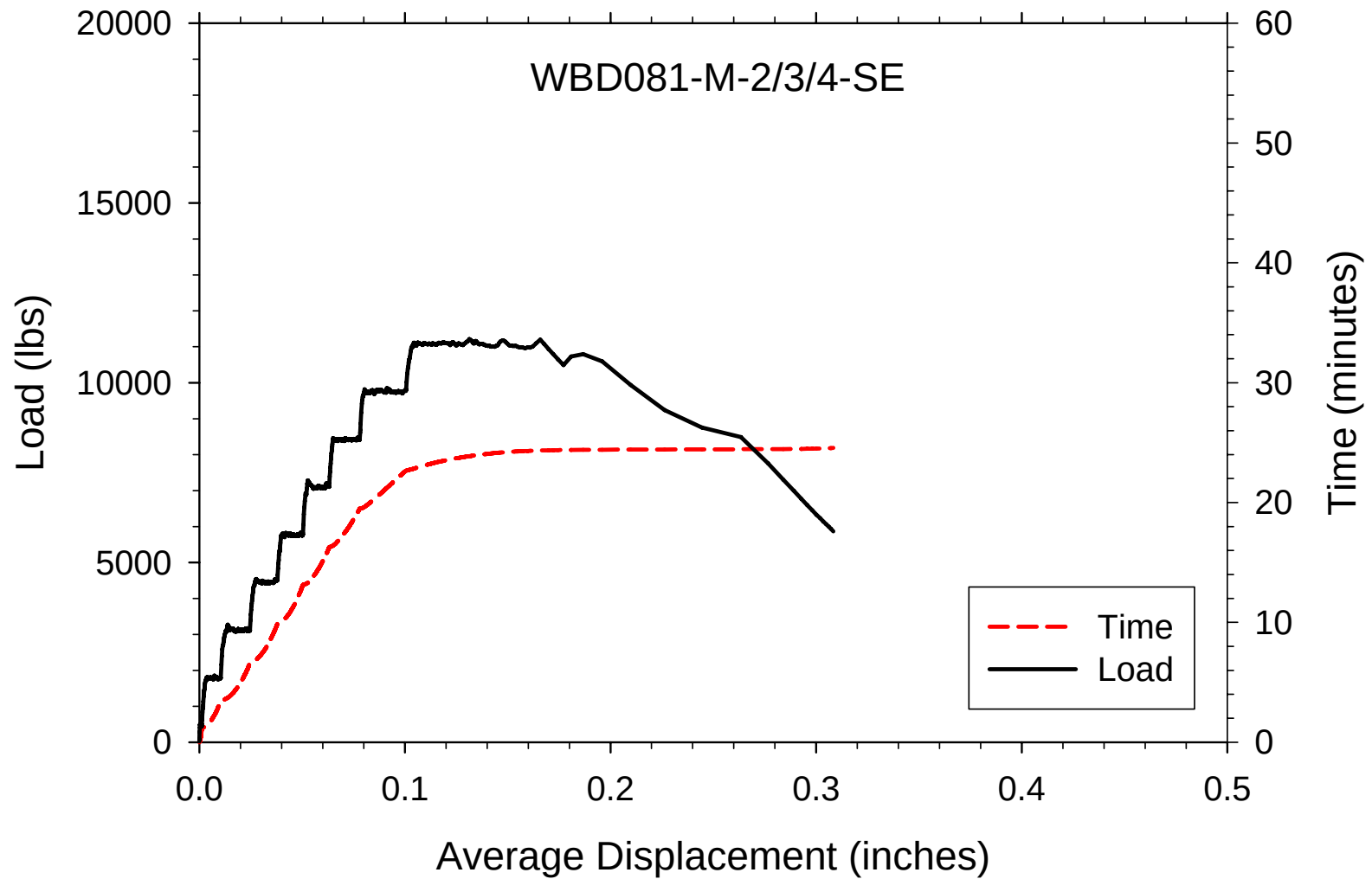


Figure 30. Load and Time vs. Displacement Plot for Anchor WBD081-M-2/3/4-SE.

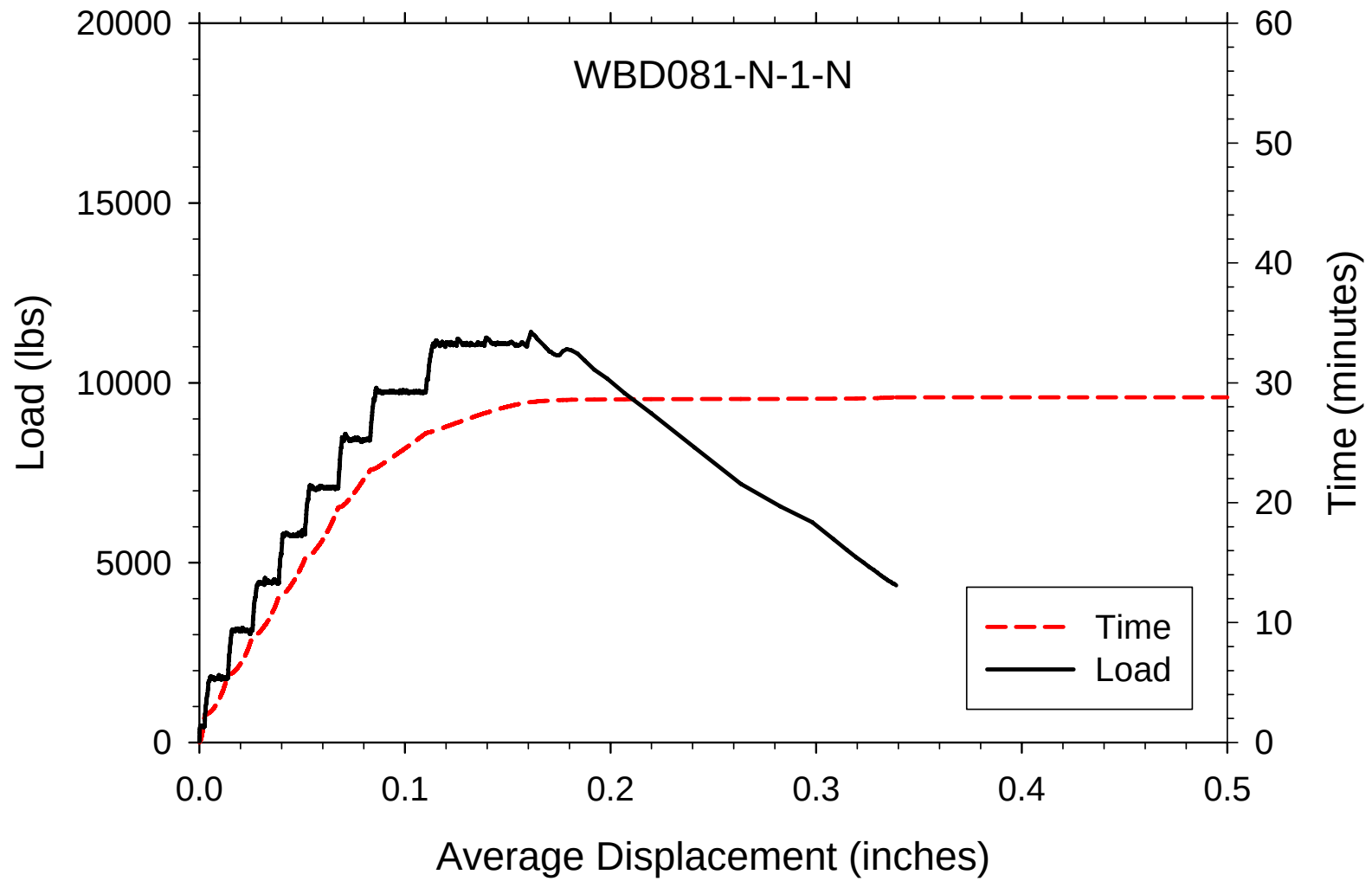


Figure 31. Load and Time vs. Displacement Plot for Anchor WBD081-N-1-N.

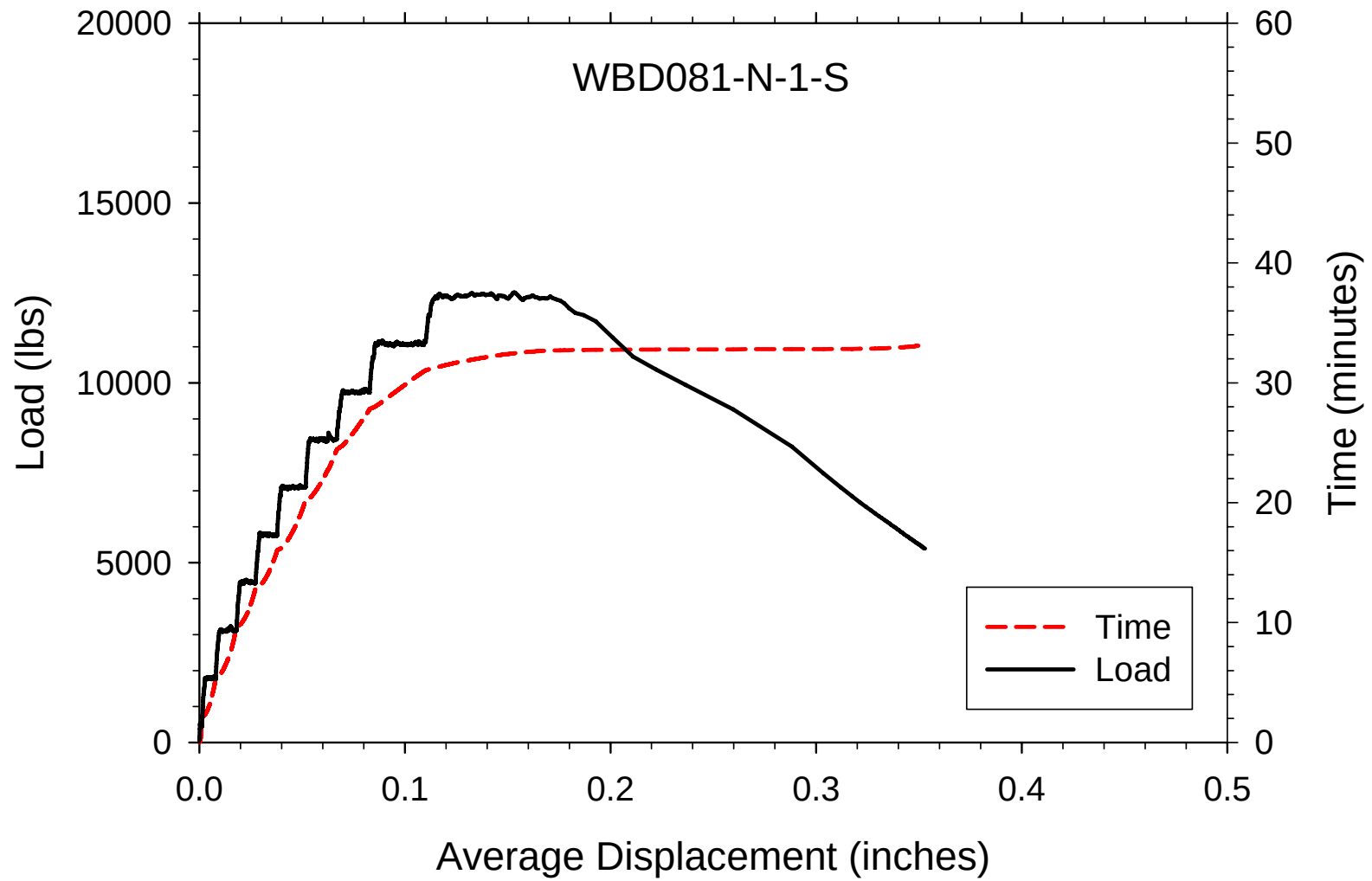


Figure 32. Load and Time vs. Displacement Plot for Anchor WBD081-N-1-S.

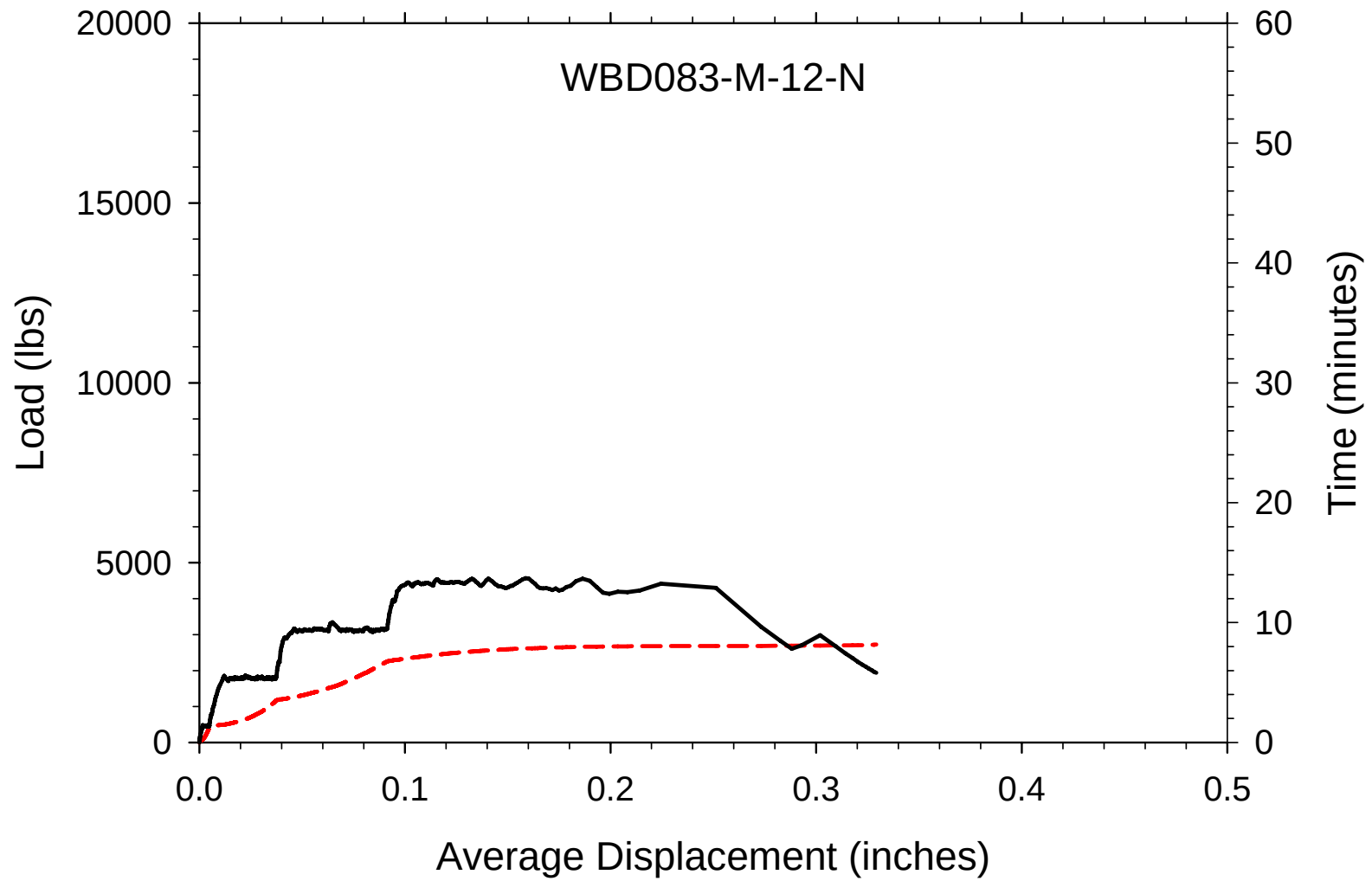


Figure 33. Load and Time vs. Displacement Plot for Anchor WBD083-M-12-N.

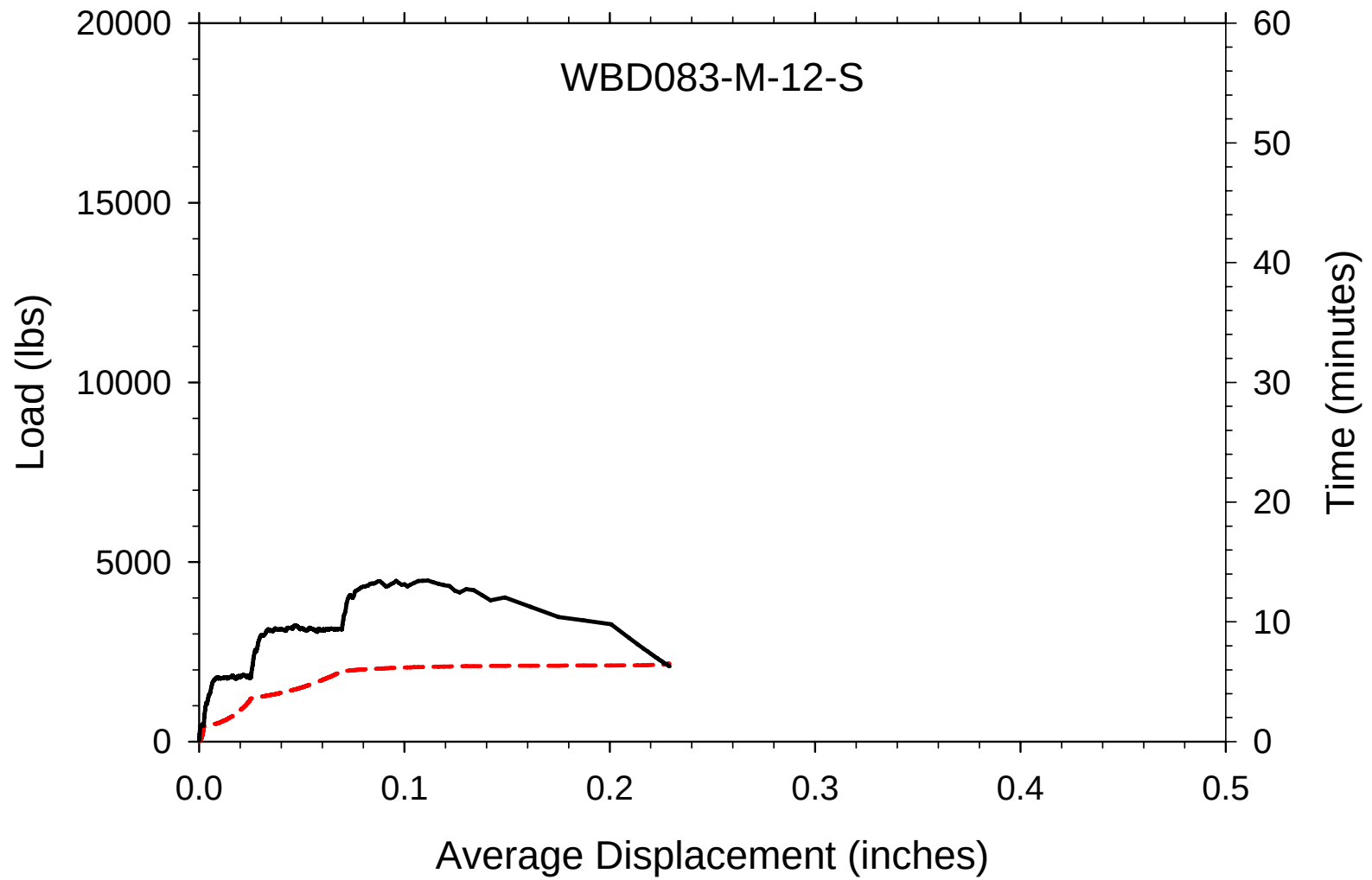


Figure 34. Load and Time vs. Displacement Plot for Anchor WBD083-M-12-S.

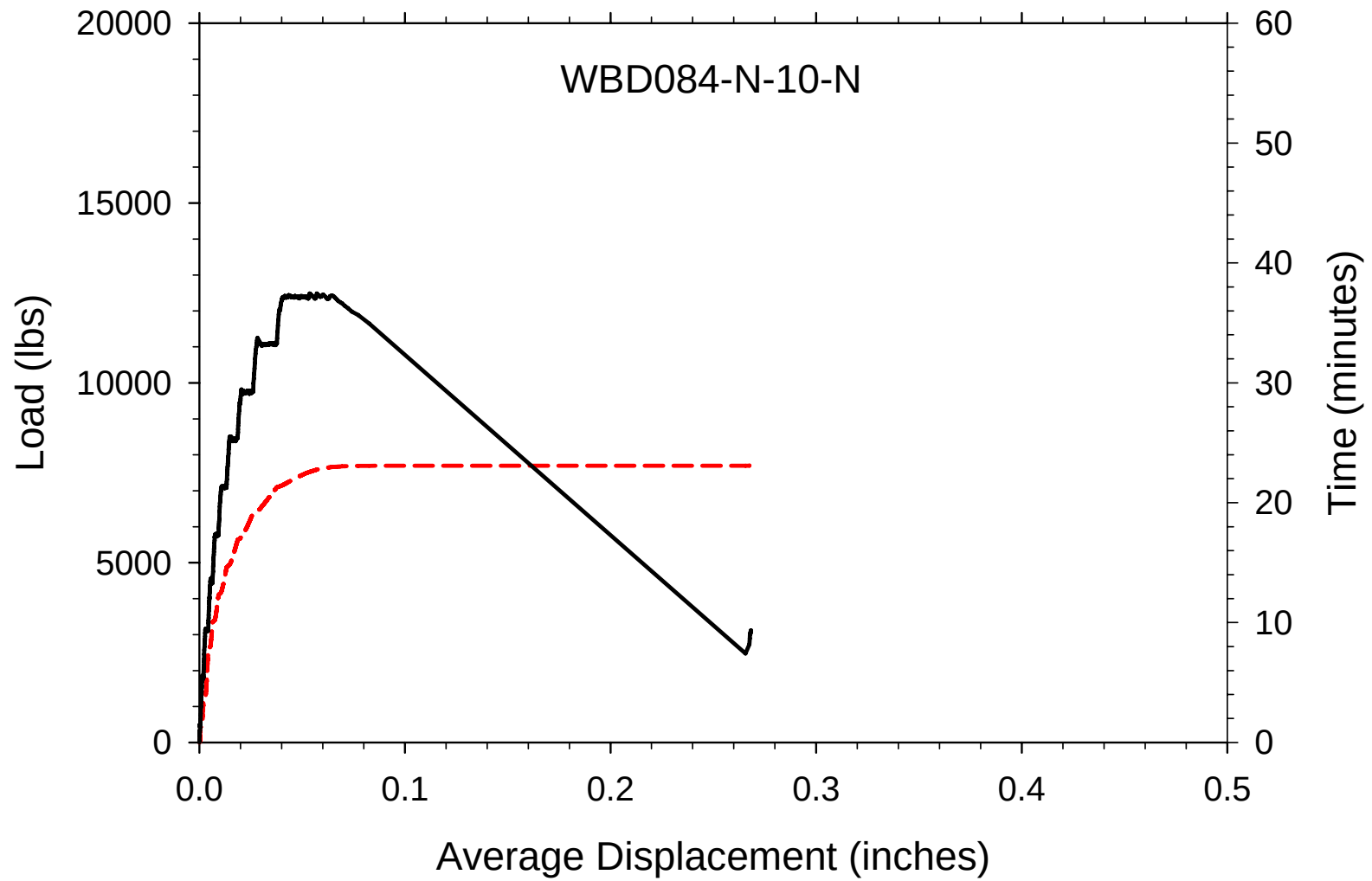


Figure 35. Load and Time vs. Displacement Plot for Anchor WBD084-N-10-N.

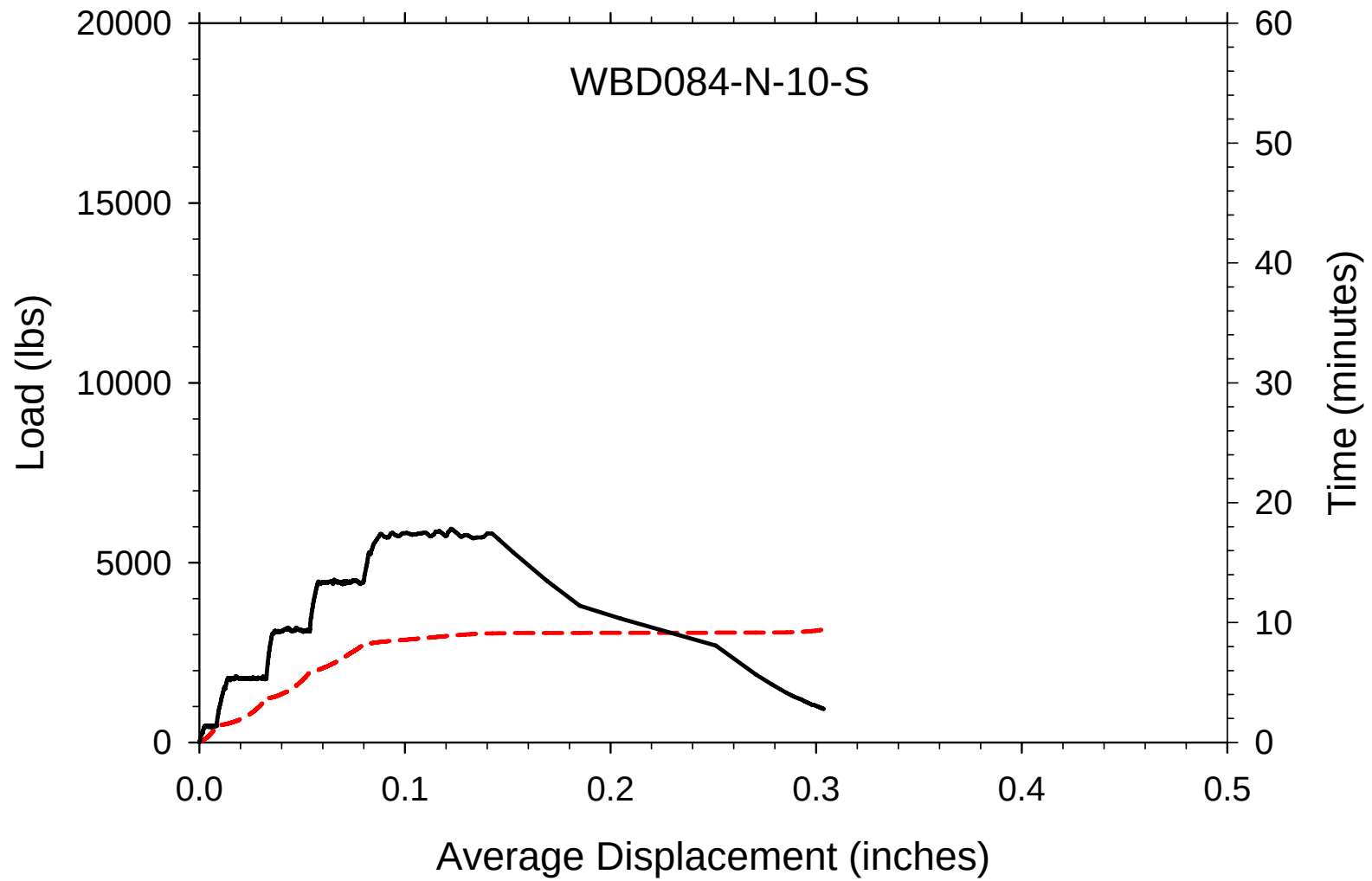


Figure 36. Load and Time vs. Displacement Plot for Anchor WBD084-N-10-S.

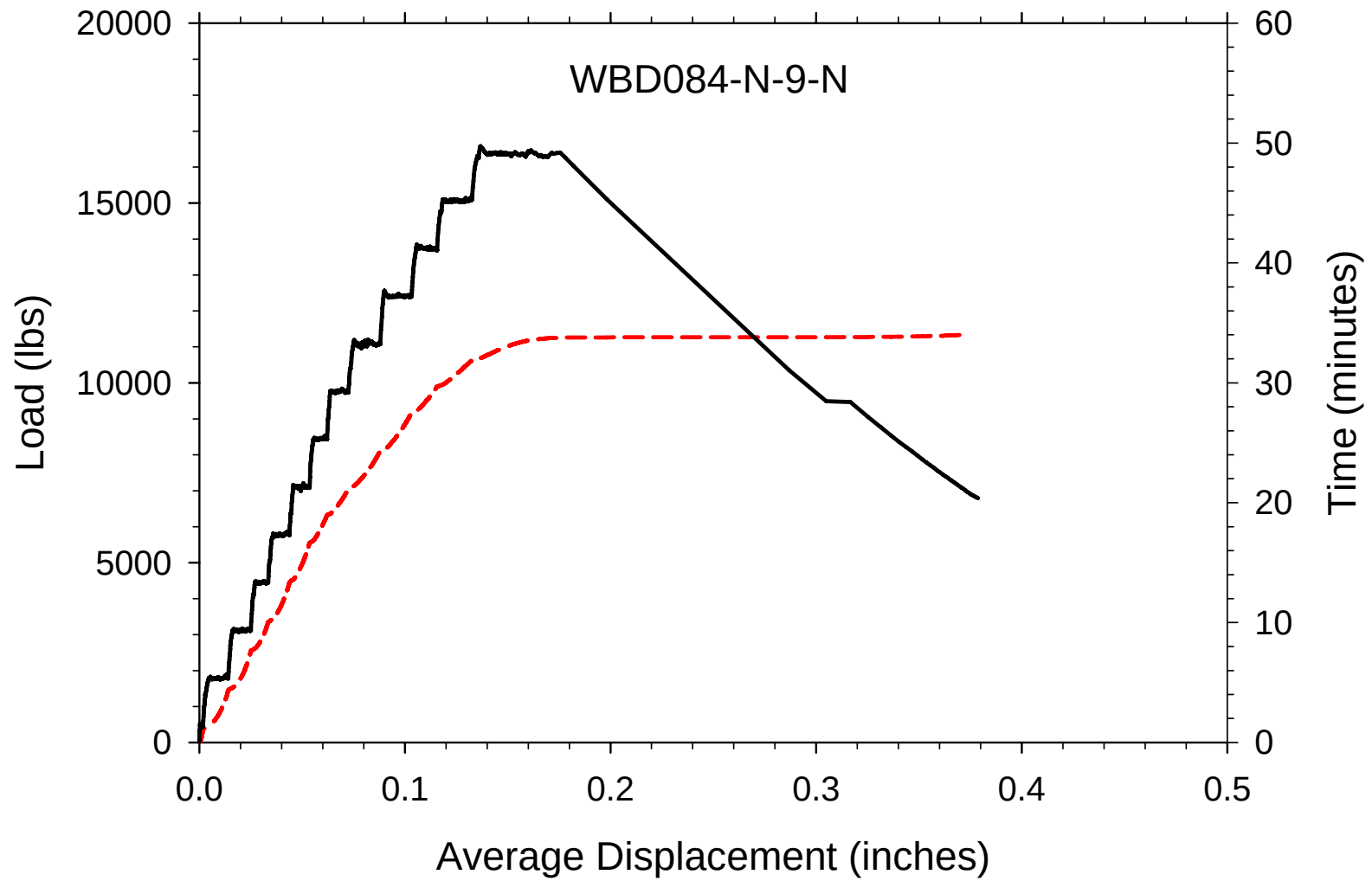


Figure 37. Load and Time vs. Displacement Plot for Anchor WBD084-N-9-N.

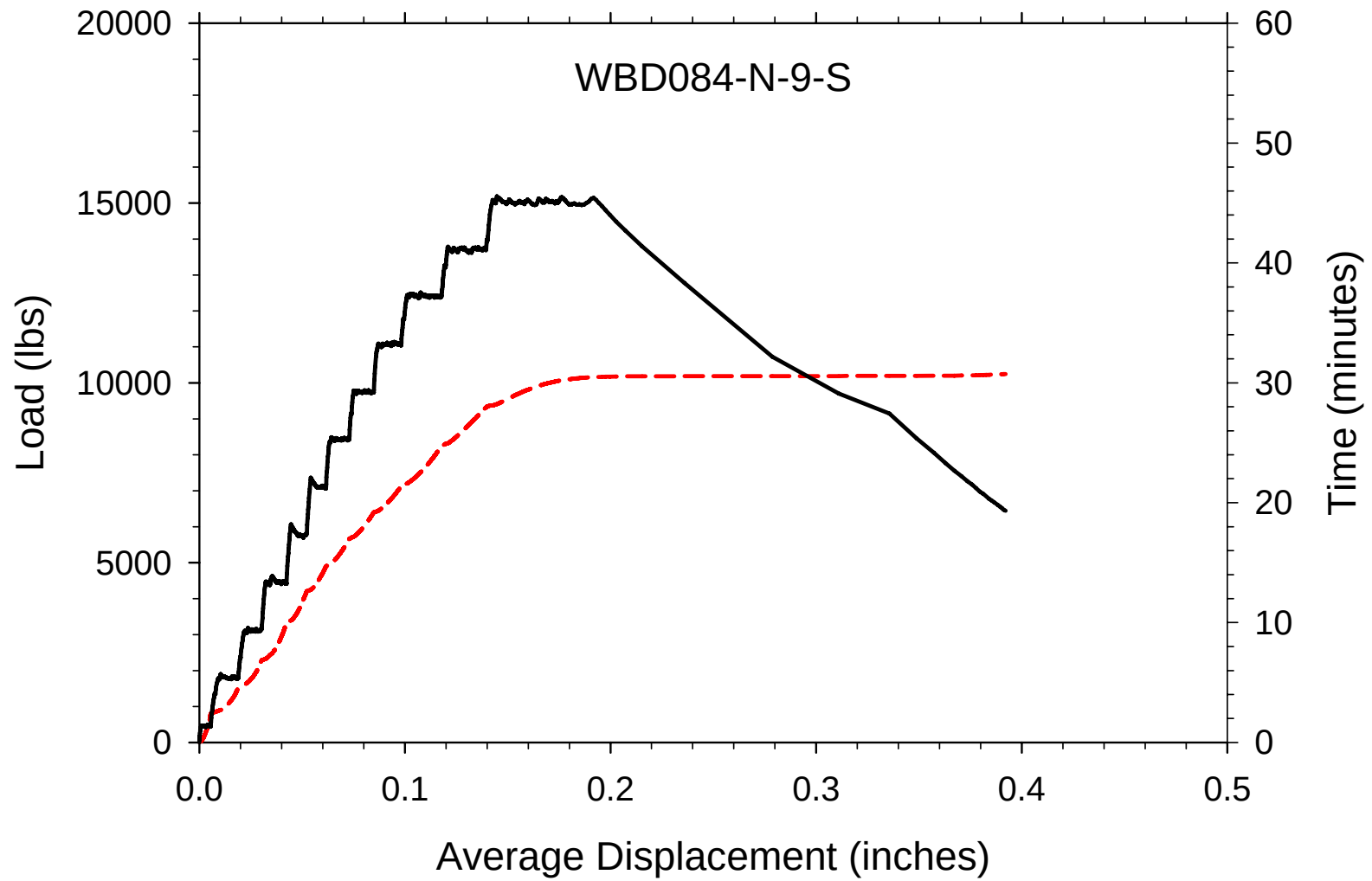


Figure 38. Load and Time vs. Displacement Plot for Anchor WBD084-N-9-S.

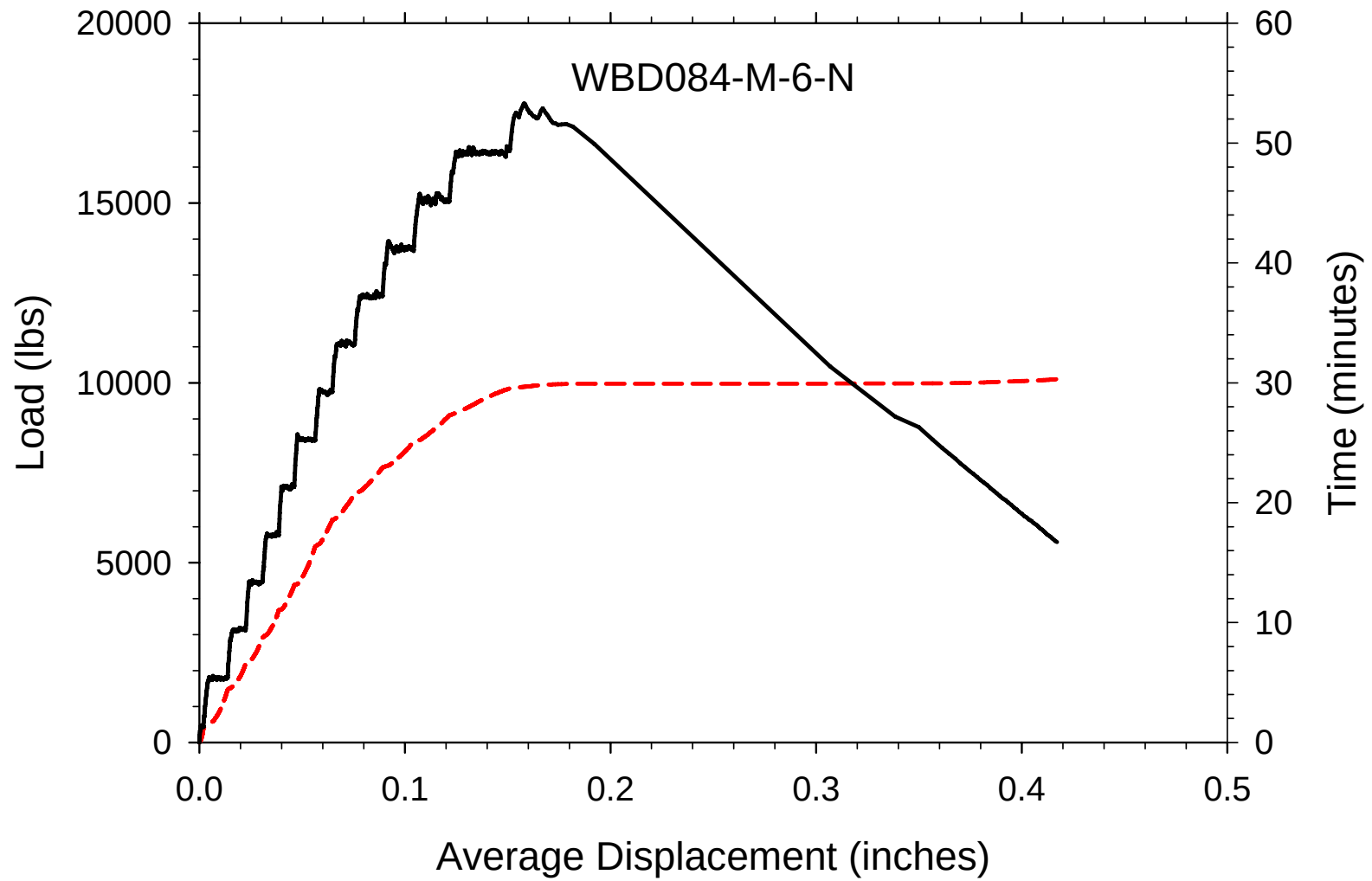


Figure 39. Load and Time vs. Displacement Plot for Anchor WBD084-M-6-N.

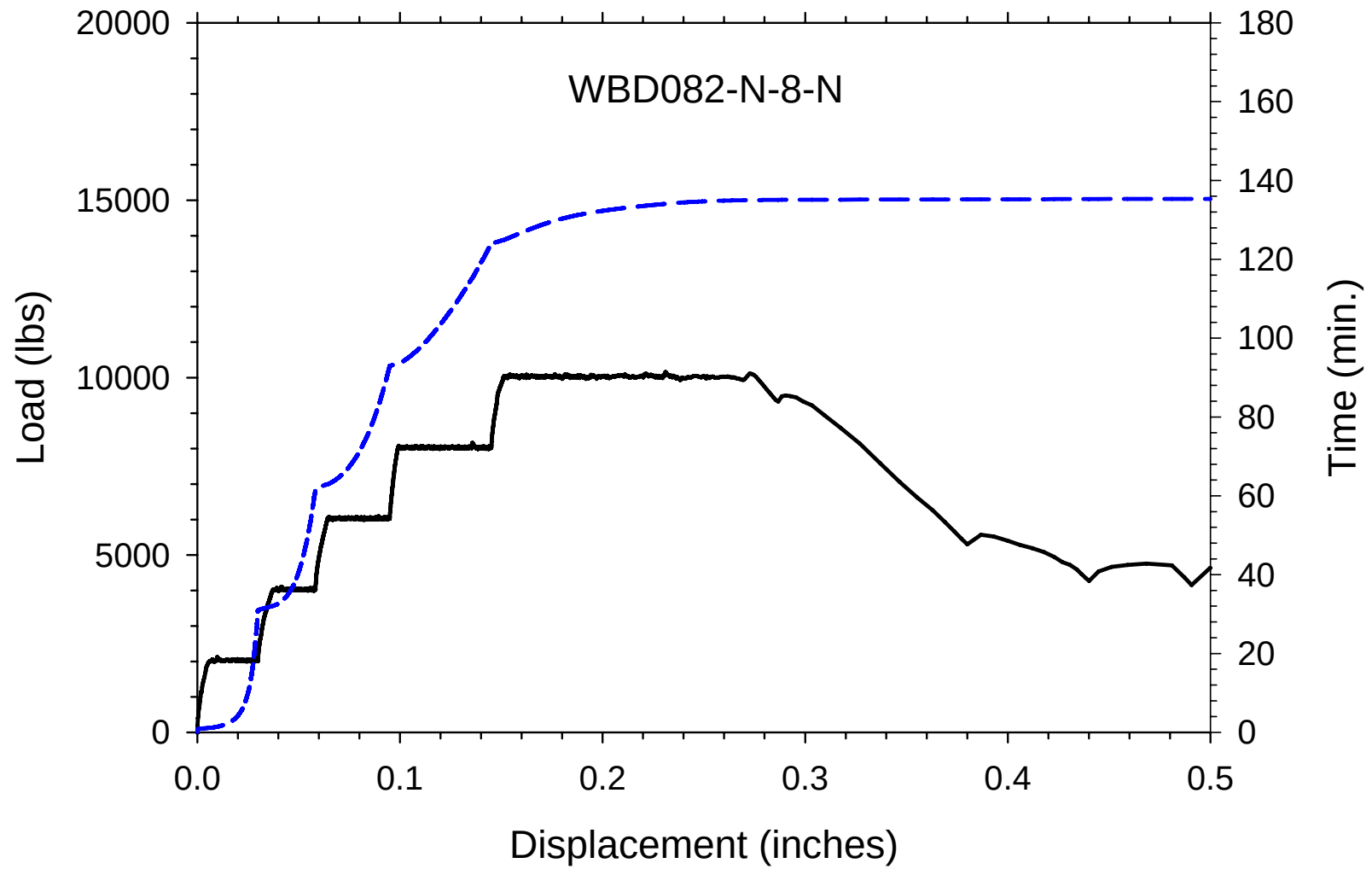


Figure 40. Load and Time vs. Displacement Plot for Anchor WBD082-N-8-N.

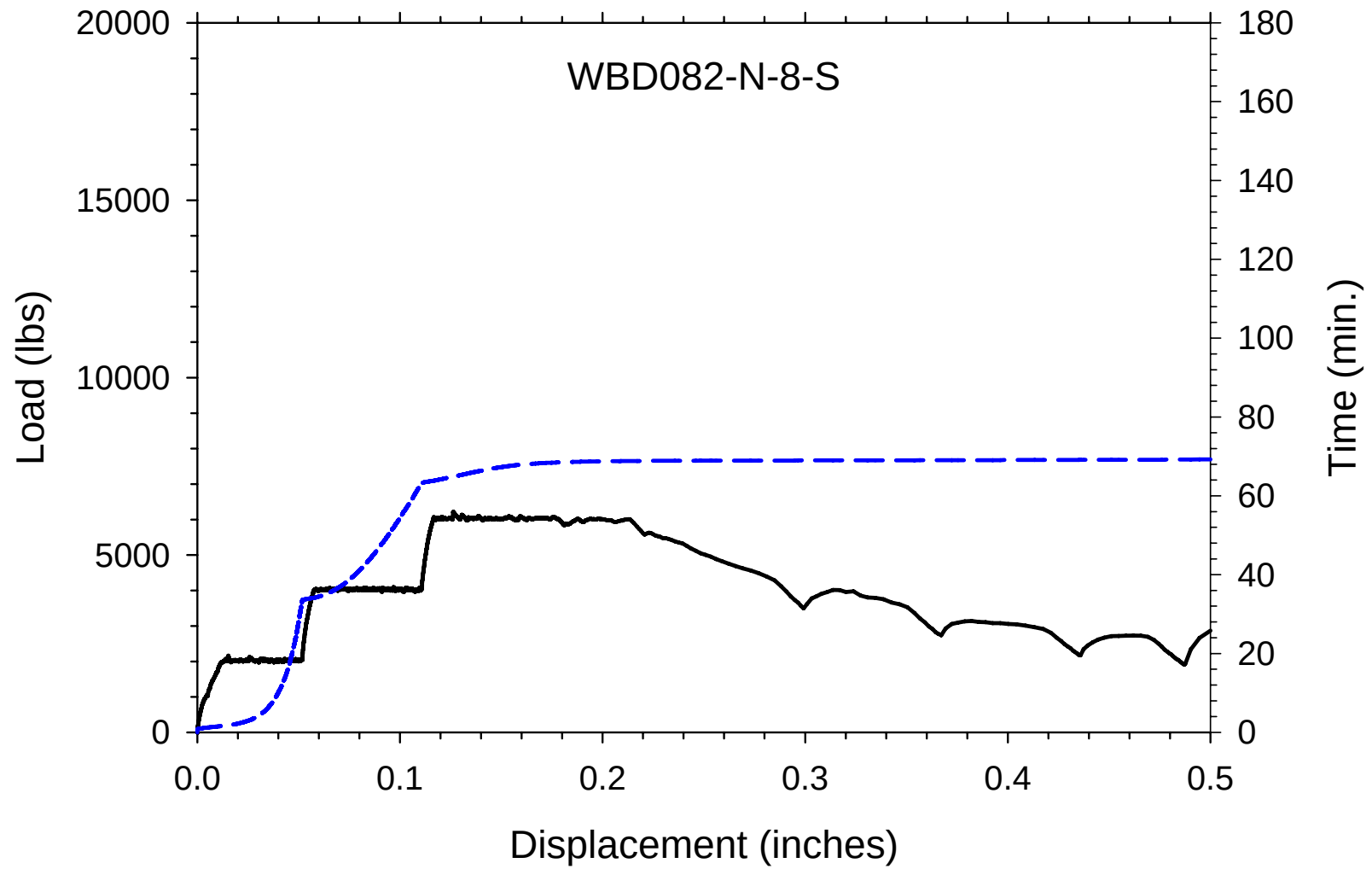


Figure 41. Load and Time vs. Displacement Plot for Anchor WBD082-N-8-S.

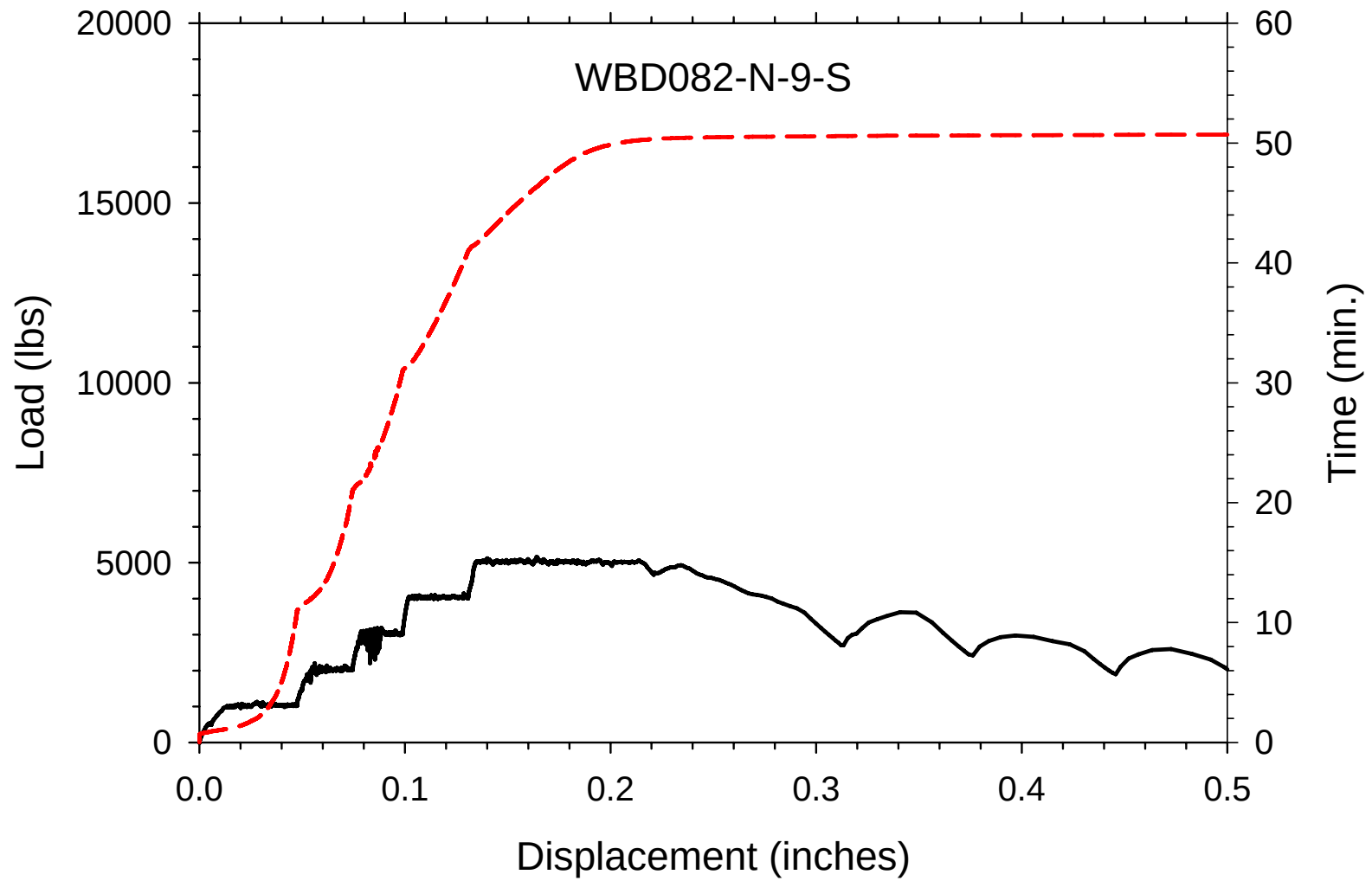


Figure 42. Load and Time vs. Displacement Plot for Anchor WBD082-N-9-S.

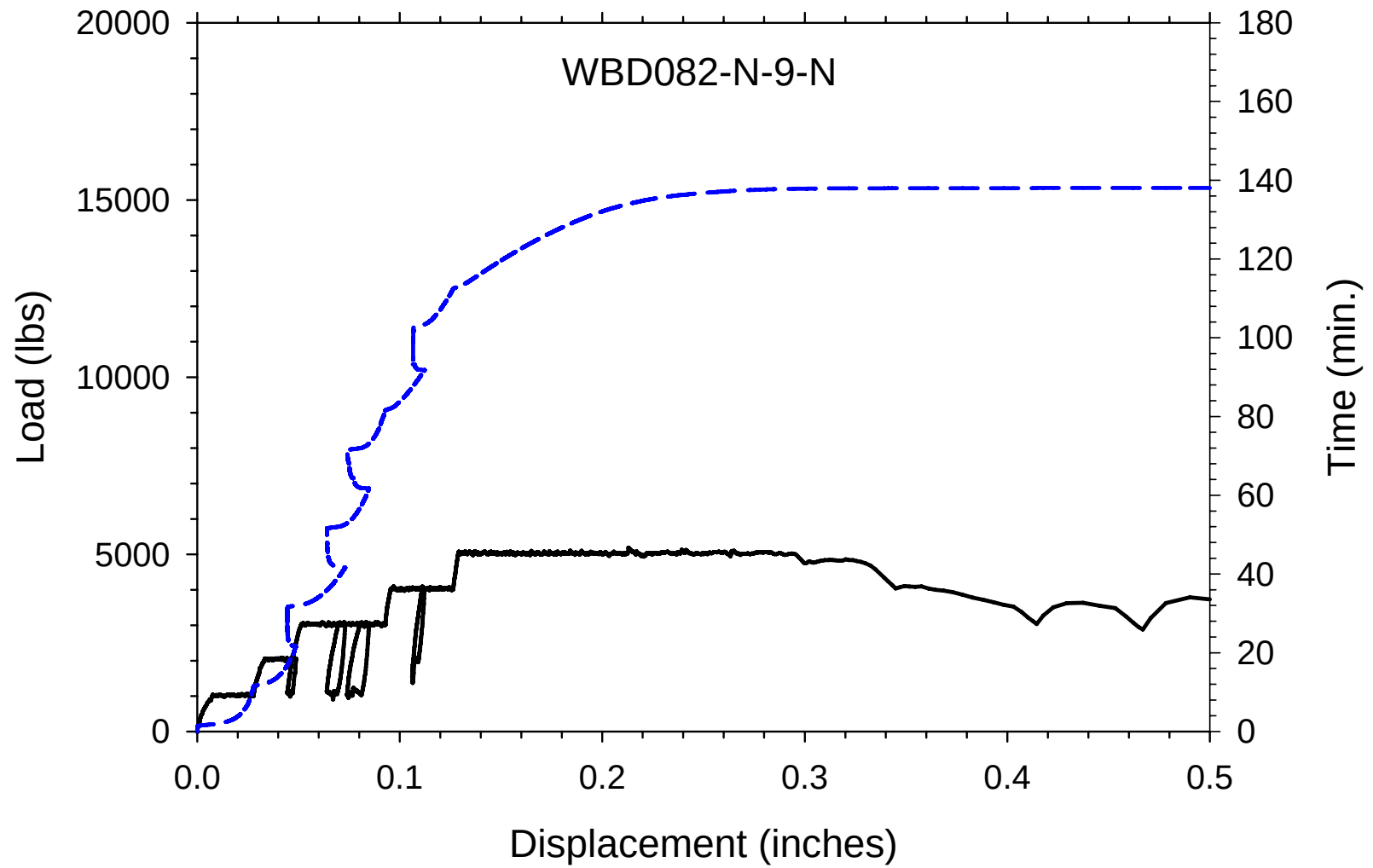


Figure 43. Load and Time vs. Displacement Plot for Anchor WBD082-N-9-N.

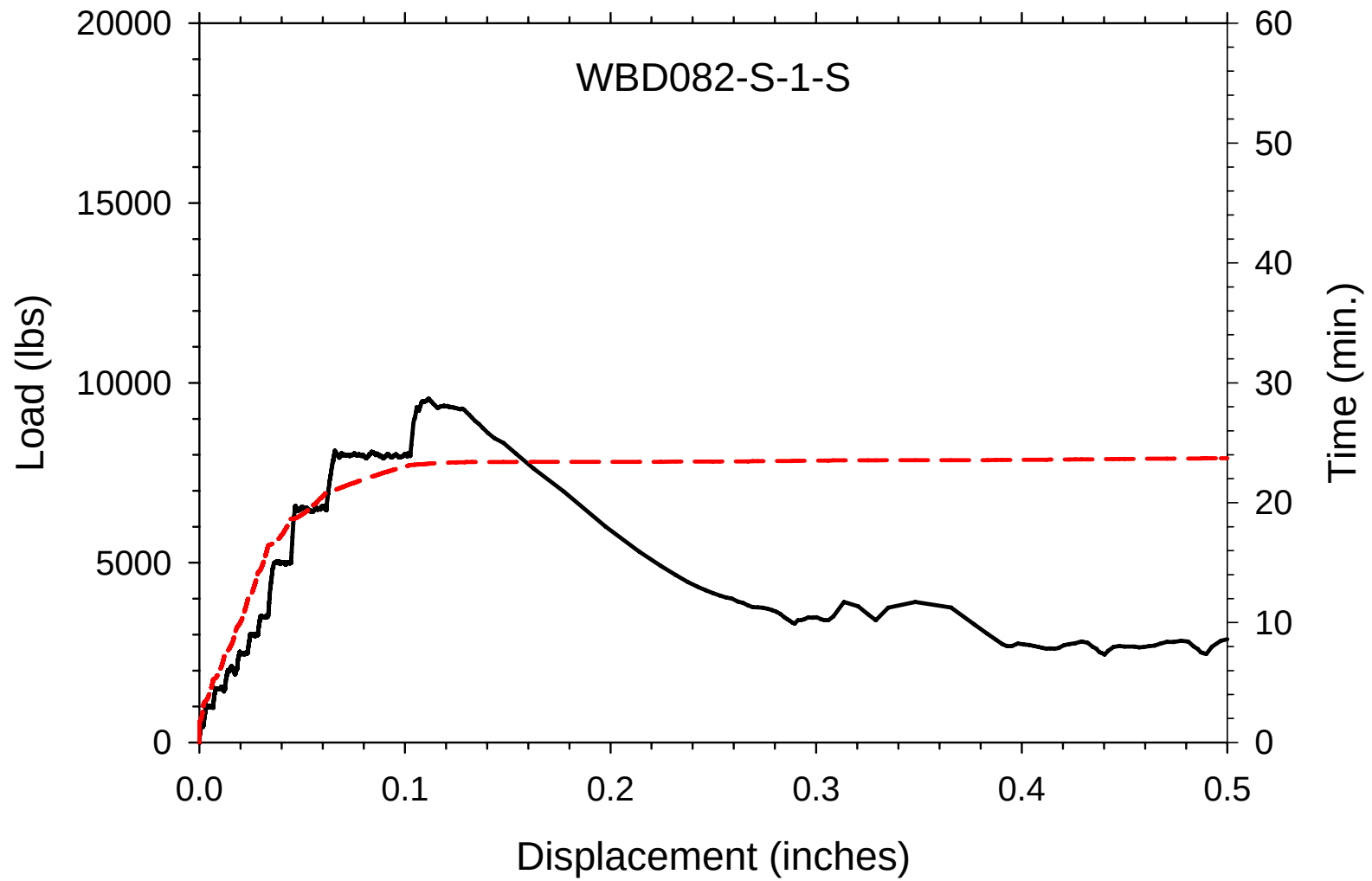


Figure 44. Load and Time vs. Displacement Plot for Anchor WBD082-S-1-S.

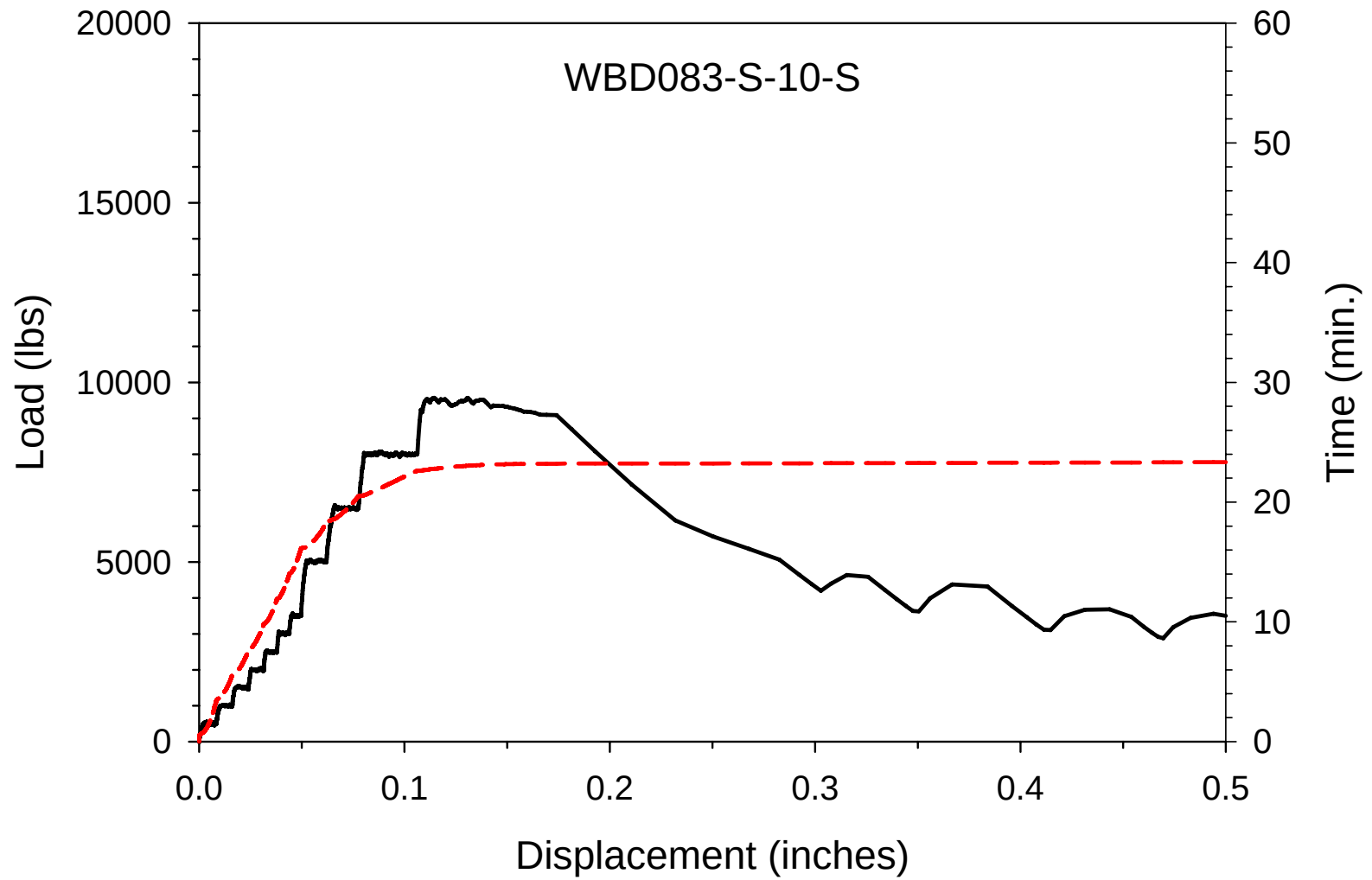


Figure 45. Load and Time vs. Displacement Plot for Anchor WBD083-S-10-S.

CHAPTER 4. SUMMARY

In support of the National Transportation Safety Board's investigation of the partial collapse of the eastbound I-90 Seaport tunnel ceiling, the Federal Highway Administration has conducted load testing and load-displacement testing on 230 anchor rods. All testing was conducted in full or substantial compliance with the static tension testing provisions of ASTM E 488 *Standard Test Method for Strength of Anchors in Concrete and Masonry Elements*. This document summarizes the equipment and procedures used, and reports the data produced during the FHWA testing.

In general, regardless of the test method or type of testing performed, each adhesive anchor tested failed within the epoxy and not at the epoxy-to-concrete interface. Also, with three exceptions, no spalling of concrete was associated with these failures.

While the peak load data from the load tests is reported herein, the results from the load-displacement testing are presented graphically due to the quantity of data produced during those experiments. A compact-disk containing the entire time-load-displacement history for each load-displacement test has been prepared to accompany this report. The individual anchor test files are contained in a Microsoft Excel Workbook.

APPENDIX A. GENERAL TESTING PROCEDURES

LOAD TESTING

General procedure for adhesive anchor load testing:

1. On a field note sheet, record the type of jack, S/N of load cell, S/N of cable, and current sensitivity of channel #1 from the strain indicator box. Insure the strain indicator box is configured correctly (i.e. 50000 full-scale value, appropriate sensitivity factor depending on load cell S/N). Balance the load cell in an unloaded state and confirm that the strain indicator box is reading near zero load.
2. Fasten wide-flange sections to ceiling with proper clearance (11 inches minimum) from any adhesive anchor to be tested.
3. Measure the initial anchor rod protrusion from the ceiling with a hand square. Document this measurement to the near 1/32" on the field note sheet.
4. While facing west take a photograph of the anchor rod with the hand square in place and a written bolt nomenclature label visible.
5. Thread the coupler over the anchor rod approximately 1 inch, to mid-depth of the coupler. Use a scaled probe to ensure embedment of the anchor rod halfway into the coupler.
6. Align and clamp the appropriate needle beam over the anchor to be tested to the wide-flange sections.
7. Position the jack and load cell assembly into place and screw the threaded loading rod extension into the remaining half of the threaded coupler.
8. Prior to hand tightening the 5/8" heavy-hex nut on the bottom of the jack/load cell assembly loading rod, center the spacer plates, jack, and load cell such that all centerlines are coincident.
9. Set the strain indicator box to record "Peak Min. Load".
10. Start the jacking operation. Slowly pump at a rate of one stroke from the hand pump every 5 seconds until the adhesive anchor no longer will resist the applied load. Continuously

observe the pressure gauge on the jack during this process to determine the peak level reached.

11. Once the adhesive anchor no longer can resist the applied load, record an approximate peak pressure (psi) displayed on the pressure gage and the peak load (pounds) captured on the strain indicator box onto the field note sheet. Also record any relevant observation or information particular to the individual test onto the field note sheet.
12. Reset the minimum peak in the strain indicator box and continue jacking at a rate of one stroke from the hand pump every five seconds to attain a secondary peak load. Record this secondary peak load on the field note sheet. *[This step was not specifically performed during the testing of July 15, 2006, and instead only an observed post failure peak load was periodically noted.]*
13. Remove the anchor rod from the ceiling using the hydraulic jack. Reset the jack as needed.

LOAD-DISPLACEMENT TESTING

General procedure for adhesive anchor load-displacement testing:

1. On a field note sheet, record the type of jack, S/N of load cell, S/N of cable, S/N of the LVTDs, and current sensitivity of channel #1 from the strain indicator box. Insure the strain indicator box is configured correctly (i.e. 50000 full-scale value, appropriate sensitivity factor depending on load cell S/N). Balance the load cell in an unloaded state and confirm that the strain indicator box is reading near zero load.
2. Fasten wide-flange sections to ceiling with proper clearance (11 inches minimum) from any adhesive anchor to be tested.
3. Measure the initial anchor rod protrusion from the ceiling with a hand square. Document this measurement to the near 1/32" on the field note sheet.
4. While facing west take a photograph of the anchor rod with the hand square in place and a written location label visible.
5. Thread a 5/8" heavy-hex nut onto the anchor rod approximately to within 1" of the tunnel ceiling. Slide the adhesive anchor through the center hole in the aluminum LVDT mounting plate until the plate bears on the heavy-hex nut installed in step 4. Double nut the LVDT mounting plate in place by threading another 5/8" heavy-hex nut onto the anchor rod and tightening it by hand.
6. Rotate the LVDT mounting plate so that the LVDT in the #1 position is approximately due north of the anchor. Use a wrench to tighten the double hex nuts holding the mounting plate

in place. Using the thumb screws, adjust the LVDTs so that they are at the near optimum position to read extension by referring to the data acquisition display.

7. Thread the coupler over the anchor rod approximately 1 inch, to mid-depth of the coupler. Use a scaled probe to ensure embedment of the anchor rod halfway into the coupler.
8. Align and clamp the appropriate needle beam over the anchor to be tested to the wide-flange sections.
9. Position the jack and load cell assembly into place and screw the threaded loading rod extension into the remaining half of the threaded coupler.
10. Prior to hand tightening the 5/8" heavy-hex nut on the bottom of the jack/load cell assembly loading rod, center the spacer plates, jack, and load cell such that all centerlines are coincident.
11. Set the strain indicator box to monitor "Min. Load" and start recording load and displacement measurements using the data acquisition system.
12. Using the hydraulic jack, apply and maintain an approximate 500 lbs load to the system for 2 minutes.
13. Proceeding in 1500 lbs increments, slowly pump at a rate of one stroke from the hand pump every 5 seconds until the anchor is resisting the load level desired or the adhesive anchor no longer will resist the applied load. Continuously observe the pressure gauge on the jack during this process to determine the peak level reached.
14. Maintain the load level desired for the 2 minute hold time or until the adhesive anchor no longer will resist the applied load.
15. After the specified load level hold time repeat steps 10 and 11 for the next desired load level.
16. Once the adhesive anchor no longer can resist the applied load, record an approximate peak pressure (psi) displayed on the pressure gage and the peak load (pounds) captured on the strain indicator box onto the field note sheet. Also record any relevant observation or information particular to the individual test onto the field note sheet.

17. Set the strain indicator box to monitor “Peak Min. Load” and continue jacking at a rate of one stroke from the hand pump every five seconds to attain a secondary peak load. Record this secondary peak load on the field note sheet.
18. Remove the anchor rod from the ceiling using the hydraulic jack. Reset the jack as needed.

APPENDIX B. TEST SPECIMEN REGISTER

Anchor 1 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-6-N	3.063	^b	^b	8682



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 46. Adhesive Anchor EB080-M2-6-N Prior to Load Test.

Anchor 2 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-6-S	3.031	b	b	9117



- ^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 47. Adhesive Anchor EB080-M2-6-S Prior to Load Test.

Anchor 3 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-3/4/5-NW	2.875			14300



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 48. Adhesive Anchor EB080-M2-3/4/5-NW Prior to Load Test.

Anchor 4 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-3/4/5-SW	3.000	b	b	11904



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Slight pre-test displacement observed but not quantified.

Figure 49. Adhesive Anchor EB080-M2-3/4/5-SW Prior to Load Test.

Anchor 5 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-3/4/5-NE	3.125			16443



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 50. Adhesive Anchor EB080-M2-3/4/5-NE Prior to Load Test.

Anchor 6 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-3/4/5-SE	3.031			6695

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 51. Adhesive Anchor EB080-M2-3/4/5-SE Prior to Load Test.

Anchor 7 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-2-N	3.125			14580



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 52. Adhesive Anchor EB080-M2-2-N Prior to Load Test.

Anchor 8 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-2-S	3.094			11648



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 53. Adhesive Anchor EB080-M2-2-S Prior to Load Test.

Anchor 9 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-1-N	3.000			16679



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 54. Adhesive Anchor EB080-M2-1-N Prior to Load Test.

Anchor 10 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M2-1-S	3.000			14562



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 55. Adhesive Anchor EB080-M2-1-S Prior to Load Test.

Anchor 11 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-12-N	2.938			11312



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 56. Adhesive Anchor EB081-M2-12-N Prior to Load Test.

Anchor 12 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-12-S	3.563			12258



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 57. Adhesive Anchor EB081-M2-12-S Prior to Load Test.

Anchor 13 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-9/10/11-NW	3.000			14126

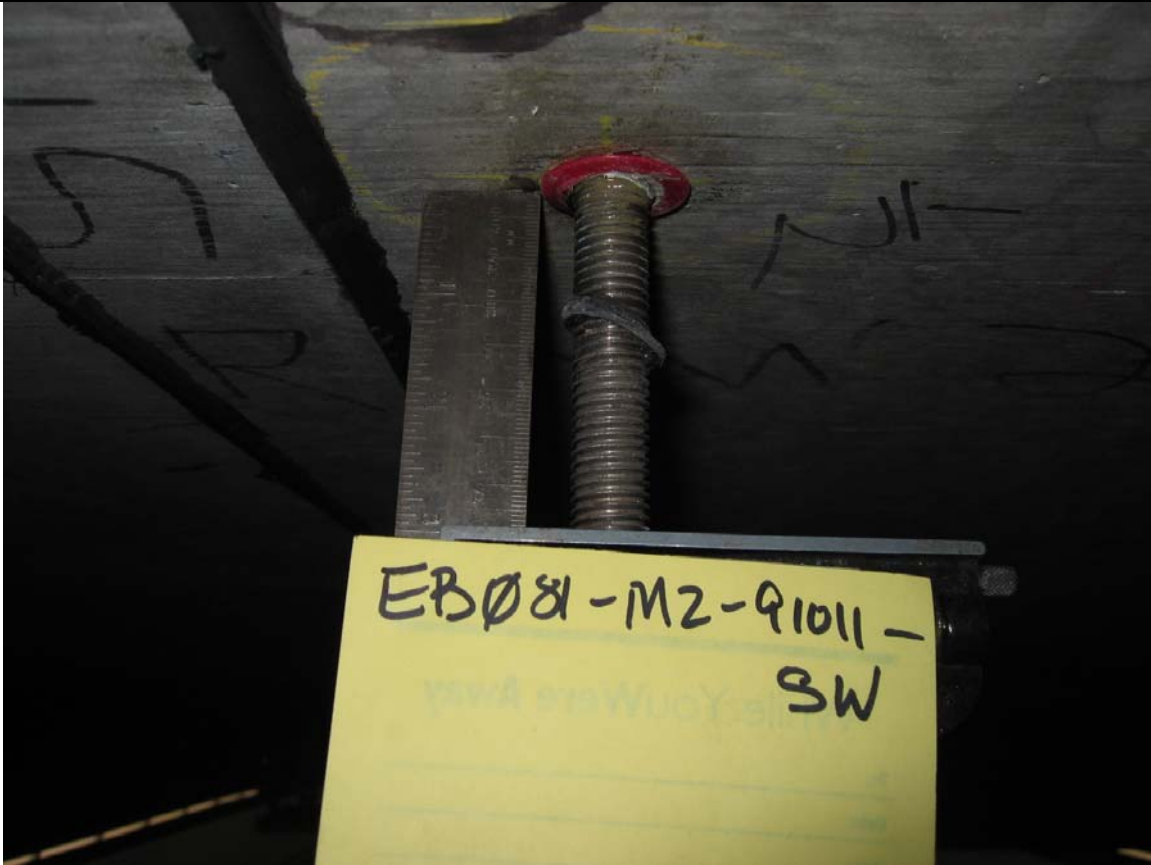


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 58. Adhesive Anchor EB081-M2-9/10/11-NW Prior to Load Test.

Anchor 14 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-9/10/11-SW	3.063			7093



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 59. Adhesive Anchor EB081-M2-9/10/11-SW Prior to Load Test.

Anchor 15 of 16 load tested July 15, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-9/10/11-NE	3.000			15803



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 60. Adhesive Anchor EB081-M2-9/10/11-NE Prior to Load Test.

Anchor 16 of 16 load tested July 15, 2006

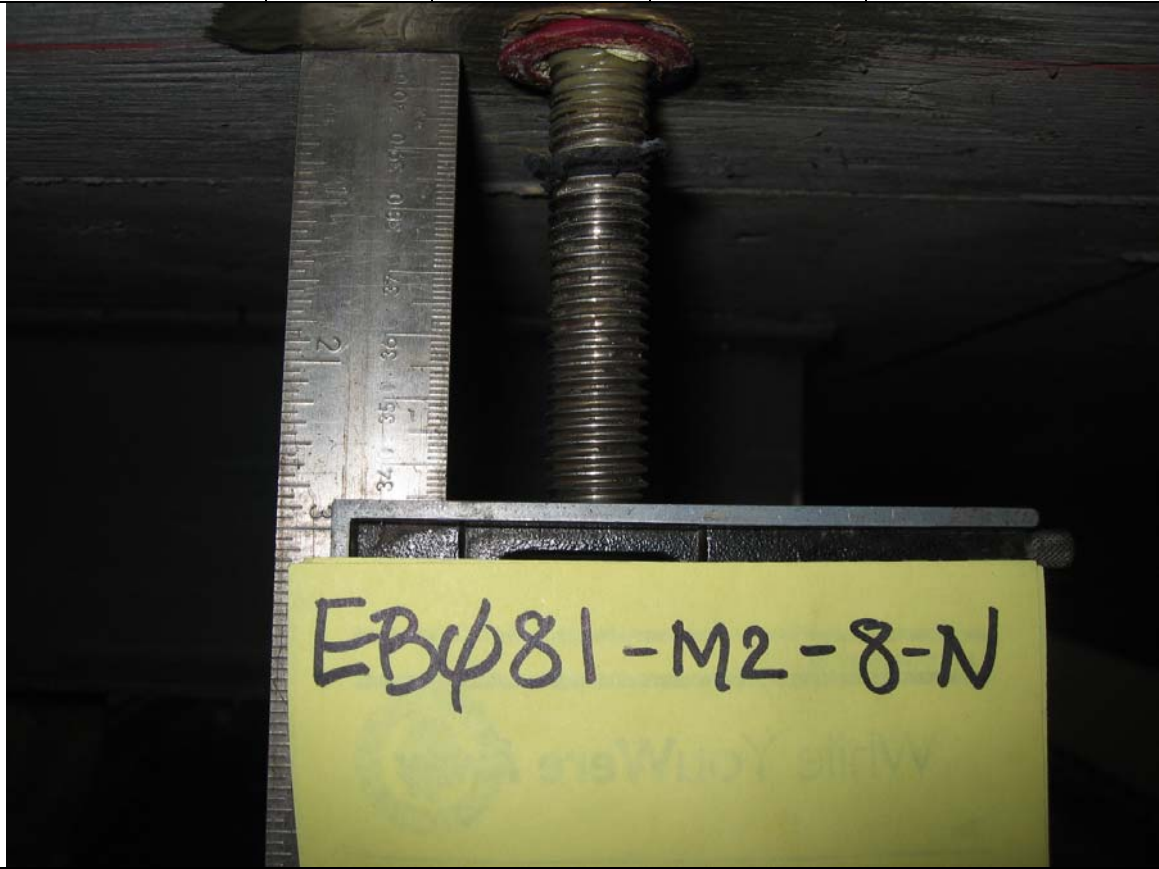
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-9/10/11-SE	3.188	^b	^b	6043

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 61. Adhesive Anchor EB081-M2-9/10/11-SE Prior to Load Test.

Anchor 1 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-8-N	2.875			11846



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 62. Adhesive Anchor EB081-M2-8-N Prior to Load Test.

Anchor 2 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-8-S	2.688			13117



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 63. Adhesive Anchor EB081-M2-8-S Prior to Load Test.

Anchor 3 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-7-N	2.875	^b	^b	11011

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 64. Adhesive Anchor EB081-M2-7-N Prior to Load Test.

Anchor 4 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-7-S	3.000	^b	^b	12175

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 65. Adhesive Anchor EB081-M2-7-S Prior to Load Test.

Anchor 5 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-6-N	2.875			10368



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 66. Adhesive Anchor EB081-M2-6-N Prior to Load Test.

Anchor 6 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-6-S	2.750			22770



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 67. Adhesive Anchor EB081-M2-6-S Prior to Load Test.

Anchor 7 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-5-N	2.813			12045



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 68. Adhesive Anchor EB081-M2-5-N Prior to Load Test.

Anchor 8 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-5-S	2.813			3179



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 69. Adhesive Anchor EB081-M2-5-S Prior to Load Test.

Anchor 9 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-1-N	2.813			13030



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 70. Adhesive Anchor EB081-M2-1-N Prior to Load Test.

Anchor 10 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-1-S	2.813			15285



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 71. Adhesive Anchor EB081-M2-1-S Prior to Load Test.

Anchor 11 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-12-N	3.250	0.375	0.469	4755



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 72. Adhesive Anchor EB082-M2-12-N Prior to Load Test.

Anchor 12 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-12-S	2.813			17196



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 73. Adhesive Anchor EB082-M2-12-S Prior to Load Test.

Anchor 13 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-11-N	3.188	0.250	0.250	6437



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 74. Adhesive Anchor EB082-M2-11-N Prior to Load Test.

Anchor 14 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-11-S	2.813			14215
No photographic record of pre-test condition.				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 75. Adhesive Anchor EB082-M2-11-S Prior to Load Test.

Anchor 15 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-7.5	2.875			7485



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 76. Adhesive Anchor EB082-M2-7.5 Prior to Load Test.

Anchor 16 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-7-N	3.125			6966



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 77. Adhesive Anchor EB082-M2-7-N Prior to Load Test.

Anchor 17 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-7-S	3.031			14155



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 78. Adhesive Anchor EB082-M2-7-S Prior to Load Test.

Anchor 18 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-6.5	2.938			13450



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 79. Adhesive Anchor EB082-M2-6.5 Prior to Load Test.

Anchor 19 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-10-N	3.500	0.438	0.375	1676



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 80. Adhesive Anchor EB083-M2-10-N Prior to Load Test.

Anchor 20 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-10-S	2.938			20626



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 81. Adhesive Anchor EB083-M2-10-S Prior to Load Test.

Anchor 21 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-11.5-N	3.000	^b	^b	6041

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 82. Adhesive Anchor EB083-M2-11.5-N Prior to Load Test.

Anchor 22 of 22 load tested July 16, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-11.5-S	2.938			7450



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 83. Adhesive Anchor EB083-M2-11.5-S Prior to Load Test.

Anchor 1 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-5-N	3.031			17793



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 84. Adhesive Anchor EB083-M2-5-N Prior to Load Test.

Anchor 2 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-5-S	3.000			19069



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 85. Adhesive Anchor EB083-M2-5-S Prior to Load Test.

Anchor 3 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-11-N	3.438	0.500	0.438	6608



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 86. Adhesive Anchor EB083-M2-11-N Prior to Load Test.

Anchor 4 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-11-S	2.969			17558



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 87. Adhesive Anchor EB083-M2-11-S Prior to Load Test.

Anchor 5 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-10-N	3.000			14515



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 88. Adhesive Anchor EB083-S-10-N Prior to Load Test.

Anchor 6 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-10-S	2.906			15022



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 89. Adhesive Anchor EB083-S-10-S Prior to Load Test.

Anchor 7 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-7-N	3.000			12522



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 90. Adhesive Anchor EB082-S-7-N Prior to Load Test.

Anchor 8 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-7-S	2.938			11107



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 91. Adhesive Anchor EB082-S-7-S Prior to Load Test.

Anchor 9 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-7.5-N	3.063			3329

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 92. Adhesive Anchor EB082-S-7.5-N Prior to Load Test.

Anchor 10 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-7.5-S	3.625	0.625	0.625	2768



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 93. Adhesive Anchor EB082-S-7.5-S Prior to Load Test.

Anchor 11 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-9-N	3.094			12058



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 94. Adhesive Anchor EB081-S-9-N Prior to Load Test.

Anchor 12 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-9-S	3.313	0.438	0.406	4489



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 95. Adhesive Anchor EB081-S-9-S Prior to Load Test.

Anchor 13 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-10-N	3.250			5633



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 96. Adhesive Anchor EB081-S-10-N Prior to Load Test.

Anchor 14 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-10-S	4.125	1.375	1.375	1639



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 97. Adhesive Anchor EB081-S-10-S Prior to Load Test.

Anchor 15 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-2-N	4.688	1.906	1.906	1600



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 98. Adhesive Anchor EB080-M1-2-N Prior to Load Test.

Anchor 16 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-2-S	3.000			19972



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 99. Adhesive Anchor EB080-M1-2-S Prior to Load Test.

Anchor 17 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-1-N ^b	3.438	0.500	0.438	9437

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight bend north

Figure 100. Adhesive Anchor EB080-M1-1-N Prior to Load Test.

Anchor 18 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-1-S	5.375	2.625	2.625	1121



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 101. Adhesive Anchor EB080-M1-1-S Prior to Load Test.

Anchor 19 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-12-N ^b	3.000	0.313	0.375	8969

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight bend north

Figure 102. Adhesive Anchor EB082-M1-12-N Prior to Load Test.

Anchor 20 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-12-S ^b	5.500	2.500	2.500	1559



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Slight bend north

Figure 103. Adhesive Anchor EB082-M1-12-S Prior to Load Test.

Anchor 21 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-11-N	2.875			16732



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 104. Adhesive Anchor EB082-M1-11-N Prior to Load Test.

Anchor 22 of 22 load tested July 17, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-11-S	4.000	1.250	1.250	3448



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 105. Adhesive Anchor EB082-M1-11-S Prior to Load Test.

Anchor 1 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-3-N ^b	3.000			12324



- ^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight bend south

Figure 106. Adhesive Anchor EB080-S-3-N Prior to Load Test.

Anchor 2 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-3-S	3.031			14736



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 107. Adhesive Anchor EB080-S-3-S Prior to Load Test.

Anchor 3 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-2-N	3.000			10104



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 108. Adhesive Anchor EB080-S-2-N Prior to Load Test.

Anchor 4 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-2-S	3.063			13428



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 109. Adhesive Anchor EB080-S-2-S Prior to Load Test.

Anchor 5 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-4/5/6-NW	2.938			12673



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 110. Adhesive Anchor EB080-S-4/5/6-NW Prior to Load Test.

Anchor 6 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-4/5/6-NE	2.906			16552



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 111. Adhesive Anchor EB080-S-4/5/6-NE Prior to Load Test.

Anchor 7 of 50 load tested July 18, 2006

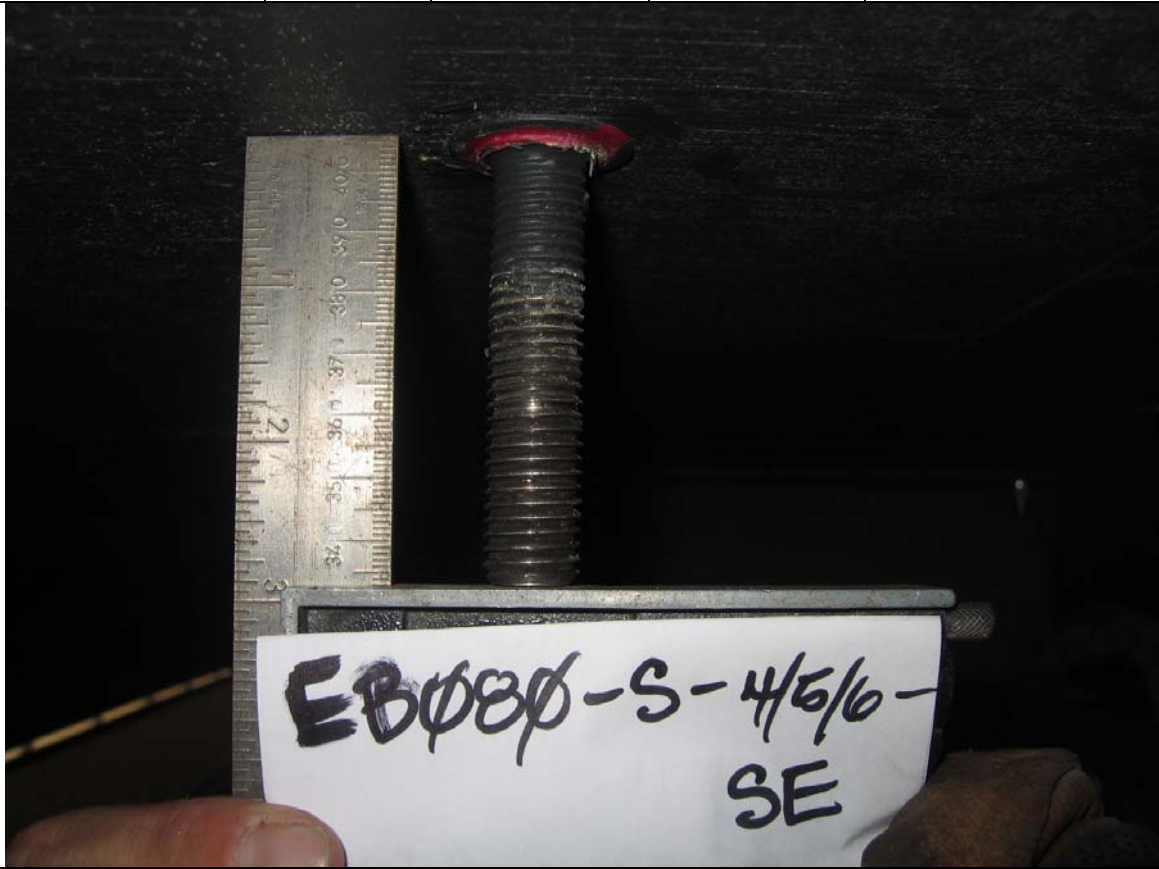
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-4/5/6-SW	3.156			8021

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 112. Adhesive Anchor EB080-S-4/5/6-SW Prior to Load Test.

Anchor 8 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-4/5/6-SE	2.906			9914



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 113. Adhesive Anchor EB080-S-4/5/6-SE Prior to Load Test.

Anchor 9 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-8-N	2.781			18622



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 114. Adhesive Anchor EB081-S-8-N Prior to Load Test.

Anchor 10 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-8-S	3.156	0.531	0.563	2793



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 115. Adhesive Anchor EB081-S-8-S Prior to Load Test.

Anchor 11 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-1-N	3.156			15039



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 116. Adhesive Anchor EB080-S-1-N Prior to Load Test.

Anchor 12 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-S-1-S	3.000			10692



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 117. Adhesive Anchor EB080-S-1-S Prior to Load Test.

Anchor 13 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-4/5/6-NW	2.906			12375



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 118. Adhesive Anchor EB081-S-4/5/6-NW Prior to Load Test.

Anchor 14 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-4/5/6-NE	3.031			12618



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 119. Adhesive Anchor EB081-S-4/5/6-NE Prior to Load Test.

Anchor 15 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-4/5/6-SW	2.938			10088



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 120. Adhesive Anchor EB081-S-4/5/6-SW Prior to Load Test.

Anchor 16 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-4/5/6-SE	3.000			7440



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 121. Adhesive Anchor EB081-S-4/5/6-SE Prior to Load Test.

Anchor 17 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-3-N	3.000			11166



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 122. Adhesive Anchor EB081-S-3-N Prior to Load Test.

Anchor 18 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-3-S	2.969			8220



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 123. Adhesive Anchor EB081-S-3-S Prior to Load Test.

Anchor 19 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-7-N	2.781			10613



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 124. Adhesive Anchor EB081-S-7-N Prior to Load Test.

Anchor 20 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-7-S	2.813			8165



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 125. Adhesive Anchor EB081-S-7-S Prior to Load Test.

Anchor 21 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-2-N	3.000			2683



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 126. Adhesive Anchor EB081-S-2-N Prior to Load Test.

Anchor 22 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-2-S	3.063			8568

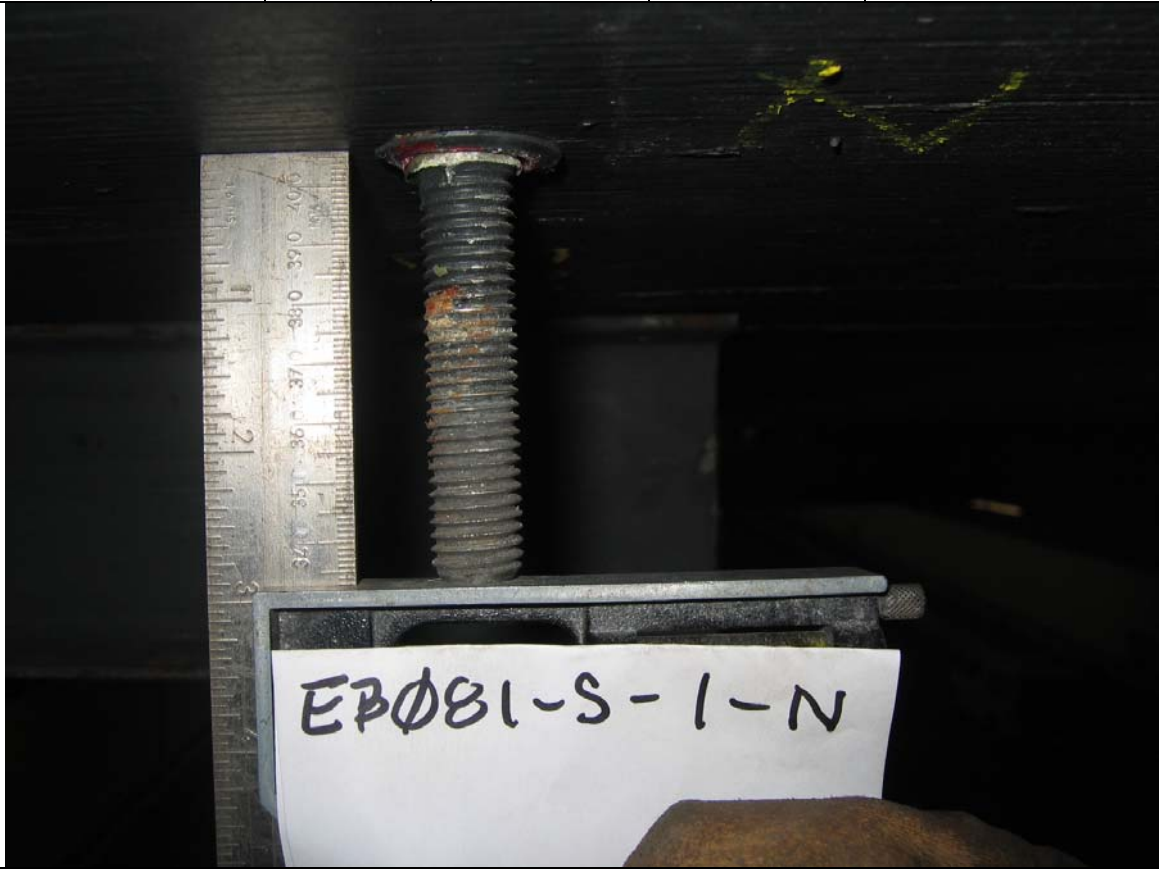


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 127. Adhesive Anchor EB081-S-2-S Prior to Load Test.

Anchor 23 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-1-N	2.906			10855



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 128. Adhesive Anchor EB081-S-1-N Prior to Load Test.

Anchor 24 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-S-1-S	2.781			9184



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 129. Adhesive Anchor EB081-S-1-S Prior to Load Test.

Anchor 25 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-10-N	2.781			9894



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 130. Adhesive Anchor EB082-S-10-N Prior to Load Test.

Anchor 26 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-10-S	2.750			12526



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 131. Adhesive Anchor EB082-S-10-S Prior to Load Test.

Anchor 27 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-9-N	2.844			14964



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 132. Adhesive Anchor EB082-S-9-N Prior to Load Test.

Anchor 28 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-9-S	2.938			13454



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 133. Adhesive Anchor EB082-S-9-S Prior to Load Test.

Anchor 29 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-9.5-N	2.969			10156



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 134. Adhesive Anchor EB082-S-9.5-N Prior to Load Test.

Anchor 30 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-9.5-S	3.094			4928



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 135. Adhesive Anchor EB082-S-9.5-S Prior to Load Test.

Anchor 31 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-4/5/6-NW	3.000			9497



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 136. Adhesive Anchor EB082-S-4/5/6-NW Prior to Load Test.

Anchor 32 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-4/5/6-NE	2.969			12191



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 137. Adhesive Anchor EB082-S-4/5/6-NE Prior to Load Test.

Anchor 33 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-4/5/6-SW	2.875			14872



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 138. Adhesive Anchor EB082-S-4/5/6-SW Prior to Load Test.

Anchor 34 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-4/5/6-SE	2.906			12820



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 139. Adhesive Anchor EB082-S-4/5/6-SE Prior to Load Test.

Anchor 35 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-3-N	3.000			14185



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 140. Adhesive Anchor EB082-S-3-N Prior to Load Test.

Anchor 36 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-3-S	3.063			23115



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 141. Adhesive Anchor EB082-S-3-S Prior to Load Test.

Anchor 37 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-2-N	2.969			10040



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 142. Adhesive Anchor EB082-S-2-N Prior to Load Test.

Anchor 38 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-2-S	3.000			9576



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 143. Adhesive Anchor EB082-S-2-S Prior to Load Test.

Anchor 39 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-1-N	3.000			18021



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 144. Adhesive Anchor EB082-S-1-N Prior to Load Test.

Anchor 40 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-1-S	3.125			1851



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 145. Adhesive Anchor EB082-S-1-S Prior to Load Test.

Anchor 41 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-9-N	2.938			11535



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 146. Adhesive Anchor EB083-S-9-N Prior to Load Test.

Anchor 42 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-9-S	3.000			18876



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 147. Adhesive Anchor EB083-S-9-S Prior to Load Test.

Anchor 43 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-8-N	2.875			17684



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 148. Adhesive Anchor EB083-S-8-N Prior to Load Test.

Anchor 44 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-8-S	3.063			8251



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 149. Adhesive Anchor EB083-S-8-S Prior to Load Test.

Anchor 45 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-7-N	2.906			9208



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 150. Adhesive Anchor EB083-S-7-N Prior to Load Test.

Anchor 46 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-7-S	2.906			12811

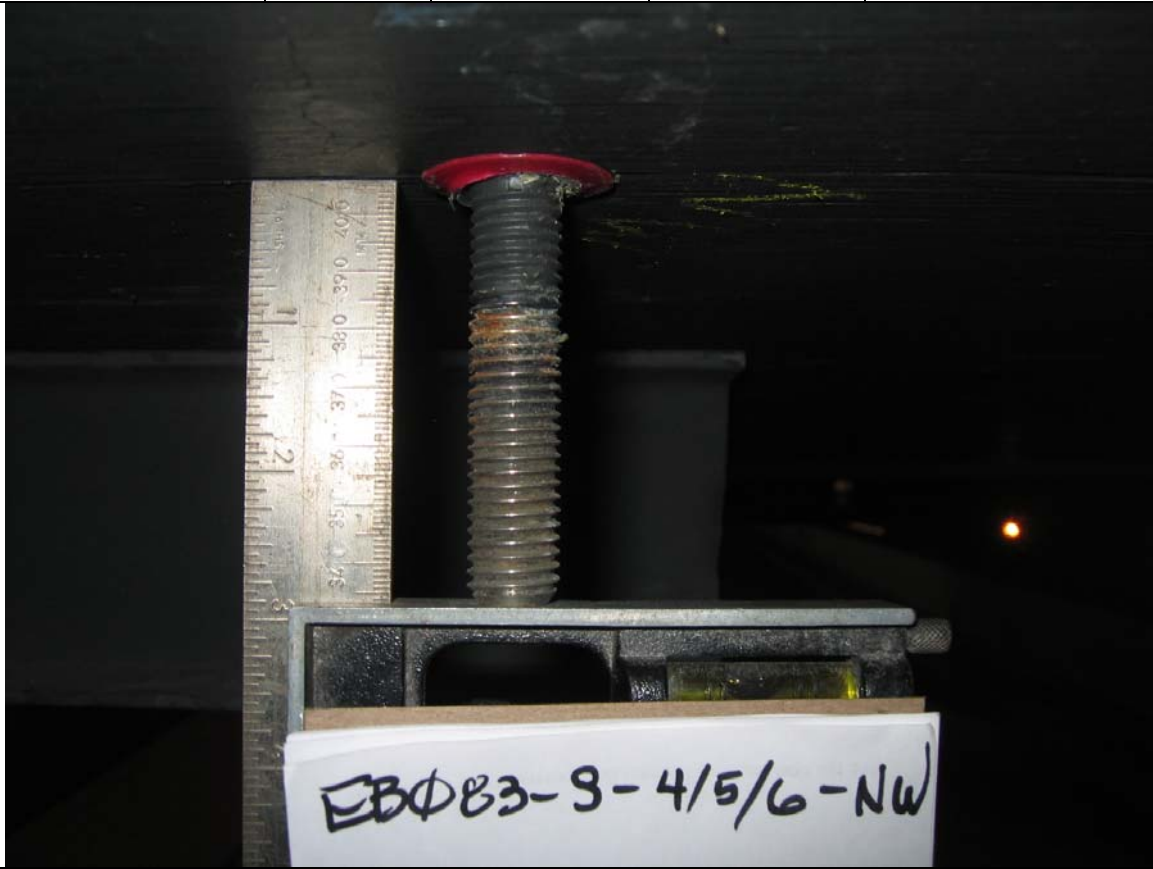


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 151. Adhesive Anchor EB083-S-7-S Prior to Load Test.

Anchor 47 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-4/5/6-NW	2.875			13373



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 152. Adhesive Anchor EB083-S-4/5/6-NW Prior to Load Test.

Anchor 48 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-4/5/6-NE	2.875			10932



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 153. Adhesive Anchor EB083-S-4/5/6-NE Prior to Load Test.

Anchor 49 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-4/5/6-SW	2.969			13328



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 154. Adhesive Anchor EB083-S-4/5/6-SW Prior to Load Test.

Anchor 50 of 50 load tested July 18, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-4/5/6-SE ^b	2.938			17395

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Shallow cone failure

Figure 155. Adhesive Anchor EB083-S-4/5/6-SE Prior to Load Test.

Anchor 1 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-3-N	2.938			9014



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 156. Adhesive Anchor EB083-S-3-N Prior to Load Test.

Anchor 2 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-3-S	2.938			14928



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 157. Adhesive Anchor EB083-S-3-S Prior to Load Test.

Anchor 3 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-2-N	3.125			15296



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 158. Adhesive Anchor EB083-S-2-N Prior to Load Test.

Anchor 4 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-2-S	3.125			5379



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 159. Adhesive Anchor EB083-S-2-S Prior to Load Test.

Anchor 5 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-1-N	3.031			8389



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 160. Adhesive Anchor EB083-S-1-N Prior to Load Test.

Anchor 6 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-S-1-S	3.438	0.250	0.219	9511



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 161. Adhesive Anchor EB083-S-1-S Prior to Load Test.

Anchor 7 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-8-N	3.156			13387



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 162. Adhesive Anchor EB082-S-8-N Prior to Load Test.

Anchor 8 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-8-S	3.063			7566



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 163. Adhesive Anchor EB082-S-8-S Prior to Load Test.

Anchor 9 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-8.5-N	2.813			17323



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 164. Adhesive Anchor EB082-S-8.5-N Prior to Load Test.

Anchor 10 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-S-8.5-S	2.813			20264



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 165. Adhesive Anchor EB082-S-8.5-S Prior to Load Test.

Anchor 11 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-1-N	3.125	0.281	0.625 ^b	8867



- ^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Possible measurement recording error as magnitude does not match photograph.

Figure 166. Adhesive Anchor EB083-M2-1-N Prior to Load Test.

Anchor 12 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-1-S	3.031	0.125	0.125	12977



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 167. Adhesive Anchor EB083-M2-1-S Prior to Load Test.

Anchor 13 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-2-N	2.875			18356



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 168. Adhesive Anchor EB083-M2-2-N Prior to Load Test.

Anchor 14 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-2-S	2.938			16914



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 169. Adhesive Anchor EB083-M2-2-S Prior to Load Test.

Anchor 15 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-3/4-NW	3.031			20198



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 170. Adhesive Anchor EB083-M2-3/4-NW Prior to Load Test.

Anchor 16 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-3/4-NE	3.000			15516



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 171. Adhesive Anchor EB083-M2-3/4-NE Prior to Load Test.

Anchor 17 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-3/4-SW	2.875			21725



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 172. Adhesive Anchor EB083-M2-3/4-SW Prior to Load Test.

Anchor 18 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-3/4-SE	3.094			18260



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 173. Adhesive Anchor EB083-M2-3/4-SE Prior to Load Test.

Anchor 19 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-6-N	3.063			9622



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 174. Adhesive Anchor EB083-M2-6-N Prior to Load Test.

Anchor 20 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-6-S	3.063			12766



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 175. Adhesive Anchor EB083-M2-6-S Prior to Load Test.

Anchor 21 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-7/8/9-NW	3.063			11340



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 176. Adhesive Anchor EB083-M2-7/8/9-NW Prior to Load Test.

Anchor 22 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-7/8/9-NE ^b	3.000			18854

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Shallow cone failure

Figure 177. Adhesive Anchor EB083-M2-7/8/9-NE Prior to Load Test.

Anchor 23 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-7/8/9-SW	3.031	0.281	0.281	9130



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 178. Adhesive Anchor EB083-M2-7/8/9-SW Prior to Load Test.

Anchor 24 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M2-7/8/9-SE	3.156	0.094	0.125	9925

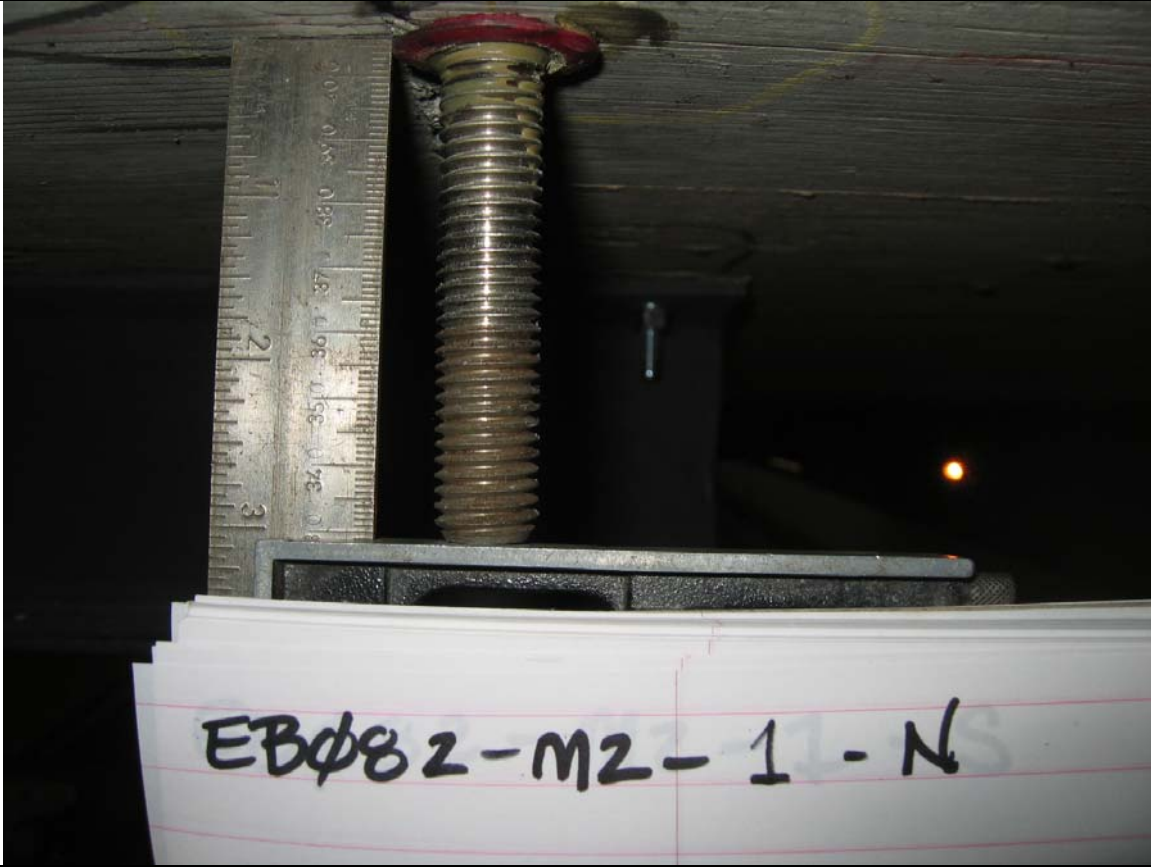


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 179. Adhesive Anchor EB083-M2-7/8/9-SE Prior to Load Test.

Anchor 25 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-1-N	3.063			10998

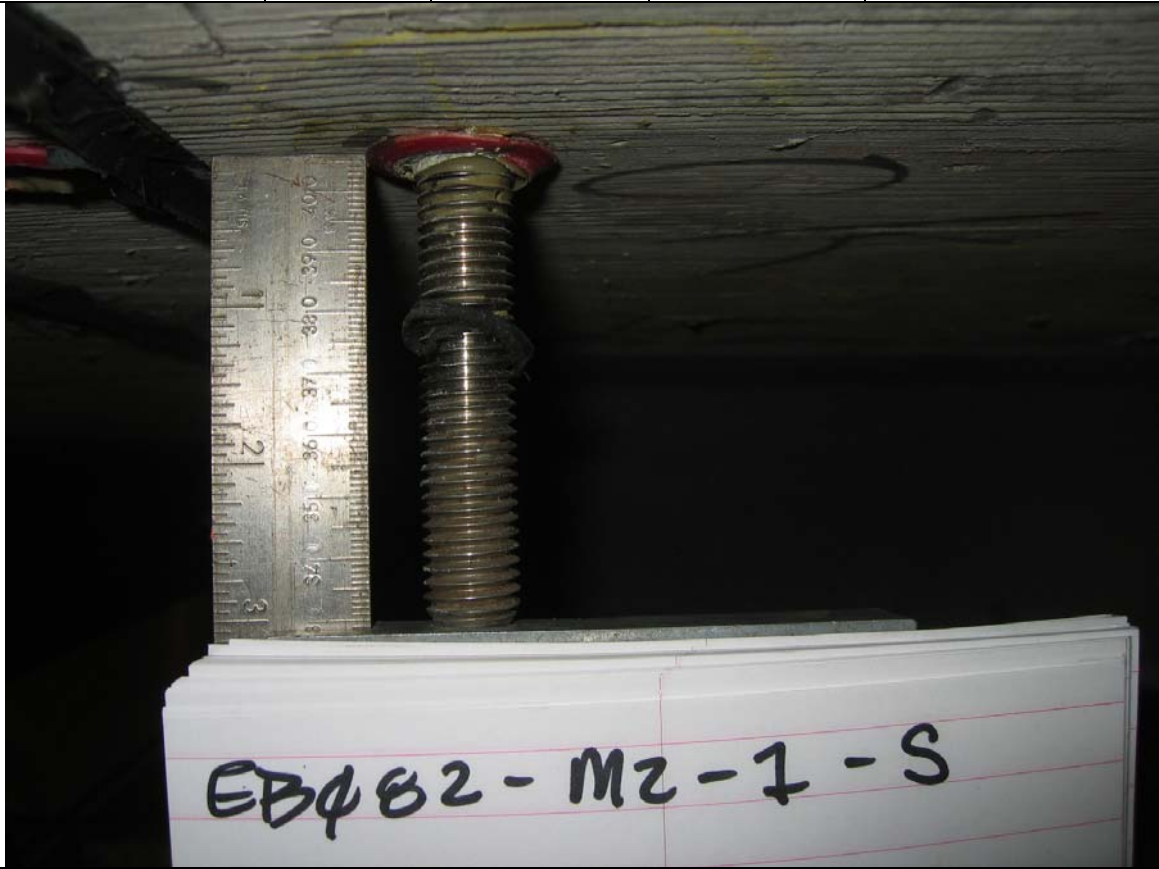


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 180. Adhesive Anchor EB082-M2-1-N Prior to Load Test.

Anchor 26 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-1-S	3.063			15552



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 181. Adhesive Anchor EB082-M2-1-S Prior to Load Test.

Anchor 27 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-2-N	3.000			19933

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 182. Adhesive Anchor EB082-M2-2-N Prior to Load Test.

Anchor 28 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-2-S	3.188			6304



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 183. Adhesive Anchor EB082-M2-2-S Prior to Load Test.

Anchor 29 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-3/4/5-NW ^b	3.000			20367

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight bend south

Figure 184. Adhesive Anchor EB082-M2-3/4/5-NW Prior to Load Test.

Anchor 30 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-3/4/5-NE	3.094	^b	^b	13083

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 185. Adhesive Anchor EB082-M2-3/4/5-NE Prior to Load Test.

Anchor 31 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-3/4/5-SW	3.000			17207



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 186. Adhesive Anchor EB082-M2-3/4/5-SW Prior to Load Test.

Anchor 32 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-3/4/5-SE	3.156	^b	^b	9740

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 187. Adhesive Anchor EB082-M2-3/4/5-SE Prior to Load Test.

Anchor 33 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-6-N	3.063			17061



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 188. Adhesive Anchor EB082-M2-6-N Prior to Load Test.

Anchor 34 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-6-S	3.125			13888



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 189. Adhesive Anchor EB082-M2-6-S Prior to Load Test.

Anchor 35 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-8/9/10-NW	3.125	0.313	0.281	4152



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 190. Adhesive Anchor EB082-M2-8/9/10-NW Prior to Load Test.

Anchor 36 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-8/9/10-NE	3.094	0.188	0.219	7444



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 191. Adhesive Anchor EB082-M2-8/9/10-NE Prior to Load Test.

Anchor 37 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-8/9/10-SW	3.000			12995



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 192. Adhesive Anchor EB082-M2-8/9/10-SW Prior to Load Test.

Anchor 38 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M2-8/9/10-SE	2.938			12956



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 193. Adhesive Anchor EB082-M2-8/9/10-SE Prior to Load Test.

Anchor 39 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-2/3/4-NW	2.719	0.063	0.063	16119



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 194. Adhesive Anchor EB081-M2-2/3/4-NW Prior to Load Test.

Anchor 40 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-2/3/4-NE	2.781			15732



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 195. Adhesive Anchor EB081-M2-2/3/4-NE Prior to Load Test.

Anchor 41 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-2/3/4-SW	2.781			13192



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 196. Adhesive Anchor EB081-M2-2/3/4-SW Prior to Load Test.

Anchor 42 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-M2-2/3/4-SE	2.813			14278



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 197. Adhesive Anchor EB081-M2-2/3/4-SE Prior to Load Test.

Anchor 43 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-N-3-N	2.969			16830



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 198. Adhesive Anchor EB083-N-3-N Prior to Load Test.

Anchor 44 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-N-3-S	3.000			24242



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 199. Adhesive Anchor EB083-N-3-S Prior to Load Test.

Anchor 45 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M1-1-N	2.969			14935



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 200. Adhesive Anchor EB083-M1-1-N Prior to Load Test.

Anchor 46 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M1-1-S	3.031			12524



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 201. Adhesive Anchor EB083-M1-1-S Prior to Load Test.

Anchor 47 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M1-2-N	3.000			16930



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 202. Adhesive Anchor EB083-M1-2-N Prior to Load Test.

Anchor 48 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB083-M1-2-S	3.000	^b	^b	9297



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 203. Adhesive Anchor EB083-M1-2-S Prior to Load Test.

Anchor 49 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-8/9/10-NW	3.438	0.625	0.594	6021



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 204. Adhesive Anchor EB082-M1-8/9/10-NW Prior to Load Test.

Anchor 50 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-8/9/10-NE	3.219	0.344	0.344	10204



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 205. Adhesive Anchor EB082-M1-8/9/10-NE Prior to Load Test.

Anchor 51 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-8/9/10-SW	3.281	0.438	0.406	9979



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 206. Adhesive Anchor EB082-M1-8/9/10-SW Prior to Load Test.

Anchor 52 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-M1-8/9/10-SE	3.188	0.156	0.188	10678



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 207. Adhesive Anchor EB082-M1-8/9/10-SE Prior to Load Test.

Anchor 53 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-3/4/5-NW	3.719	0.750	0.750	3562



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 208. Adhesive Anchor EB080-M1-3/4/5-NW Prior to Load Test.

Anchor 54 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-3/4/5-NE	3.938	0.875	0.875	3762



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 209. Adhesive Anchor EB080-M1-3/4/5-NE Prior to Load Test.

Anchor 55 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-3/4/5-SW	3.156	0.313	0.344	10740



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 210. Adhesive Anchor EB080-M1-3/4/5-SW Prior to Load Test.

Anchor 56 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-M1-3/4/5-SE	3.250	0.438	0.438	7369



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 211. Adhesive Anchor EB080-M1-3/4/5-SE Prior to Load Test.

Anchor 57 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-N-4/5/6-NW	2.969			14716



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 212. Adhesive Anchor EB080-N-4/5/6-NW Prior to Load Test.

Anchor 58 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-N-4/5/6-NE	3.250	0.313	0.281	1854



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 213. Adhesive Anchor EB080-N-4/5/6-NE Prior to Load Test.

Anchor 59 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-N-4/5/6-SW	3.344			19955

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 214. Adhesive Anchor EB080-N-4/5/6-SW Prior to Load Test.

Anchor 60 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB080-N-4/5/6-SE	3.188	0.313	0.313	5737



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 215. Adhesive Anchor EB080-N-4/5/6-SE Prior to Load Test.

Anchor 61 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-10-N ^b	5.125	2.250	2.219	8605

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight bend south

Figure 216. Adhesive Anchor EB081-N-10-N Prior to Load Test.

Anchor 62 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-10-S ^b	3.063			5020



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Slight bend south

Figure 217. Adhesive Anchor EB081-N-10-S Prior to Load Test.

Anchor 63 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-9-N ^b	2.969			18822



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Slight bend north

Figure 218. Adhesive Anchor EB081-N-9-N Prior to Load Test.

Anchor 64 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-9-S	3.000			19721

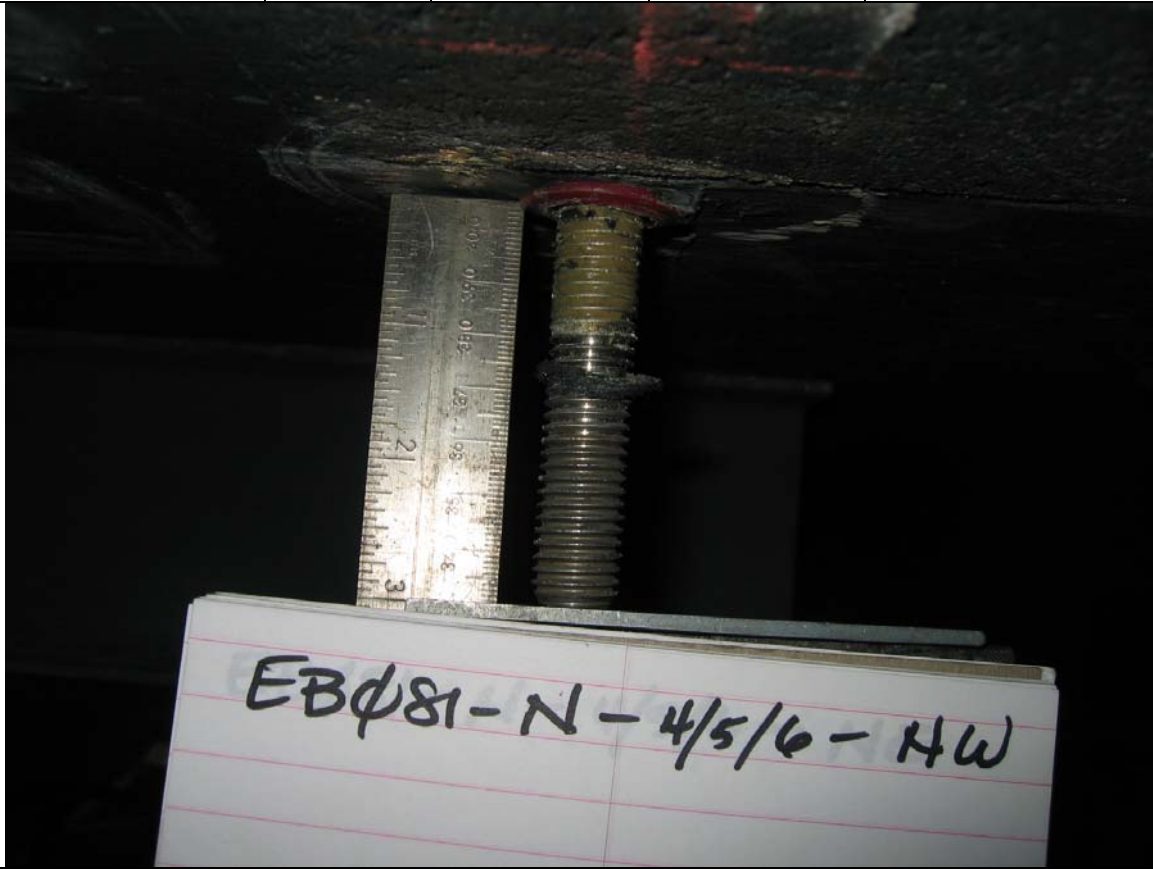


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 219. Adhesive Anchor EB081-N-9-S Prior to Load Test.

Anchor 65 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-4/5/6-NW	3.000	0.125	0.125	17658



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 220. Adhesive Anchor EB081-N-4/5/6-NW Prior to Load Test.

Anchor 66 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-4/5/6-NE	2.938	0.125	0.125	8264



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 221. Adhesive Anchor EB081-N-4/5/6-NE Prior to Load Test.

Anchor 67 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-4/5/6-SW	2.813			15017



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 222. Adhesive Anchor EB081-N-4/5/6-SW Prior to Load Test.

Anchor 68 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB081-N-4/5/6-SE	2.938			11653



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 223. Adhesive Anchor EB081-N-4/5/6-SE Prior to Load Test.

Anchor 69 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-N-10-N	2.813			13595



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 224. Adhesive Anchor EB082-N-10-N Prior to Load Test.

Anchor 70 of 70 load tested July 19, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
EB082-N-10-S	2.875			16962



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 225. Adhesive Anchor EB082-N-10-S Prior to Load Test.

Anchor 1 of 8 load tested July 20, 2006

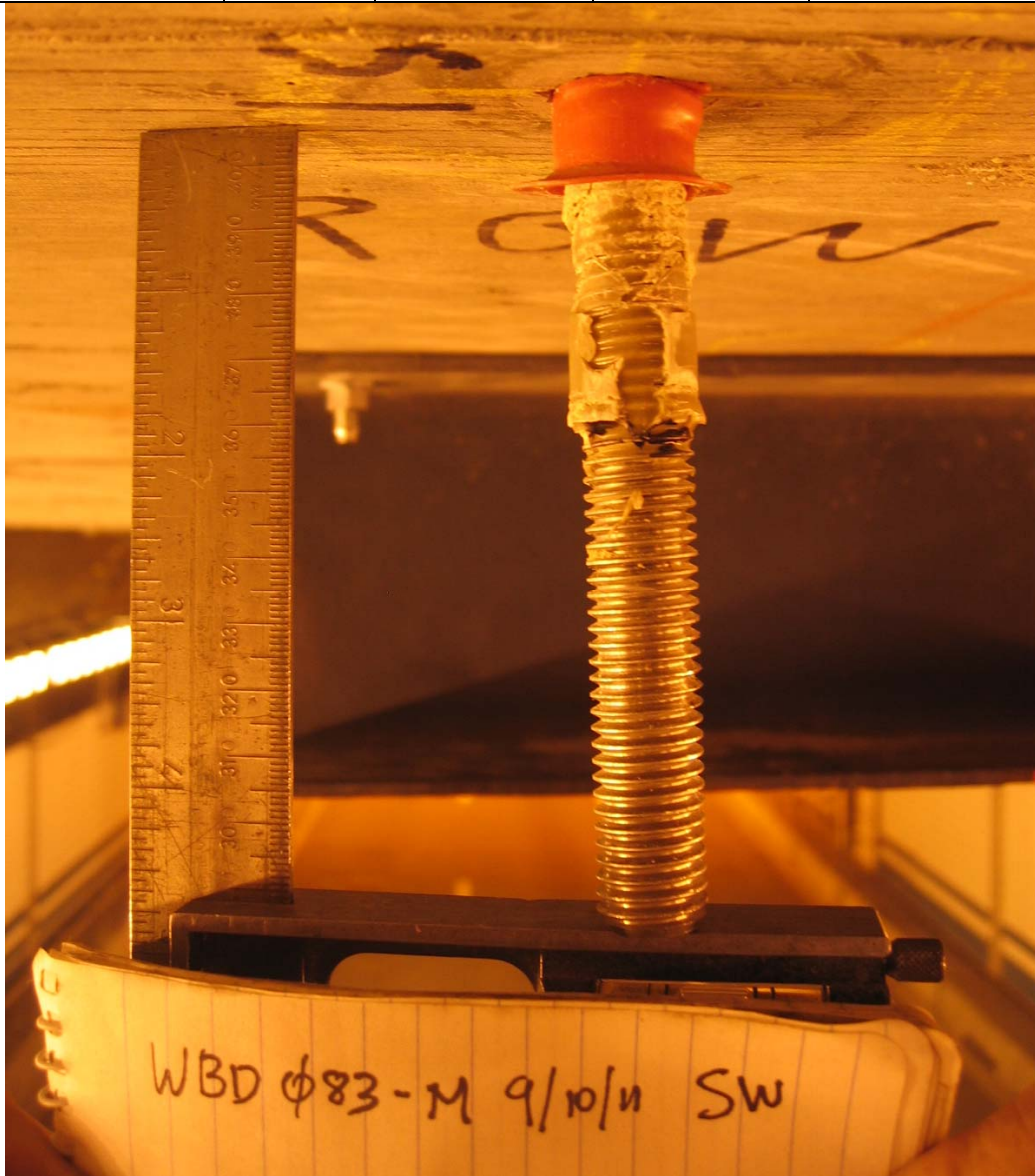
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD083-M-9/10/11-NW	3.063	^b	^b	9280

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Slight pre-test displacement observed but not quantified.

Figure 226. Adhesive Anchor WBD083-M-9/10/11-NW Prior to Load Test.

Anchor 2 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD083-M-9/10/11-SW	4.688	1.875	1.875	1403



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 227. Adhesive Anchor WBD083-M-9/10/11-SW Prior to Load Test.

Anchor 3 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD083-M-9/10/11-SE	4.625	1.813	1.813	1679

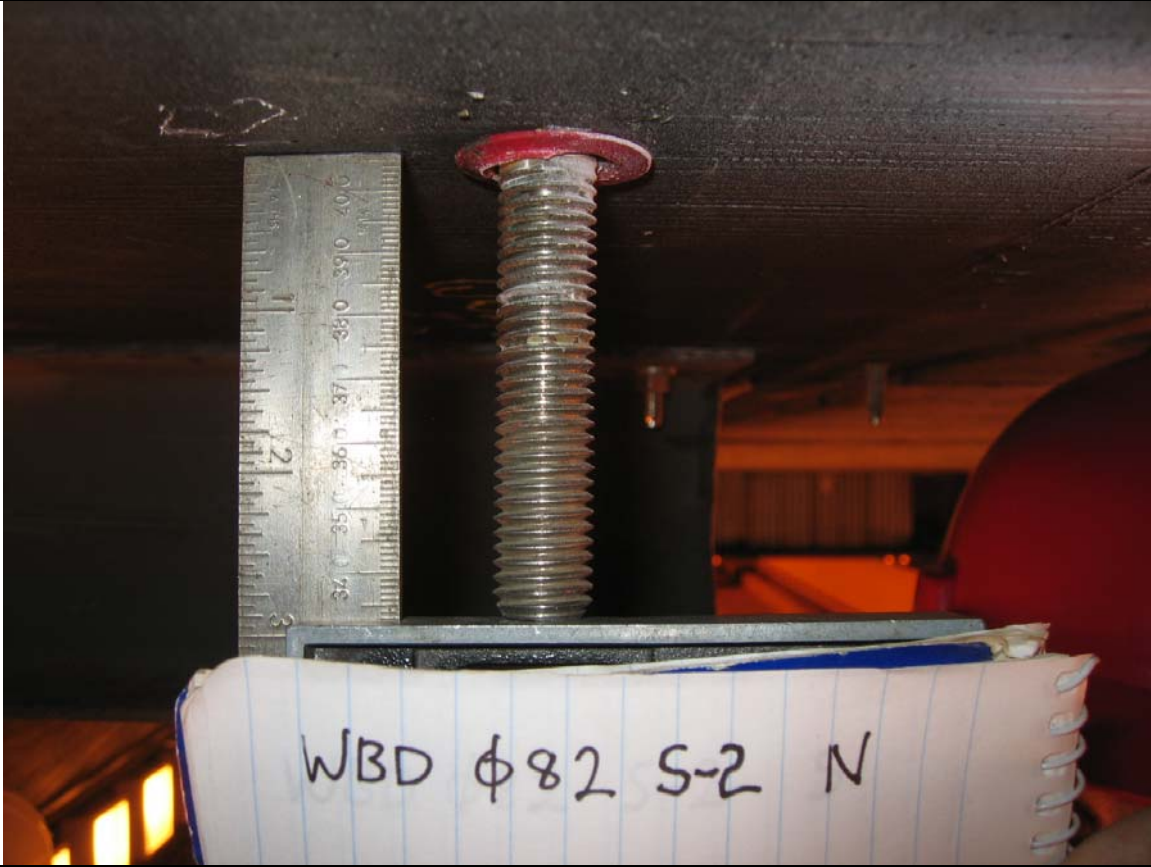


^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 228. Adhesive Anchor WBD083-M-9/10/11-SE Prior to Load Test.

Anchor 4 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD082-S-2-N	2.875			12201



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 229. Adhesive Anchor WBD082-S-2-N Prior to Load Test.

Anchor 5 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD082-S-2-S	2.469 ^b			12164



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Possible measurement recording error as magnitude does not match photograph.

Figure 230. Adhesive Anchor WBD082-S-2-S Prior to Load Test.

Anchor 6 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD082-M-6-N	3.000	0.125	0.125	5598



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 231. Adhesive Anchor WBD082-M-6-N Prior to Load Test.

Anchor 7 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD082-M-6-S	4.000	1.188	1.188	2727



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 232. Adhesive Anchor WBD082-M-6-S Prior to Load Test.

Anchor 8 of 8 load tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD082-M-5-S	3.813	1.031	1.031	470 ^b



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Suspect value, load indicated was much lower than corresponding pressure value (approximately 3500 lbs based on recorded pressure of 1400 psi).

Figure 233. Adhesive Anchor WBD082-M-5-S Prior to Load Test.

Anchor 1 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD084-N-4/5/6-NW	2.938			3674
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 234. Adhesive Anchor WBD084-N-4/5/6-NW Prior to Load-Displacement Test.

Anchor 2 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD084-N-4/5/6-NE	3.063			9074
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 235. Adhesive Anchor WBD084-N-4/5/6-NE Prior to Load-Displacement Test.

Anchor 3 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD084-N-4/5/6-SW	3.031			13197
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 236. Adhesive Anchor WBD084-N-4/5/6-SW Prior to Load-Displacement Test.

Anchor 4 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD084-N-4/5/6-SE	3.000			7883
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 237. Adhesive Anchor WBD084-N-4/5/6-SE Prior to Load-Displacement Test.

Anchor 5 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD082-M-5-N	2.875			12498
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 238. Adhesive Anchor WBD082-M-5-N Prior to Load-Displacement Test.

Anchor 6 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD083-N-9-N	3.156	0.469	0.438	3388
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 239. Adhesive Anchor WBD083-N-9-N Prior to Load-Displacement Test.

Anchor 7 of 7 load-displacement (dial gage) tested July 20, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Peak Load (lbs) ^a
		East Side (inches)	West Side (inches)	
WBD083-N-9-S	2.875			7130
				

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 240. Adhesive Anchor WBD083-N-9-S Prior to Load-Displacement Test.

Anchor 1 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-6-S	3.625	0.500	0.625	7000 / 2

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 241. Adhesive Anchor WBD084-M-6-S Prior to Load-Displacement Test.

Anchor 2 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-8-N	3.563	0.750	0.750	5700 / 2
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

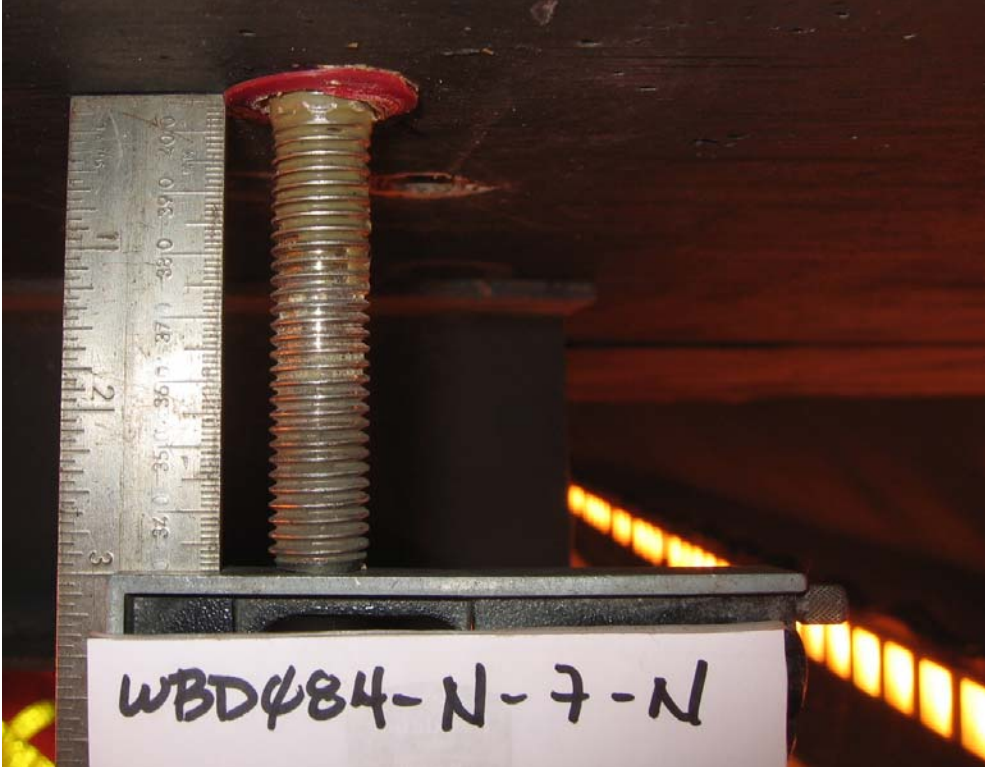
Figure 242. Adhesive Anchor WBD084-M-8-N Prior to Load-Displacement Test.

Anchor 3 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-8-S	3.063			16500 / 2
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 243. Adhesive Anchor WBD084-M-8-S Prior to Load-Displacement Test.

Anchor 4 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-N-7-N	2.969			10000 / 2
				

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 244. Adhesive Anchor WBD084-N-7-N Prior to Load-Displacement Test.

Anchor 5 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WDB084-N-7-S	2.906			7000 / 1
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 245. Adhesive Anchor WDB084-N-7-S Prior to Load-Displacement Test.

Anchor 6 of 16 load-displacement tested July 26, 2006

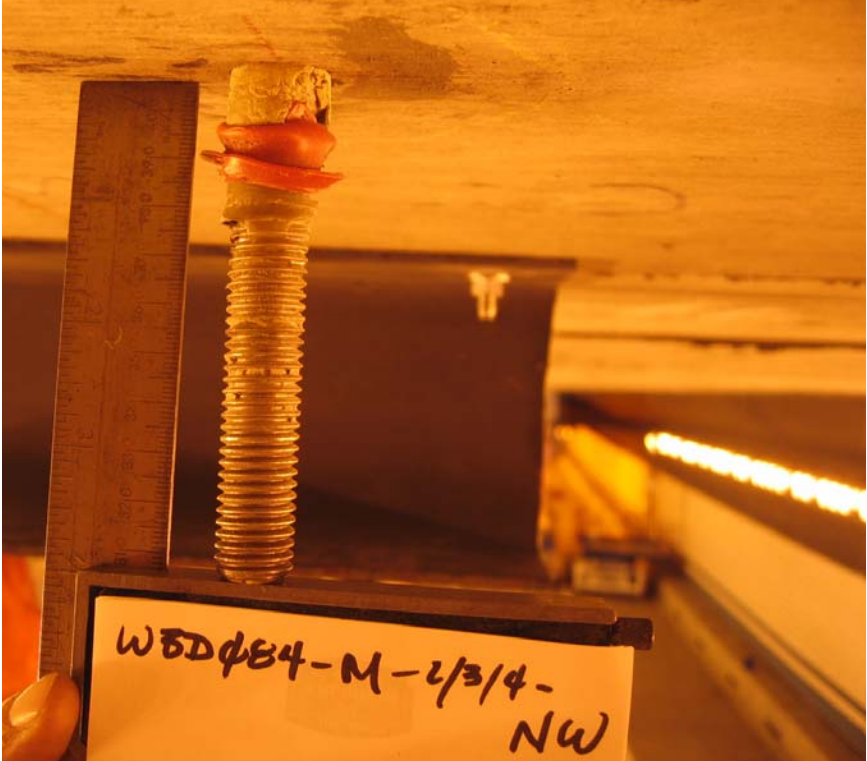
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-2/3/4-NW	3.969	1.281	1.281	1500 / 2
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 246. Adhesive Anchor WBD084-M-2/3/4-NW Prior to Load-Displacement Test.

Anchor 7 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-2/3/4-NE	4.000	1.188	1.188	5800 / 2
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 247. Adhesive Anchor WBD084-M-2/3/4-NE Prior to Load-Displacement Test.

Anchor 8 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-2/3/4-SW	3.188	0.188	0.188	4200 / 0
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 248. Adhesive Anchor WBD084-M-2/3/4-SW Prior to Load-Displacement Test.

Anchor 9 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-2/3/4-SE	3.312	0.344	0.344	5800 / 1
				

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 249. Adhesive Anchor WBD084-M-2/3/4-SE Prior to Load-Displacement Test.

Anchor 10 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-S-10-N	3.281	0.188	0.188	7000 / 1

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 250. Adhesive Anchor WBD084-S-10-N Prior to Load-Displacement Test.

Anchor 11 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-S-10-S	4.000	1.000	1.000	1000 / 2

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 251. Adhesive Anchor WBD084-S-10-S Prior to Load-Displacement Test.

Anchor 12 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-M-2/3/4-SE	2.875 ^b			6000 / 10

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.
^b – Measurement scaled from picture.

Figure 252. Adhesive Anchor WBD082-M-2/3/4-SE Prior to Load-Displacement Test.

Anchor 13 of 16 load-displacement tested July 26, 2006

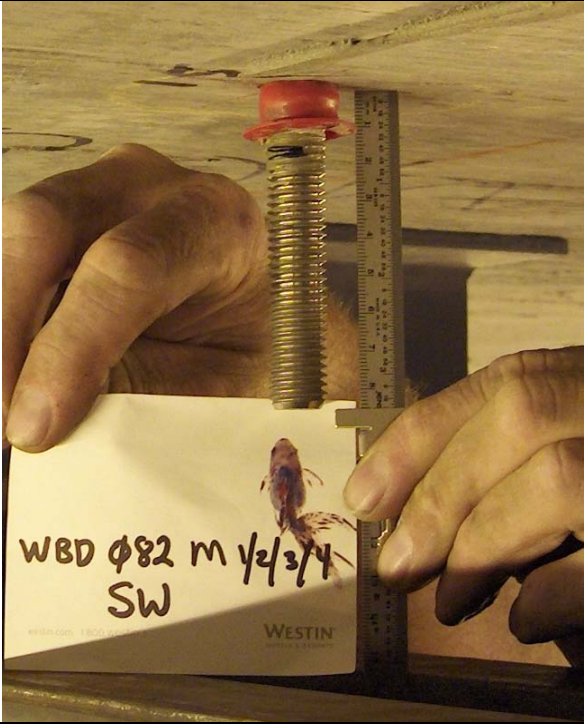
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-M-2/3/4-SW	3.500	0.625 ^b	0.625 ^b	3000 / 3.5
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C. ^b – Measurement scaled from picture.				

Figure 253. Adhesive Anchor WBD082-M-2/3/4-SW Prior to Load-Displacement Test.

Anchor 14 of 16 load-displacement tested July 26, 2006

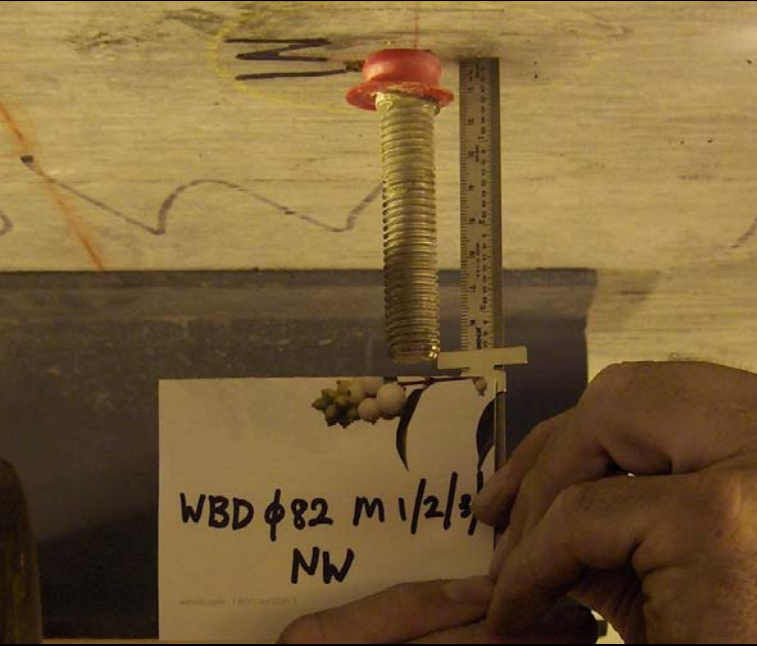
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-M-2/3/4-NW	3.540	0.433	0.433	3000 / 1.5
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 254. Adhesive Anchor WBD082-M-2/3/4-NW Prior to Load-Displacement Test.

Anchor 15 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD083-M-8-N	4.500	1.500 ^b	1.500 ^b	1500 / 1



^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

^b – Measurement scaled from picture.

Figure 255. Adhesive Anchor WBD083-M-8-N Prior to Load-Displacement Test.

Anchor 16 of 16 load-displacement tested July 26, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD083-M-8-S	3.125			7000 / 8

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 256. Adhesive Anchor WBD083-M-8-S Prior to Load-Displacement Test.

Anchor 1 of 17 load-displacement tested July 27, 2006

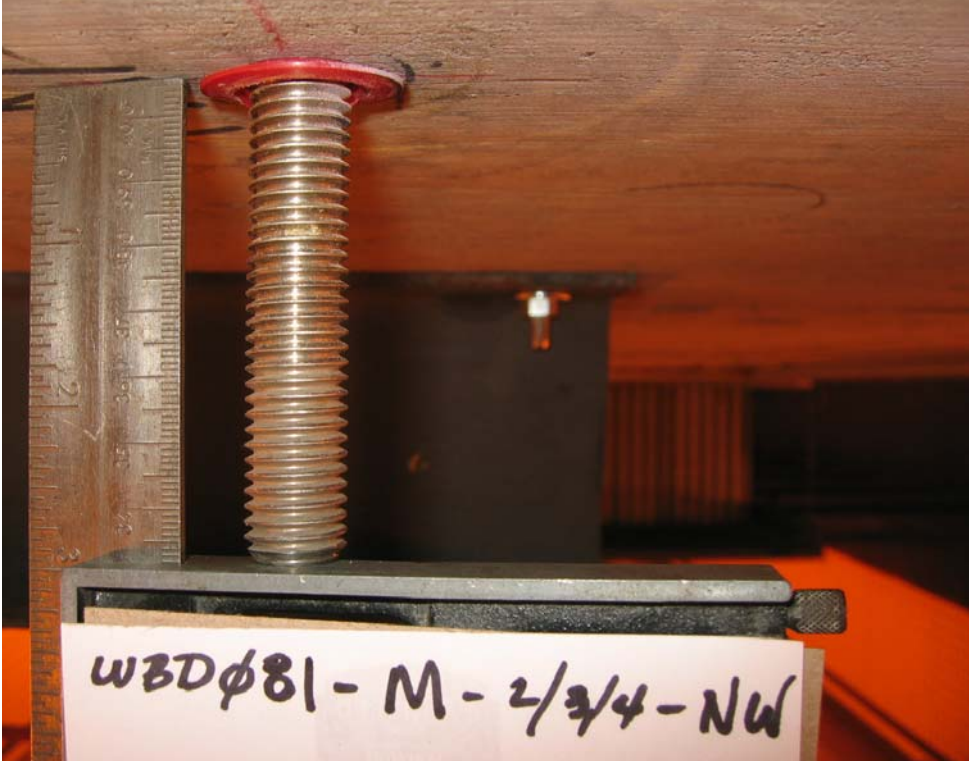
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD081-M-2/3/4-NW	2.938			7000 / 1
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 257. Adhesive Anchor WBD081-M-2/3/4-NW Prior to Load-Displacement Test.

Anchor 2 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD081-M-2/3/4-NE	3.031	0.219	0.219	12500 / 1.5
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 258. Adhesive Anchor WBD081-M-2/3/4-NE Prior to Load-Displacement Test.

Anchor 3 of 17 load-displacement tested July 27, 2006

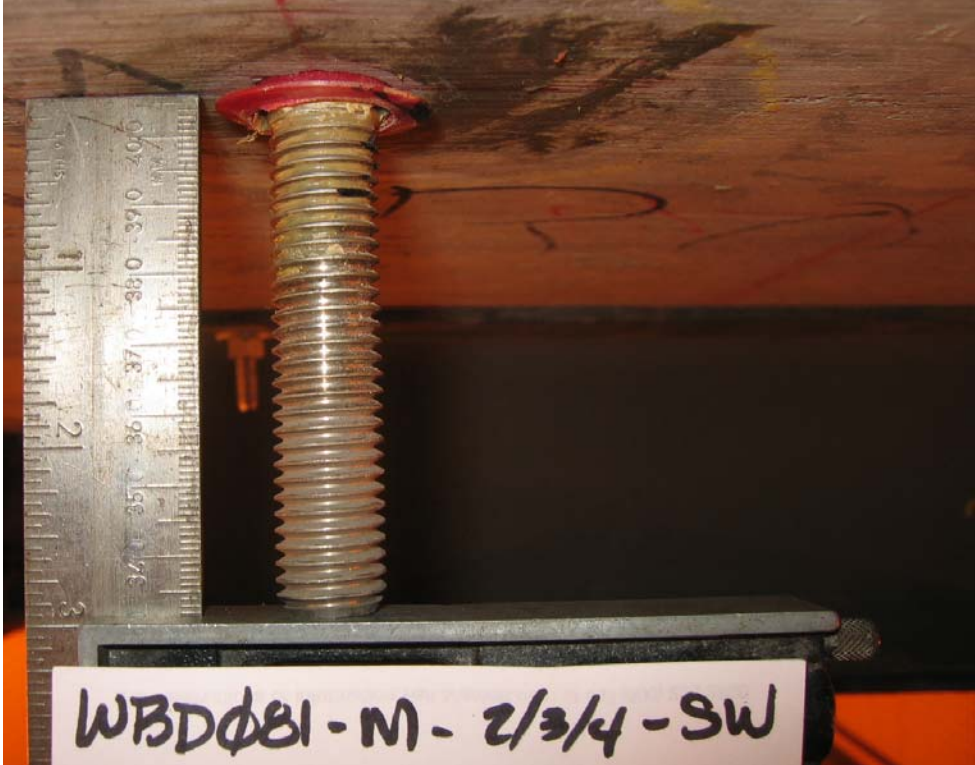
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD081-M-2/3/4-SW	2.938	0.156	0.156	12500 / 1.5
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 259. Adhesive Anchor WBD081-M-2/3/4-SW Prior to Load-Displacement Test.

Anchor 4 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD081-M-2/3/4-SE	2.875	0.188	0.188	11000 / 1.5
				

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 260. Adhesive Anchor WBD081-M-2/3/4-SE Prior to Load-Displacement Test.

Anchor 5 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD081-N-1-N	2.875			11000 / 1.5

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 261. Adhesive Anchor WBD081-N-1-N Prior to Load-Displacement Test.

Anchor 6 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD081-N-1-S	2.844			12500 / 1.5

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 262. Adhesive Anchor WBD081-N-1-S Prior to Load-Displacement Test.

Anchor 7 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD083-M-12-N	3.500	0.438	0.406	4500 / 1
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 263. Adhesive Anchor WBD083-M-12-N Prior to Load-Displacement Test.

Anchor 8 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD083-M-12-S	4.250	1.188	1.188	4500 / 0.33
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 264. Adhesive Anchor WBD083-M-12-S Prior to Load-Displacement Test.

Anchor 9 of 17 load-displacement tested July 27, 2006

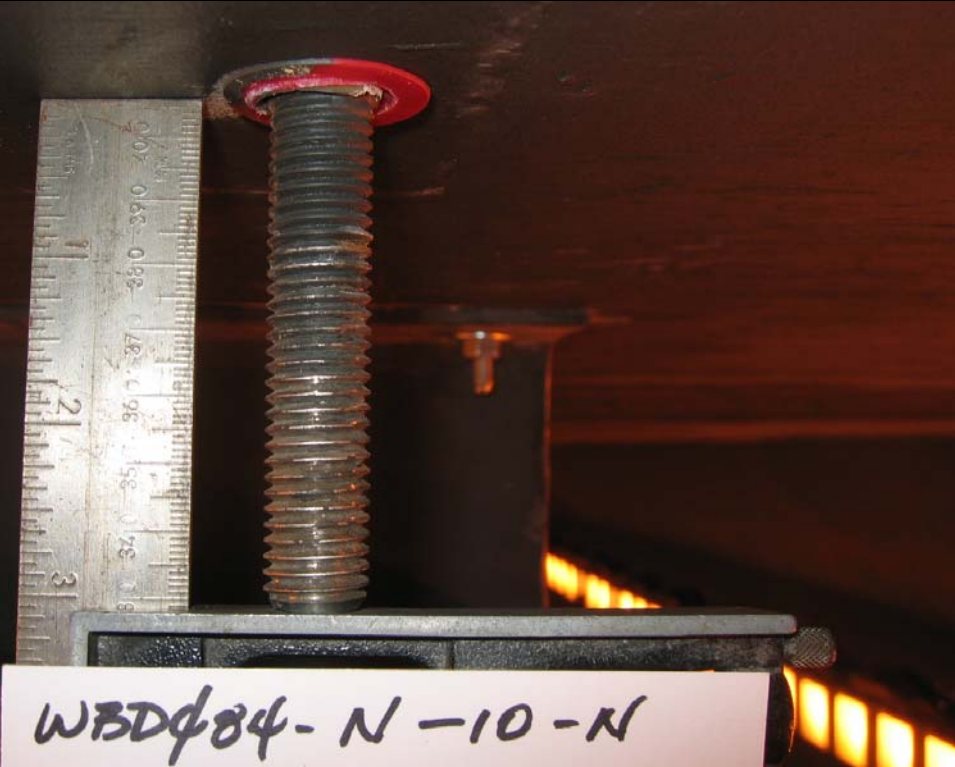
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-N-10-N	3.063			12500 / 1
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 265. Adhesive Anchor WBD084-N-10-N Prior to Load-Displacement Test.

Anchor 10 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-N-10-S	3.031			5800 / 1
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 266. Adhesive Anchor WBD084-N-10-S Prior to Load-Displacement Test.

Anchor 11 of 17 load-displacement tested July 27, 2006

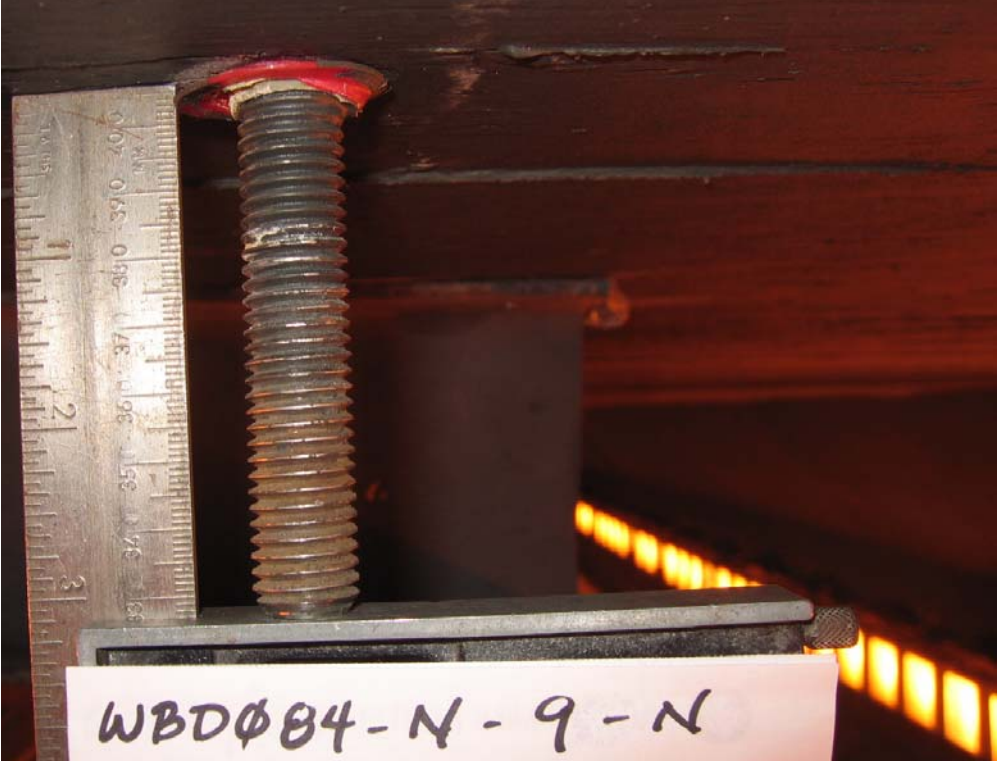
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-N-9-N	3.125			16500 / 1.5
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 267. Adhesive Anchor WBD084-N-9-N Prior to Load-Displacement Test.

Anchor 12 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-N-9-S	3.031	0.125	0.125	15000 / 1.5

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 268. Adhesive Anchor WBD084-N-9-S Prior to Load-Displacement Test.

Anchor 13 of 17 load-displacement tested July 27, 2006

Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD084-M-6-N	3.063	0.125	0.125	17500 / 1
				

^a – Represents the adjusted load value based on load cell calibration, see Appendix C.

Figure 269. Adhesive Anchor WBD084-M-6-N Prior to Load-Displacement Test.

Anchor 14 of 17 load-displacement tested July 27, 2006

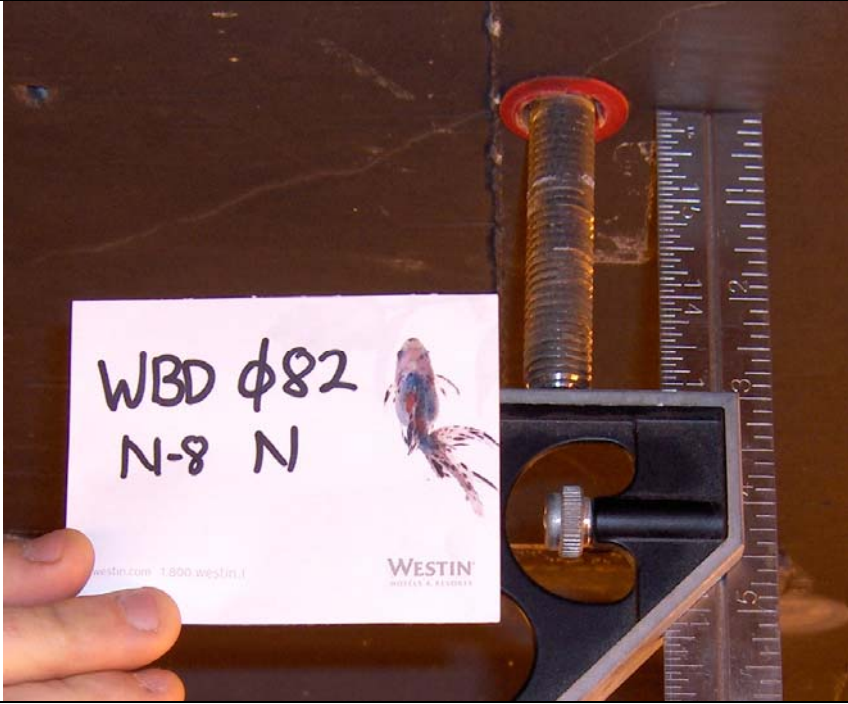
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-N-8-N	3.063			10000 / 10
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 270. Adhesive Anchor WBD082-N-8-N Prior to Load-Displacement Test.

Anchor 15 of 17 load-displacement tested July 27, 2006

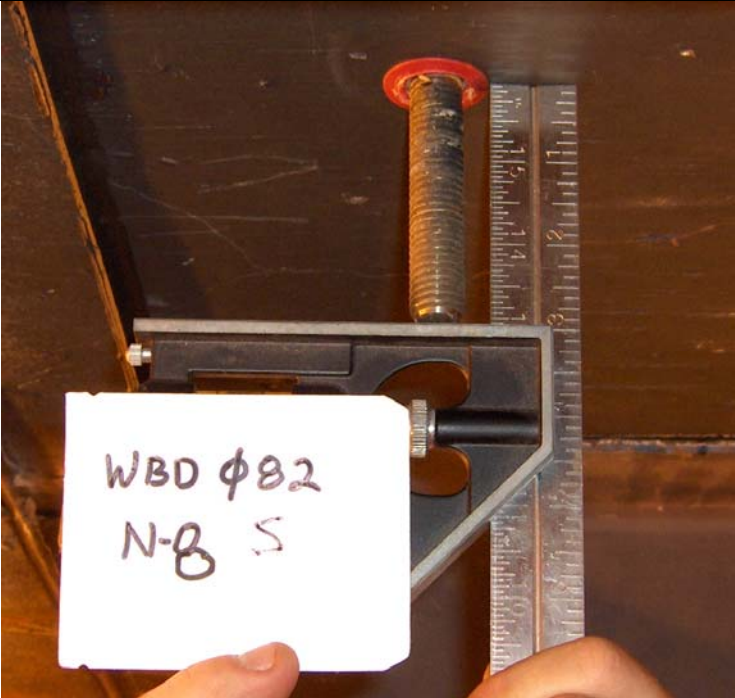
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-N-8-S	3.063			6000 / 5
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 271. Adhesive Anchor WBD082-N-8-S Prior to Load-Displacement Test.

Anchor 16 of 17 load-displacement tested July 27, 2006

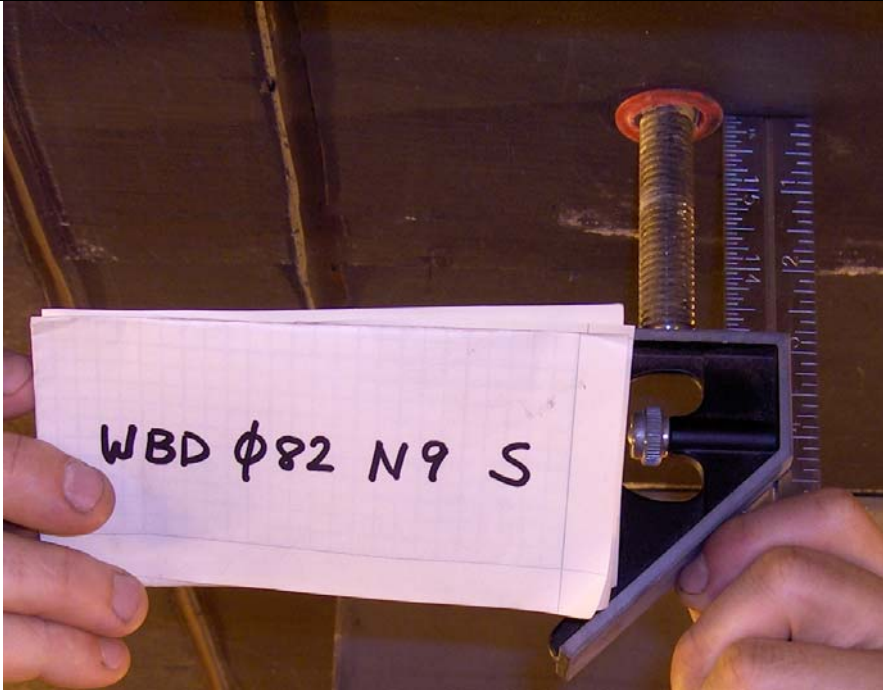
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-N-9-S	3.125			5000 / 9
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 272. Adhesive Anchor WBD082-N-9-S Prior to Load-Displacement Test.

Anchor 17 of 17 load-displacement tested July 27, 2006

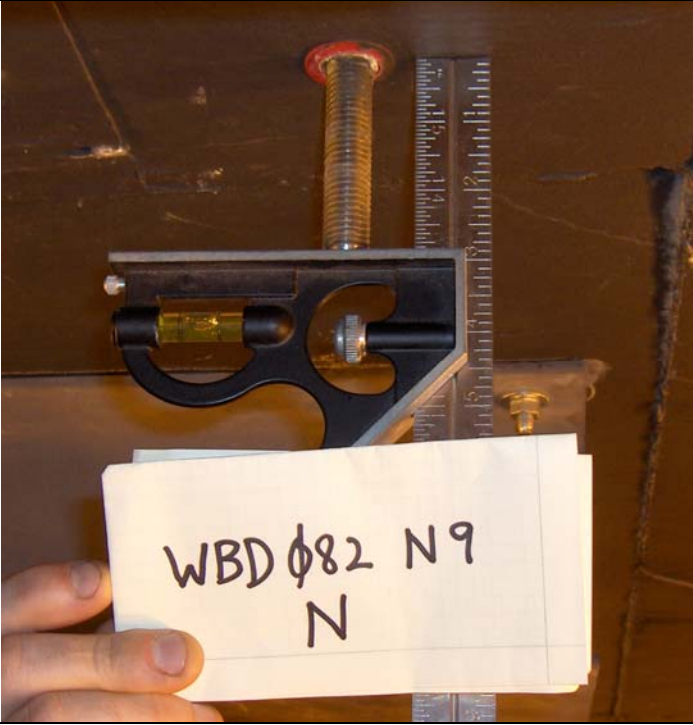
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-N-9-N	3.094			5000 / 25
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 273. Adhesive Anchor WBD082-N-9-N Prior to Load-Displacement Test.

Anchor 1 of 2 load-displacement tested July 28, 2006

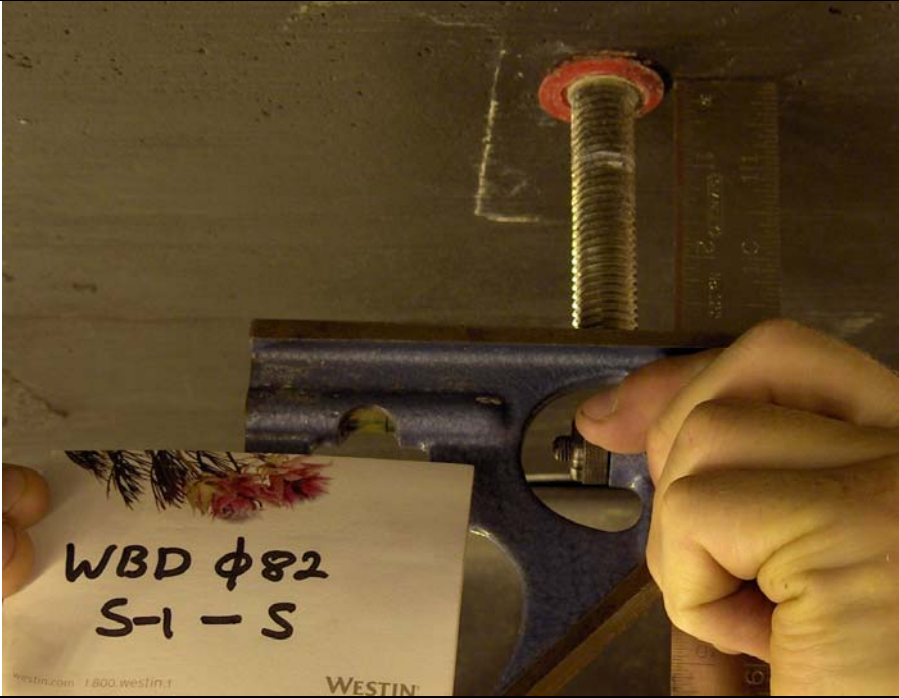
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD082-S-1-S	3.063			9500 / 1
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 274. Adhesive Anchor WBD082-S-1-S Prior to Load-Displacement Test.

Anchor 2 of 2 load-displacement tested July 28, 2006

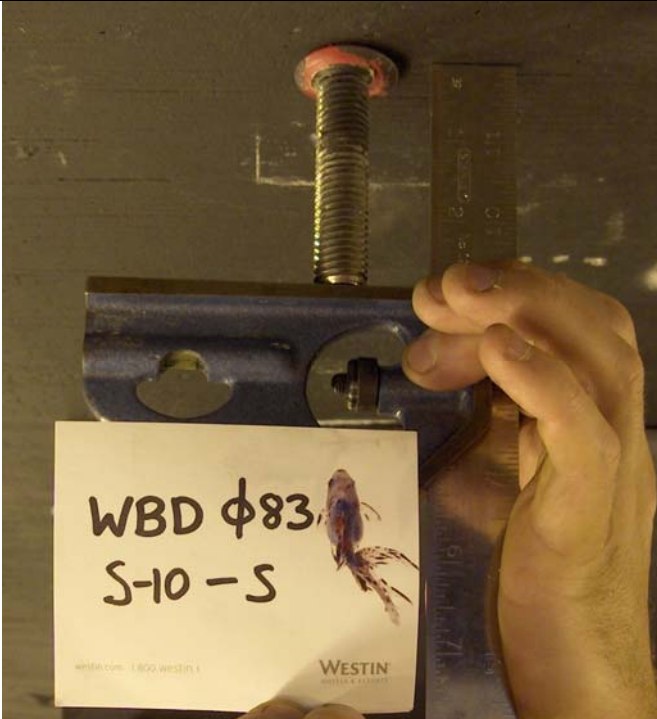
Anchor Label	Initial Protrusion (inches)	Pre-test Displacement		Approximate Peak Load/Hold (lbs) ^a / (min.)
		East Side (inches)	West Side (inches)	
WBD083-S-10-S	2.969			9500 / 0
				
^a – Represents the adjusted load value based on load cell calibration, see Appendix C.				

Figure 275. Adhesive Anchor WBD083-S-10-S Prior to Load-Displacement Test.

APPENDIX C. LOAD CELL CALIBRATION

LOAD CELL USED JULY 15-20, 2006

Load measurements during July 15-20 were made with Wagner Instruments (WI) 50 kip load cells. A formal load cell calibration against a certified load standard was not possible prior to testing. Therefore, a shunt calibration¹ was performed to obtain a sensitivity factor for each load cell used. The shunt calibration utilized a Vishay Model P3 strain indicator box as the external conditioner. The sensitivity factor for each load cell was determined by shunting the circuit with a decade resistor to produce an 80% full-scale reading on the strain indicator box display. The results are tabulated in Table 7. The load cell sensitivities shown in 0 were used for all testing the week of July 15-20, 2006.

Table 11. Model P3 Sensitivity Factors Attained on July 14, 2006

Load Cell S/N	Shunt Resistance (k Ω)	Sensitivity (mV/V)
3349	62.2	2.395
3350	61.8	2.400
3348	62.4	2.385
All three load cells calibrated with Wagner cable S/N 3347, used Channel 1 on the strain indicator, output pound force (lbf), used a 50000 full-scale value, and used an undetermined full-bridge setup.		

A formal calibration of the WI load cell S/N 3348 was performed subsequent to July 15-20 testing on July 24, 2006. Separate calibrations were performed on the S/N 3348 load cell in combination with each of the WI cables, numbers 3347 and 3348, used during testing. The wire in cable number 3347 that failed during testing, was first repaired before it was used in the calibration.

The calibration was conducted by connecting the load cell to the Vishay Model P3 strain indicator configured identically as it was used in the field. The load cell was then loaded in a 1000 kip capacity Forney concrete cylinder testing machine at TFHRC. The Forney machine is calibrated yearly against NIST standards and its last calibration occurred on August 9, 2005. The load cell was loaded in seven increments between approximately 3000 and 20,000 lbs and the display readings from the strain indicator box and the Forney were recorded by hand at each

¹ A load cell relies on a Wheatstone bridge circuit to amplify the resistance changes of strain gauges affixed within the load cell. The resistance changes of the strain gauges are directly related to the load within the load cell. An external conditioner is needed to excite the Wheatstone bridge and provide useful output. As a means to quickly check the calibration of a load cell, it can be "shunted". Shunting applies an artificially high resistance across one leg of the Wheatstone bridge, causing an artificially high load reading from the conditioner. The usefulness of shunt calibrating assumes the load cell has been calibrated against a known load standard, then a specified shunt resistor can always produce the same load reading to ensure accuracy.

level of load. The results of these two calibrations are graphed in 0. Also shown in this figure are the equations resulting from a linear regression performed on each data set.

The calibration shows that exchanging cable 3347 for 3348 had only a small effect on the output of the load cell (~1% difference). More importantly, the calibration data indicates that the resistance applied during the shunt calibration previously performed should have been adjusted to produce a full scale reading from the load cell and not the 80% used. However, since the load cell's performance was confirmed by the calibration, regression equations were used to determine a linear scalar with which the recorded loads were amended. Specifically, load measurements made between July 15-20, 2006 using load cell 3348, were revised by scaling the recorded values by 1.261 and 1.278 respectively for WI cables 3347 and 3348.

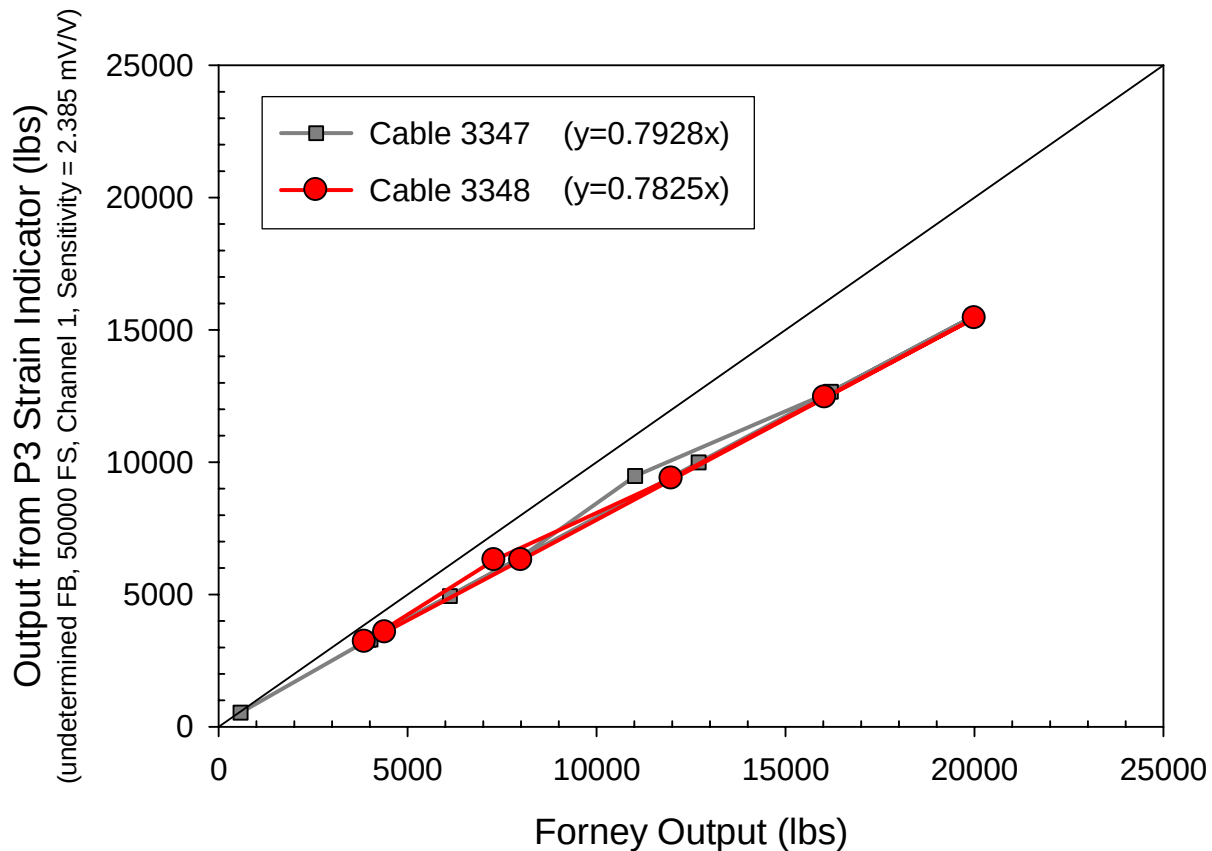


Figure 276. Calibration results of Wagner load cell 3348 on July 24, 2006.

LOAD CELLS USED JULY 26-28, 2006

Load measurements were made July 26-28 with two WI load cells; numbers 3350 and 3348. Prior to testing these load cells were calibrated on Channel 1 of the Vishay Model P3 strain indicator setup with a 50000 lbs full-scale value, undetermined full-bridge orientation, and used WI cable number 3346. The load cells were compressed in a 1000 kip capacity Forney concrete cylinder testing machine which is itself calibrated yearly with a NIST load standard. The load cells were loaded to 50 kips and the sensitivity factor within the Model P3 strain indicator was

adjusted until the indicated load output from the Model P3 and Forney matched. The output from each monitoring system was then checked at five different load levels to verify linear behavior. Both load cells yielded correct output with a sensitivity of 1.807. The analog output from the Model P3 was monitored through the loading process and was found to output 1.25 volts per 45 kips of loading with the analog output in its normal range. The analog output was monitored with a digital voltmeter.

Testing July 26-28 paired load cell 3350 with a MEGADAC data acquisition system, and load cell 3348 with a Data Translation model 9804 USB Data Acquisition Function Module linked with a Dell Latitude C810 laptop computer. Both systems also monitored linearly variable displacement transducers (LVDT) which measured displacement during testing. The LVDTs were calibrated onsite using a table top stand and digital caliper. The voltage output from each LVDT was monitored with the data acquisition systems as a fixed displacement was imposed by the digital caliper. This process generated a calibration factor which was entered into the data collection software to transform voltage into an equivalent magnitude of displacement.

A post-testing formal calibration of load cell 3350 and the MEGADAC system was performed on July 31, 2006. This process was conducted using a 1000 kip capacity Forney concrete cylinder testing machine which is itself calibrated yearly against a NIST load standard. During the calibration process, the MEGADAC was used to monitor the load cell as well as the voltage output from the Forney machine so that concurrent data from both sources could be captured throughout the loading spectrum. The load cell was loaded to 20 kips at a rate of approximately 200 lbs/sec. The calibration graph can be seen in 0. This figure indicates that the output recorded by the MEGADAC is approximately 13.0% higher than the load indicated by the Forney machine which is considered to be a standard reference. Therefore to revise the load data collected with the MEGADAC July 26-28, the recorded loads have been scaled by a factor of 0.885.

The primary source of the discrepancy was found to be initial misconfiguration of the MEGADAC data acquisition system. The MEGADAC was programmed to convert load cell voltage output to load at a proportion of 1.25 volts to 50 kips. However, as reported earlier, the Model P3 strain indicator was found to output 1.25 volt per 45 kips of load. The ratio of 50 to 45 accounts for 11.1% of the 13.0% calibration change. The remaining 1.9% change was due to monitoring the voltage with a digital voltmeter and the MEGADAC and was expected.

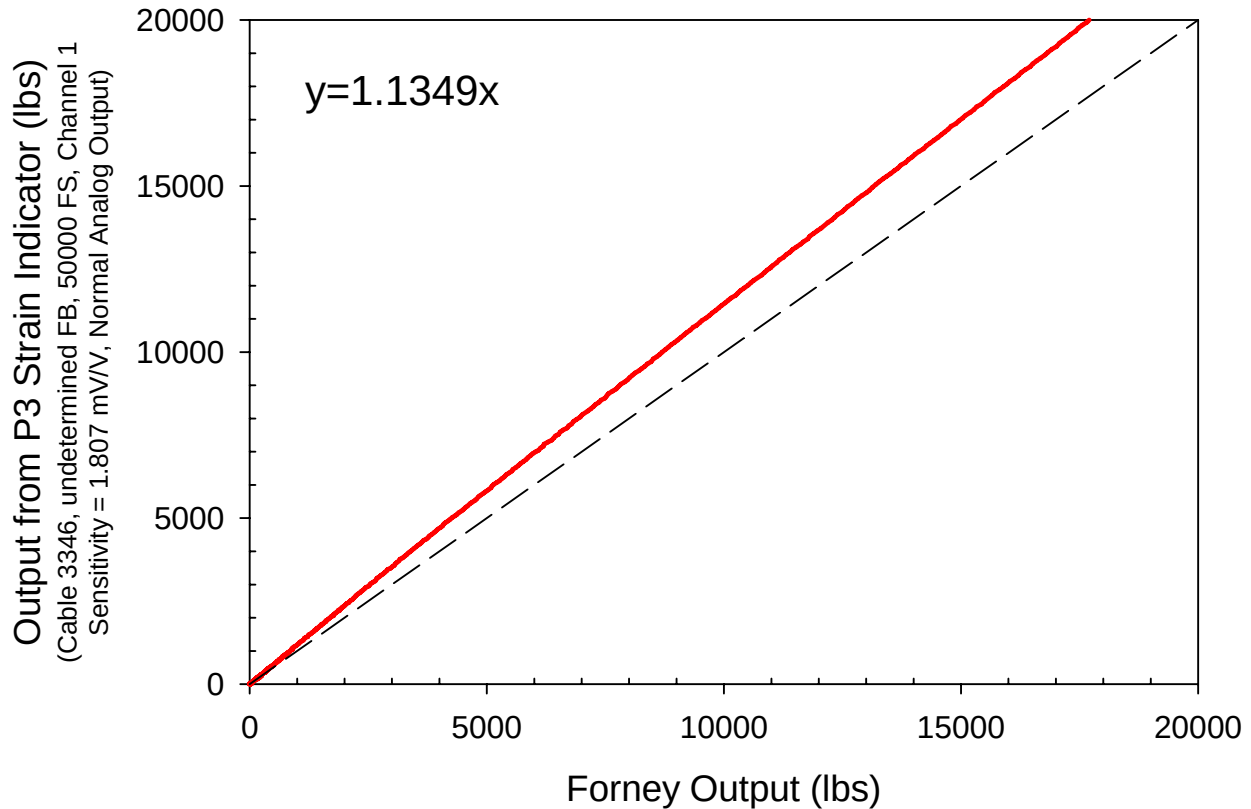


Figure 277. Calibration of Wagner Instruments 3346 with MEGADAC data acquisition system.

A post-testing formal calibration of load cell 3348 and the Data Translation model 9804 USB Data Acquisition Function Module was performed on August 1, 2006. This process was conducted using a 1000 kip capacity Forney concrete cylinder testing machine which is itself calibrated yearly against a NIST load standard. During the calibration process, the Data Translation module was used to monitor the load cell as well as the voltage output from the Forney machine so that concurrent data from both sources could be captured throughout the loading spectrum. The load cell was loaded to 20 kips at a rate of approximately 200 lbs/sec. The calibration graph can be seen in Figure 51. This figure indicates that the load measurement recorded by the Data Translation module is essentially identical to the load indicated by the Forney machine; the data plot almost directly over the 1 to 1 reference line. Therefore, the load data collected with the Data Translation module July 26-28, did not need to be revised.

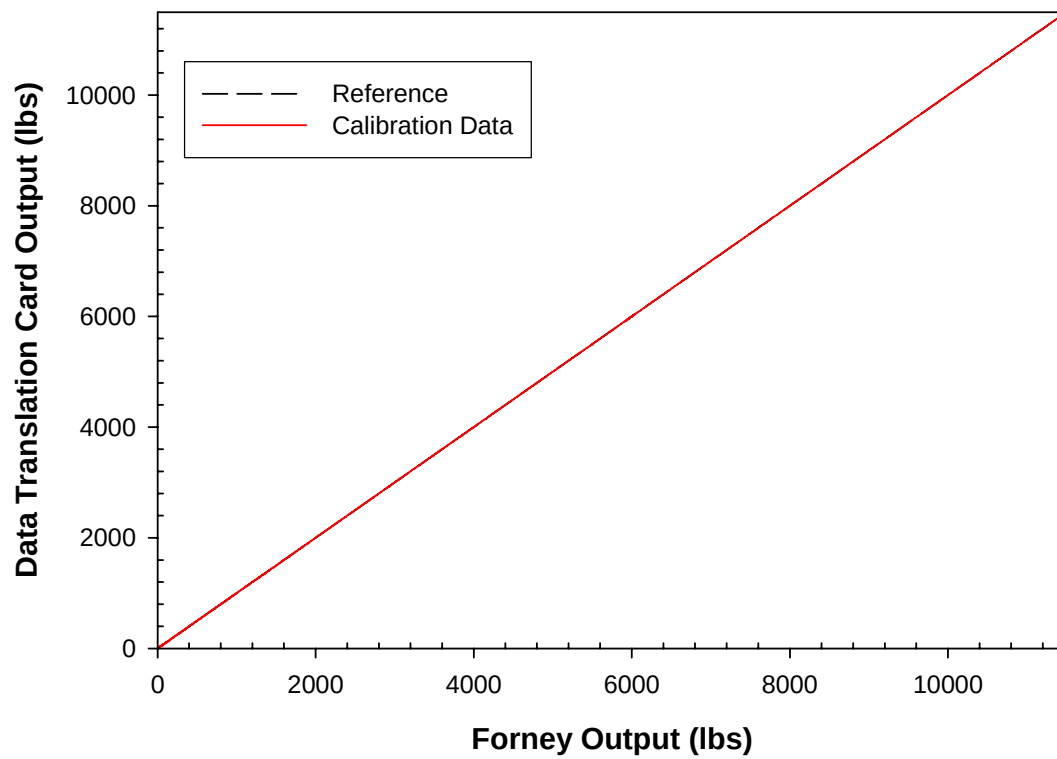


Figure 278. Calibration of Wagner Instruments 3348 with the Data Translation module.